

GE Healthcare

BrightSpeed Elite, Edge, Excel

Pre-Installation Manual

OPERATING DOCUMENTATION



5141177-100

Rev 11

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IMPORTANT PRECAUTIONS

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)	• Това упътване за работа е налично само на английски език. • Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. • Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. • Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациент в резултат на токов удар или механична или друга опасност.
警告 (ZH-CN)	• 本维修手册仅提供英文版本。 • 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。
VÝSTRAHA (CS)	• Tento provozní návod existuje pouze v anglickém jazyce. • V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. • Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. • V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	• Denne servicemanual findes kun på engelsk. • Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. • Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået. • Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	• Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. • Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. • Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is. • Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	• This Service Manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.
HOIATUS (ET)	• Käesolev teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	• Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	• Ce manuel de service n'est disponible qu'en anglais. • Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
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FIGYELMEZTETÉS (HU)	• Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. • Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	• Þessi þjónustuhandbók er eingöngu áanleg á ensku. • Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálapjónustu. • Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. • Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnaða eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	• Il presente manuale di manutenzione è disponibile soltanto in inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. • Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.
警告 (JA)	• このサービスマニュアルには英語版しかありません。 • サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 • このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	• 본 서비스 지침서는 영어로만 이용하실 수 있습니다 . • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 . • 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 . • 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS (LV)	• Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu. • Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	• Šis eksploatavimo vadovas yra prieinamas tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė. • Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote. • Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.
ADVARSEL (NO)	• Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	• Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego. • Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT-BR)	• Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode por em perigo a segurança do técnico, operador ou paciente devido a choques elétricos, mecânicos ou outros.

ATENÇÃO (PT-PT)	• Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutra linguagem, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	• Acest manual de service este disponibil numai în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depănatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	• Данное руководство по обслуживанию предлагается только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	• Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilacke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanimanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehanickih i drugih opasnosti.
UPOZORNENIE (SK)	• Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION (ES)	• Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	• Den här servicehandboken finns bara tillgänglig på engelska. • Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DIKKAT (TR)	• Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION
Risk of
Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION
Danger
d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
11	11/04/08	Chapter 3: updated option list in Section 1.0 Chapter 4: added new FWS Table (part 5168666) illustration in Section 6.4 Section 7.0 updated Injector AC power cable. Section 8.0 updated Floor Levelness Chapter 5: Added the Notice for environmental requirement at beginning of Section 1.0 Section 1.1: Updated Temperature (Scan Room & Control Room) according to SRS requirement. Section 3.0: Updated Altitude values according to SRS requirement. Section 4.0 updated Figure 5-1 of EMI requirement Chapter 6: updated floor loading specification in Section 1.0 added FWS seismic info in section 2.0 Chapter 8, section 4.0: updated Figure 8-1 System Ground Map Chapter 9, section 3.0: updated System Interconnect Diagram Section 3.4: updated table of GEMS Supplied Cables for Options Section 5.0 Updated the Fuse Kit BOM
10	12/03/07	Important Precautions: updated content for Damage In Transportation Chapter 4: updated gantry width Section 2.1.2 updated Regulated Minimum Working Clearance by Major Subsystem Section 6.4 updated for FWS Seismic information Added note to Section 8.4 Floor levelness Added network specs to Section 9.0 Network Connections Chapter 5: Section 4.0 added note Section 4.2 added EMI of color monitor updated Figure 5-1 Sample Room Layout, showing approximate EMI requirements Chapter 6: updated gantry width Chapter 9: added LOTO requires to Section 7.1 Primary Power Disconnect
9	8/13/07	Updated measurements to metric to meet European standards and compliance. General revision of manual. Important Precautions: updated by adding Bulgarian warnings Chapter 4: Added Section 8.7: Ceiling Requirement Removed FWS extended cables Table to Chapter 9, Section 3.2 Chapter 5: updated Section 2.0 Cooling Requirement Chapter 7: Updated section 7.0/8.0/9.0 shipping considerations Updated Section 6.0 System transportation temperature. Chapter 9: Updated Section 5.0 for Fuse Kit
8	6/28/07	Important Precautions: updated multi-language warnings Chapter 4: Added Section 4.0 for Short Footprint consideration
7	4/29/07	Chapter 8: Updated Power Source Configuration
6	12/12/06	Chapter 3: Updated option catalog numbers Appendix: Changed FCT to CT

Revision	Date	Reason for change
5	9/30/06	Chapter 1: Added Section 4 Medical Electrical Equipment for EMC. Chapter 4: Added Limited Access Room information. Chapter 5: Deleted Section 4.8 EMC Edition 2 Compliance - duplicate with newly added Chapter 1, Section 4. Chapter 9: Updated Figure 9-1 System Interconnect Diagram to include Options. Added Section 3.3 Cable list of options, Section 5 Fuse. Added Note: Use dry cleaning for electro components.
4	6/27/06	Chapter 3: Updated Table 3-1 BrightSpeed System Options Chapter 4: Updated recommended room layout and minimum room layout. Updated Section 8.4 - Floor Levelness. Chapter 6: Updated seismic mounting bracket in Figure 6-5 Power Distribution Unit. Appendix: Updated <i>CT Installation Site Ready Form</i> .
3	3/31/06	Chapter 4: Updated room size dimensions and typical room layout; Updated Figure 4-4: Typical Control Room Layout with Freedom Workspace (FWS) Desktop; Chapter 6: Updated Table 6-1: BrightSpeed System Floor Loading; Chapter 8: Added note that 75KVA rating is for BrightSpeed Excel (4 slice) only.
2	2/15/06	Chapter 4: Added Freedom Workspace (FWS) extended cables length information.
1	1/24/06	Initial Release.

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Chapter 1

Introduction

This direction contains physical and electrical data necessary for planning and preparing a site. Pre-installation work is defined as site preparation for installation of the GE CT scanner. It is the responsibility of the purchaser to arrange and pay for this work. Pre-installation work includes:

- Installation of electrical conduit, junction boxes, ducts, outlets, and line safety switches.
- Installation of interconnection wiring that is AWG stranded copper. The electrical contractor shall ring out and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Ground wires must conform to local codes.
- Any site renovation.
- Alterations and modifications to products not specifically included in the sales contract.

All work must conform to local building and safety codes. Unless specifically mentioned, GE Healthcare does not provide or install wires, conduits, junction boxes, and ducts as illustrated in this publication.

All CT site plans, preliminary concepts and final working drawings must be reviewed by General Electric Headquarters Architectural Planning prior to construction or approval.

Contact your local General Electric sales representative for complete information regarding your site-specific room layout.

Section 1.0

Site Readiness

Site ready is a requirement that must be achieved to install a CT product. For your convenience, a site ready visit inspection shall be performed at least three (3) days prior to the installation date. The site inspection must conclude with a minimum of a conditional pass status to be ready on the requested installation delivery date. Site ready inspections on the delivery date will not be acceptable unless prior arrangements have been made.

Pre-Installation and Site Ready Tools:

- BrightSpeed Floor template
- Pre-Installation check List
- Pre-Installation Block Diagram
- Site Room Layouts
- Power and Grounding Inspection
- Pre-Installation Support

Section 2.0

Responsibility of Purchaser

2.1 Customer Room Prep Items

The CT air intake is near the bottom of the gantry. Fine dust as listed below will be deposited throughout the gantry, table, console and PDU electronics. This fine dust cannot be completely removed and can be damaging to electronic components.

For these reasons, the scanner should be the last item installed in your CT suite area.

“Pre-installation” is work necessary to plan and prepare a site for installation of equipment.

Pre-installation work helps the user (customer) avoid:

- Application delay and scheduling
- Surprise siting discoveries
- Installation confusion
- Waste of manpower

The following **MUST** be completed before installation work can begin for a GE CT scanner:

- Completely finished:
 - Wall painted or have final wall covering
 - Ceiling tiles installed and no remaining ceiling work is required
 - Final floors covering installed with no remaining dust causing floor work required
 - All room millwork installed as shown on the site print
 - All plumbing work in the CT suite is completed
 - No construction in or around the scan suite AREA that will produce:
 - * Concrete dust
 - * Drywall dust
 - * Ceiling tile dust
 - * Wood sawdust or shaving
 - * Dust tracked into the CT suite area
- Active Broad Band connection
 - A completed network connection is required for ALL CT installations.
 - A GE Healthcare network specialist may be required to complete the VPN connection. This may take a week or longer to schedule.
- Power available to A1, with provision for Lockout/Tagout at the A1 disconnect
 - If a UPS is required, a GE A1 breaker* will be needed to complete this installation. Refer to the electrical section for more details.



NOTICE **SERVICE NOTICE: An improperly prepared site (i.e., one that is in a state of construction) can result in increased installation time.**

A CT scanner installed in a dirty environment is more prone to contamination, which can result in decreased reliability and increased scanner downtime.

2.2 Purchaser Site Preparation Work

This list below will describe many of the items to consider when planning for a system replacement or designing a room for new equipment

- Determine room dimensions and verify that doorways are large enough for the scanner system.
- Install appropriate conduits and duct work for system cables. If additional components are required in the CT suite, their connection consideration must be determined and completed.
- Install junction boxes of correct size with covers at locations shown in installation plan.
- A1 main disconnect installation
- Install power supply of correct voltage output and adequate KVA rating.
- Install local disconnects, including proper over-current protection.
- Install “steelwork” or other suitable support work for mounting equipment on walls or from ceiling.
- Camera should be on-site at the time of installation.
- Complete all suite and room alterations and modifications.
- Verify that room shielding is adequate for the system being installed
- Review structural requirement - including floor vibration, levelness, and thickness
- Review HVAC requirements including system regulation and patient comfort.
- Review operational clearances to see if your daily used items fit, such as beds and carts.
- Emergency medical equipment should also be considered
- Storage cabinets and sink (if required) must be shown on the site print
- These contractors and others may be required to help confirm that the site meet all installation requirements:
 - Structural Engineer and /or Architect
 - HVAC contractor
 - Electrical contractor
 - Qualified radiological health physicist

The above items can be found in chapter 2 through 9 in this manual.

It is suggested that this work be completed at least three days prior to delivery

2.3 Manufacturer’s System Level Siting Requirements

These siting requirements are the minimum that must be met in order to install a new or replacement system.

- Network Communication in place and active
- Meets all scan room regulatory and service requirements
- Meets all minimum scan room structural requirements
- Meets minimum scan room HVAC requirements
- Meets minimum scan room electrical requirements
- Reviewed radiation protection section in the Pre-Installation manual
- All in room items shown on the final GE Healthcare site print and the final print is on site
- No construction in the scan room or neighboring suite areas

It is suggested that this work be completed at least three days prior to delivery

2.3.1 Meeting Site Ready Requirements

The site ready visit will take place at least three days prior to the delivery date. The site ready visit is intended to verify that all of the siting requirement are met and the site is ready for installation.

The site ready visit will result in a report to the project manager indicating one of the following:

Pass - All required items are present, completed and the site is ready for installation.

Conditional Pass - is issued when 80% of all of the task are completed and all parties agree that the 20% will be completed by the installation delivery date.

If a "Conditional Pass" is granted on the inspection date, the project manager must present conclusive evidence that unfinished tasks are completed and that the site is ready for delivery one business day prior to delivery.

Fail - is issued when less than 80% of the task are completed and all parties cannot agree that the remaining work will be completed by the requested installation delivery date. Failed sites will be rescheduled when all items are completed.

2.3.2 Quick Installs

Quick installations are described as sites with minimum room improvements required. These include, but are not limited to the following items:

- Existing electrical disconnect device, wire size and grounds meet all of the above requirements.
- Existing structural items including floor thickness meet all of the above requirements
- Existing HVAC capacity and regulation meet all of the above requirements
- Existing CT suite meets all of the above regulatory and minimum size requirements
- Existing facility can accommodate the delivery and meet all of the above delivery requirements

Quick Installs are subject to the following restrictions:

- Quick installs must have a new room print that accurately reflects the rooms to be upgraded.
- Existing floor anchors from a non-BrightSpeed system CANNOT be reused.
- New floor anchors must be a minimum of 4" from any existing floor penetrations.

Quick Installs typically involve a weekend de-install and room prep completion, with a next business day delivery and install.

2.3.3 Site Ready Inspection Visit

Must meet all in section 1 and 2 plus these additions reviewed at the site ready visit.

The GE Healthcare project manager will review the site delivery process with you to determine how to best transfer the equipment from the transportation truck to your room.

This site ready inspection will review and check these items:

Delivery information

- Determine delivery route into the scan room
- Determine if tilt dollies or riggers are required
- Determine if elevators, doorways and hallways are adequate for delivery
- Determine if floor protection is required
- Determine if a tilt bed truck is required for ground delivery and ordered.

Regulatory Requirement

- Room size meets the minimum requirements
- Site print is present and accurately reflects the room size and layout.
- No grounded walls are present in the regulatory clearance areas
- All regulatory clearances space is met
- Room meets all local codes

Manufacturer Requirements: As listed in section 2 - all requirements are met

Purchaser's Site Preparation Work: As listed in section 1 - all actions are completed

Section 3.0

Pre-Installation Checklist

GE	CUST	DATES
Y N	Y N	
	☐ ☐	Project schedule verified with contractor, facilities department, and GE?
	☐ ☐	Can you meet the committed site ready date?
☐ ☐	☐ ☐	Construction completion date matches delivery date?
☐ ☐	☐ ☐	Power & Ground survey complete, date _____ Hospital contact _____
	☐ ☐	Delivery date is scheduled for: _____ (after site ready visit)
	☐ ☐	First Use date is scheduled for: _____
☐ ☐	☐ ☐	Does the delivery and / or installation date need to be adjusted?
☐ ☐	☐ ☐	Applications dates: On Site scheduled for: _____, Education Center Training: _____
	☐ ☐	Person assigned to review and verify that all installation requirements are met? (Site Ready Visit)
☐ ☐	☐ ☐	Site Ready Visit is Scheduled for: _____

GE	CUST	GENERAL / SITE REQUIREMENTS
Y N	Y N	
		<i>Must be completed 5 week prior to the delivery</i>
☐ ☐	☐ ☐	Final drawings distributed to the contractors and reviewed with GE?
	☐ ☐	Have the Radiologist Health Physician review the room layout.
	☐ ☐	Final print(s) "signed off" that approve equipment layout / orientation?
	☐ ☐	Have the specific site requirements been discussed with the contractor? Refer to the GE final drawings specifications. (See Table 1-1, below)
	☐ ☐	Has the responsibility of cabling, installing, interfacing accessories not on the order been discussed? (Refer to service price pages for GE's support of 3rd party accessories)
	☐ ☐	All 3 rd party vendors identified, notified and scheduled? (i.e., Netcom, 3M, Kodak, Medrad, etc.)
	☐ ☐	Will existing network, broadband, and camera cable drops reach new locations / requirements / function with BrightSpeed?
	☐ ☐	Have the following requirements (in Table 1-1) been reviewed by your contractor

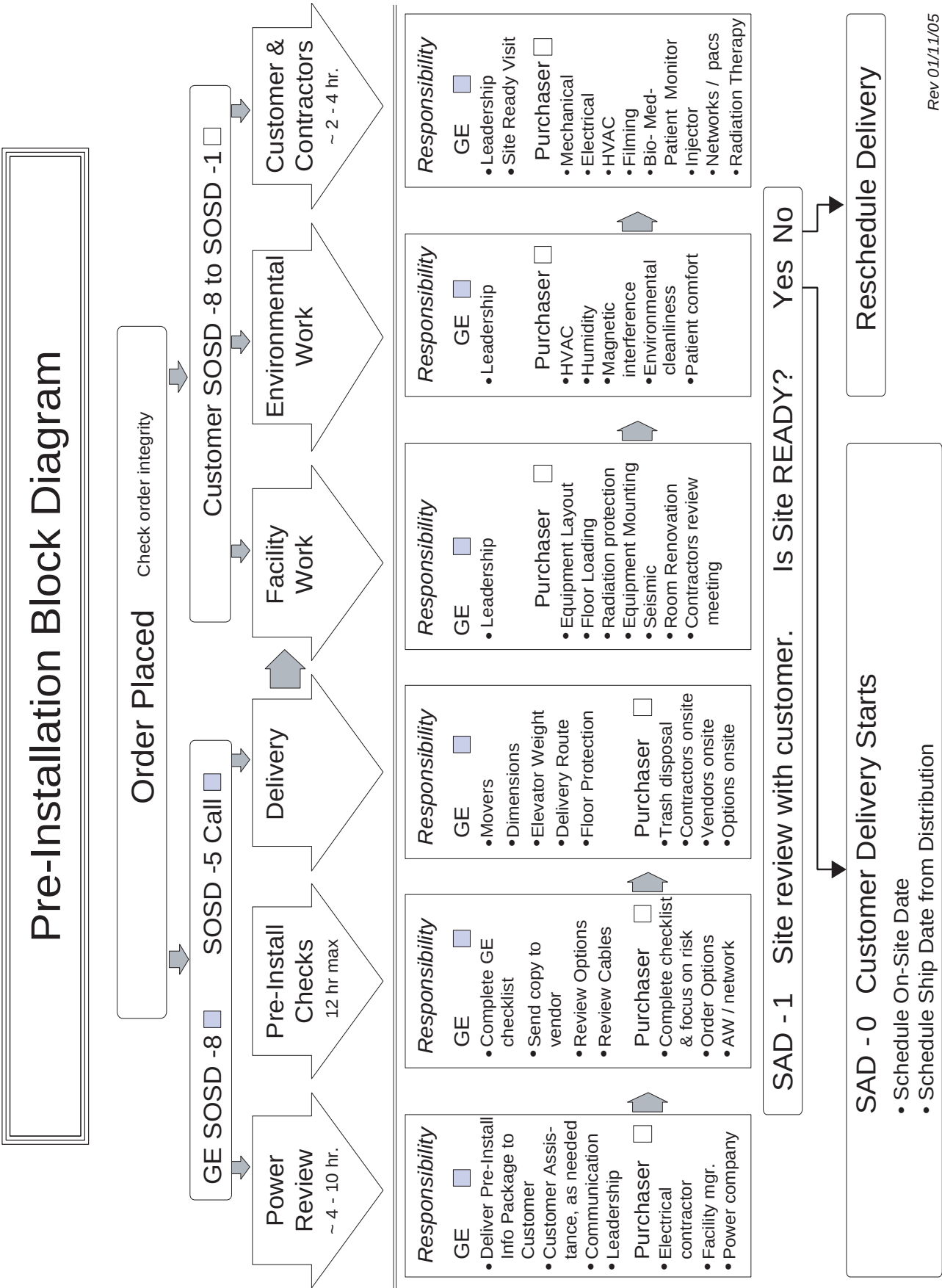
Requirements	
• Air Conditioning	• Structural
• Electrical	• Radiation Protection
All work by contractors must be completed prior to GE delivery	

Table 1-1 Specific Site Requirements

GE	CUST	EQUIPMENT
Y N	Y N	<i>Must be completed 5 weeks prior to the delivery</i>
☐ ☐	☐ ☐	Order reviewed for completeness and compatibility with existing equipment.
	☐ ☐	Accessories on the order verified for compatibility (i.e. DASMs, cable lengths, laser cameras, AW work station, injector and other NL items, etc.) Verify these will work with your CT System
	☐ ☐	Interfaces to existing and/or new accessories ordered and planned for accordingly.
	☐ ☐	Has the location of peripherals been determined and planned accordingly with GE representatives? EKG monitor___ Injector control___ Laser camera ___ UPS ___ 2 nd Monitor___
☐ ☐	☐ ☐	Have any additional services to be provided by GE been discussed and agreed upon?
	☐ ☐	Remote monitors ___ AW relocation ___ Wall mounted accessories ___
☐ ☐		Are correct length cables on Order?

GE	CUST	NETWORKING
Y N	Y N	<i>Must be completed 5 weeks prior to the delivery</i>
☐ ☐	☐ ☐	Have required IP addresses and Host Names been obtained?_____
	☐ ☐	Will a network camera be used?_____
☐ ☐	☐ ☐	Mandatory Item: Broad-band installed? _____
☐ ☐	☐ ☐	Optional Item: CT service telephone line identified and installed for the InSite (<i>Electrical, mechanical, etc.</i>)
☐ ☐	☐ ☐	Mandatory Item: Network installed? _____ Network jacks installed and tested?
☐ ☐	☐ ☐	Mandatory Item: Network options ordered _____ HIS RIS option ___ DICOM print ___ AW _____
	☐ ☐	Test network connections.
☐ ☐	☐ ☐	Optional Item: Has modem option ordered? _____ (Requires a Deviation)

GE	CUST	OTHER
Y N	Y N	<i>Must be completed prior to the delivery</i>
☐ ☐	☐ ☐	Review: Arrangements made in the schedule to allow for remodeling, if required. (i.e., wall, floor, or ceiling repair work, painting, other cosmetic finishes)
☐ ☐	☐ ☐	Have arrangements been made to clean the floor <i>after</i> equipment removal and <i>prior</i> to reinstall?
☐ ☐	☐ ☐	Is de-installation required? No___ Yes ___ CT removal date _____
☐ ☐	☐ ☐	Delivery route identified, and verified with the proper hospital personal? _____
☐ ☐	☐ ☐	Will acceptance testing or Bio-Medical testing be required?
☐ ☐	☐ ☐	Trash bins available for the removal of papers/ boxes/ etc. during the installation.
	☐ ☐	Is floor protection required? _____
	☐ ☐	Is this a dock level delivery? _____
	☐ ☐	If a ground delivery is required, are a tilt bed truck and/or riggers required?
	☐ ☐	Is a tilt table or zero clearance dollies required?
		If yes, order from the UMI web site: http://www.umi-dollyshop.com/default1.htm



Section 4.0

Medical Electrical Equipment for EMC

4.1 General Scope

This equipment complies with IEC60601-1-2 Edition 2 EMC standard for medical electrical equipment.

The BrightSpeed System is suitable to be used in the electromagnetic environment, as per the limits & recommendations described in the tables hereafter:

- Emission Compliance level & limits (see [Table 4-2](#))
- Immunity Compliance level & recommendations to maintain equipment clinical utility (see [Table 4-3](#), [Table 4-4](#), and [Table 4-5](#)).

Note: This system complies with above mentioned EMC standard when used with supplied cables up to maximum lengths referenced in the MIS MAPS or system cable interconnect diagrams.

4.2 Electromagnetic Emission

Table 4-2 Emission Declaration

EMC Emissions Guidance & Declaration for BrightSpeed System		
The BrightSpeed System is intended for use in the electromagnetic environment specified below. The customer or the user of the BrightSpeed System should assure that it is used in such an environment.		
Emissions Test	Compliance	Electromagnetic Environment Guidance
RF emissions CISPR 11	Group 1	The BrightSpeed System uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class A	
Harmonic emissions IEC 61000-3-2	Not applicable	The BrightSpeed System is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Voltage fluctuations/ flicker emissions IEC 61000-3-2	Not applicable	

4.3 Electromagnetic Immunity

Table 4-3 Immunity Declaration

EMC Immunity Guidance & Declaration for BrightSpeed System			
The BrightSpeed System is intended for use in the electromagnetic environment specified below. The customer or the user of the BrightSpeed System should assure that it is used in such an environment.			
Immunity Test	IEC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	± 6 kV contact ± 8 kV air	± 6 kV contact ± 8 kV air	Floors should be wood, concrete, or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	± 2 kV for power supply lines ± 1 kV for input/output lines	± 2 kV for power supply lines ± 1 kV for input/output lines	Mains power quality should be that of typical commercial or hospital environment.
Surge IEC 61000-4-5	± 1 kV line-line ± 2 kV line-earth	± 1 kV line-line ± 2 kV line-earth	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	< 5 % U_T (> 95% dip in U_T) for 5 sec	< 5 % U_T (> 95% dip in U_T) for 5 sec	Mains power quality should be that of a typical commercial or hospital environment. If the user of the BrightSpeed System requires continued operation during power mains interruptions, it is recommended that the BrightSpeed System be powered from an uninterruptible power supply or a battery.
Powerfrequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
Note : U_T is the a.c. mains voltage prior to application of the test level.			

Table 4-4 Immunity Declaration con't

EMC Immunity Guidance & Declaration for BrightSpeed System			
The BrightSpeed System is intended for use in the electromagnetic environment specified below. The customer or the user of the BrightSpeed System should assure that it is used in such an environment.			
Immunity Test	IEC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Conducted RF IEC 61000-4-6	3 V_{RMS} 150kHz to 80 MHz	3 V 150kHz to 80 MHz	Portable and mobile RF communications equipment should be used no closer to any part of the BrightSpeed System, including cables, than the recommended separation distance calculated from the equation appropriate for the frequency of the transmitter. Recommended Separation Distance $d = \left[\frac{3.5}{3} \right] \sqrt{P}$ (see Table 4-5)
Radiated RF IEC 61000-4-3 (alternative method: IEC 61000-4-21)	3 V/m 80 MHz to 2.5 GHz	3 V/m 80 MHz to 2.5 GHz	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$ 80 MHz to 800 MHz (see Table 4-5) $d = \left[\frac{7}{3} \right] \sqrt{P}$ 800 MHz to 2.5GHz (see Table 4-5) where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey^a, should be less than the compliance level in each frequency range^b. Interference may occur in the vicinity of equipment marked with the following symbol: 
a Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast, and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the BrightSpeed System is used exceeds the applicable RF compliance level above, the BrightSpeed System should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the BrightSpeed System. b Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m. Note : These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.			

Table 4-5 Separation Distances

Recommended separation distances between portable and mobile RF communications equipment and the BrightSpeed System.			
The BrightSpeed System is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the BrightSpeed System can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the BrightSpeed System as recommended below, according to the maximum output power of the communications equipment.			
Rated Maximum Output Power (P) of Transmitter Watts (W)	Separation distance according to frequency of transmitter		
	150 kHz to 80 MHz	80 MHz to 800MHz	800 MHz to 2.5 GHz
	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$	$d = \left[\frac{7}{3} \right] \sqrt{P}$
	Separation Distance meters	Separation Distance meters	Separation Distance meters
0.01	0.12	0.12	0.23
0.1	0.37	0.37	0.74
1	1.17	1.17	2.33
10	3.69	3.69	7.38
100	11.7	11.7	23.3
For transmitters rated at a maximum output power not listed above, the separation distance can be estimated using the equation in the corresponding column, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer. Note 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies. Note 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.			

4.3.1 Limitations Management :

Adhering to the distance separation recommended in [Table 4-5](#), between 150kHz & 2.5GHz, will reduce disturbances recorded at the image level but may not eliminate all disturbances. However, when installed and operated as specified herein, the system will maintain its essential performance by continuing to acquire, display, and store diagnostic quality images safely.

(*) For example, a 1W mobile phone (800MHz to 2.5GHz carrier frequency) shall be put 2.3 meters apart from the BrightSpeed System (in order to avoid image interference risks).

4.4 Use Limitation :

4.4.1 External components

The use of accessories, transducers, and cables other than those specified may result in degraded ELECTROMATHNETIC COMPATIBILITY of the BrightSpeed System .

4.5 Installation Requirements & Environment Control :

In order to minimize interference risks, the following requirements shall apply.

4.5.1 Cable Shielding & Grounding

All interconnect cables to peripheral devices must be shielded and properly grounded. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

4.5.2 This product complies the radiated emission as per CISPR11 Group 1 Class A standard limits

The BrightSpeed System is predominantly intended for use, in non-domestic environments, and not directly connected to the Public Mains Network. The BrightSpeed System is predominantly intended for use (e.g. in hospitals) with a dedicated supply system, and with a X-ray shielded room. In case of using in a domestic environment (e.g. doctor's offices), in order to avoid interferences, it is recommended to use a separated AC power distribution panel & line with a X-ray shielded room.

4.5.3 Subsystem & Accessories Power Supply Distribution

All components, accessories subsystems, systems which are electrically connected to the BrightSpeed System, must have all AC power supplied by the same power distribution panel & line.

4.5.4 Stacked Components & Equipment

The BrightSpeed System should not be used adjacent to or stacked with other equipment; if adjacent or stacked use is necessary, the BrightSpeed System should be observed in order to verify normal operation in the configuration in which it will be used.

4.5.5 Low Frequency Magnetic Field

In case of a digital BrightSpeed System, the Gantry (digital detector) shall be apart 1meter from the generator cabinet, and 1meter apart from the analog (CRT) monitors. These distance specifications will minimize the low frequency magnetic field interference risk.

4.5.6 Static Magnetic Field Limits

In order to avoid interference on the BrightSpeed system, static field limits from the surrounding environment are specified.

Static field is specified less than <1 Gauss in Examination room, and in the Control Area.

Static field is specified less than <3 Gauss in the Technical Room.

4.5.7 Electrostatic Discharge Environment & Recommendations

In order to reduce electrostatic discharge interference, install a charge dissipative floor material to avoid electrostatic charge buildup.

The relative humidity shall be at least 30 percent.

The dissipative material shall be connected to the system ground reference, if applicable.

Chapter 2

Pre-Installation Overview

Before a BrightSpeed system can be installed, all pre-installation requirements must be complete.

- Chapter 4, Sections [8.0](#) and [9.0](#)
Structural Requirements
- Chapter 4, Section [10.0](#)
Radiation Protection
- Broadband standard
- Site Ready Visit
- [Chapter 5, Environmental Conditions](#),
Sections [1.0](#) & [2.0](#) Temp, Humidity & Cooling
- [Chapter 8, Power Requirements](#)
(Site Power Audit Required)
- [Chapter 9, Interconnection Data](#)

Section 1.0

Site Ready for Installation

Site-specific items must be verified before the installation can begin.

1.1 Dust/Dirt Contamination

The BrightSpeed systems (consisting of: Console, PDU, Table and Gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site.



NOTICE Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.

1.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of BrightSpeed Series components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration's should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

1.3 Walls, Ceiling, and Floor

All walls, ceiling, and flooring must be completed before installation can begin.

For structural requirement, refer to Chapter 4, [Section 8.0, on page 65](#).

1.4 Broad-band

For information on Broad-band requirements, refer to Chapter 4, [Section 9.0, on page 67](#).

1.5 Phone Line (for optional modem)

Two phone lines must be installed at or near the console and be operational prior to installation.

- 1-Analog line (for modem use)
- 1-Voice line

1.6 Review

The BrightSpeed Series systems use adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads. The table has five (5) pads used for leveling it.

Using the GE print to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the template (P/N 5160024) shipped with the system, locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

- 1.) Lay out the 2 floor templates.
- 2.) Start with the Gantry template—align per the GE print.
- 3.) Place the table template over the top of the Gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.
- 4.) Check the level of the floor (See [Figure 2-1](#)) across the templates. This should be measured on the template over the table/gantry area, as shown in [Figure 2-1](#) below.

FLOOR LEVELNESS SPECIFICATION: 1/8" (0.125") over 10'; 3mm over 3048mm

Note: Tiles (or other resilient flooring) around all holes will be cut during the installation process.

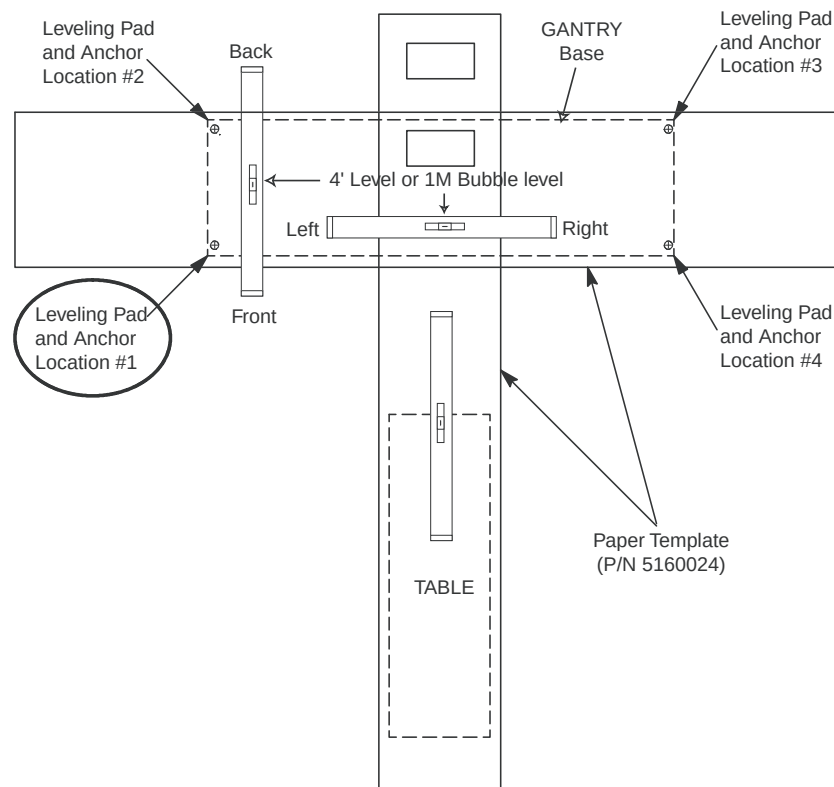


Figure 2-1 Hole Locations

Chapter 3

System Catalog

Section 1.0

Option Catalog Numbers

The following is a list of system options requiring site planning work for the BrightSpeed system. Contact your local GE Medical Systems Sales representative for a complete list of all system options or visit us at www.gehealthcare.com. Refer to the instruction manuals supplied with specific options for respective details.

CATALOG NUMBER	OPTION DESCRIPTION
B7850LD/B78502LD	International Dolly Set
B79302CA	Single-piece desktop kit
B79312CA	Side-by-side desktop kit (Freedom WorkSpace)
B7866AB	SmartScore Option, EKG Monitor and Recording Device
B7500PL	ConnectPro Option (SW & Bar Code Reader), provides a direct interface to HIS/RIS
B7540RB	Bar Code Reader
B7710LN	Boom in a Room monitor
B71182CA	Long Cable Set
B71172CA	Short Cable Set
B7700MG	Global Modem Kit
B77302CA	BSD Global USB Modem
B7850TC	Rear Cable Cover Assembly
B7700SB	Limited Access Option
B75792CA/ B75022BS	Gantry Accessory I/F HW
B75802CA	AC Outlet Box for OC
B75822CA/ B75002BE	Xtream Injector
B75832CA/ B75012BE	Enhanced Xtream Injector
B75812CA	16FPS Upgrade

Table 3-1 BrightSpeed System Options

Section 2.0

Base Scanner System

2.1 Application

The CT scanner system includes hardware and software to support patient data acquisition and image analysis for whole-body computed tomography.

2.2 Configuration

The base scanner system is configured as shown. All scan and analysis functions are controlled from the operator's console (not shown).



Figure 3-1 Base Scanner System

Chapter 4

Room Planning

Section 1.0 Required Systems Clearances

Consult your local GE Sales and Service Representative about your specific needs.

Some possible room size dimensions are shown in the tables below.

Recommended Scan Room Size	Typical Scan Room Size	Clearance Requirements
3962 x 6299 mm (13ft. x 20ft. 8in.)	3912 x 5868 mm (12ft. 10in. x 19ft. 3in.)	see Figure 4-1

Table 4-1 List of Scan Room Layouts

Limited Access Rooms (width only):

Recommended Ctrl Room Size (Standard Console Table)	Recommended Ctrl Room Size (Freedom Workspace Console Table)
3962 x 2743 mm (13ft. x 9ft.)	3861 x 2500 mm (12ft. 8in. x 8ft. 2in.)

Table 4-2 List of Control Room Layouts

Minimum Width: 3353 mm (11ft.); 356 mm (14in.)cover to wall

Average Width: 3708 mm (12ft. 2in.); 711 mm (28in.) cover to wall

Short Footprint (length only):

Minimum length: 5480 mm (18ft.)

Additional component dimensions are available in [Figure 4-6](#) through [Figure 4-10](#) of this document. Consult your local General Electric Project Manager of Installation for your appropriate room specifications.

For equipment clearance requirements, refer to [Section 2.0](#). Remember, sufficient Regulatory and Service clearances must be maintained around equipment for full operation, service and safety.

Cable length is an important consideration in room layout. A set of short cables (B71172CA) and a set of long cables (B71182CA) are optional for BrightSpeed system.

Note, also, that where possible, the cables should enter the gantry from the rear, utilizing the rear cable cover assembly. Alternate cable entry is possible at the center of the gantry (refer to the Installation template).

- Excess cable length cannot be stored behind the console or PDU.
- Long cable must not be cut or shortened.
- Excess cable may be stored in conduits, a cable storage box if present, or the floor duct, provided sufficient space is available. Observe the fill rate for each option. If there are questions regarding local electrical or building codes, consult the project electrical contractor or electrician.
- All NEC 70-E Electrical Regulations regarding conduit or duct fill must be observed.

Section 2.0 Regulatory Clearances

2.1 Regulatory Clearances

MINIMUM CLEARANCES UNDER U.S. FEDERAL REGULATIONS AND NATIONAL STANDARDS: 29 CFR 1910 (OSHA), NFPA 70E (STANDARD FOR ELECTRICAL SAFETY IN THE WORKPLACE), AND NFPA 101 (LIFE SAFETY CODE):

Figure 4-1 is a map of clearance requirements for U.S. regulatory compliance. See clearance tables on the following pages for detailed dimensional clearances. Please note all systems installed in the United States must comply with all Federal and local regulations. For installations outside the United States, country-specific or other local regulatory clearance requirements must be met.

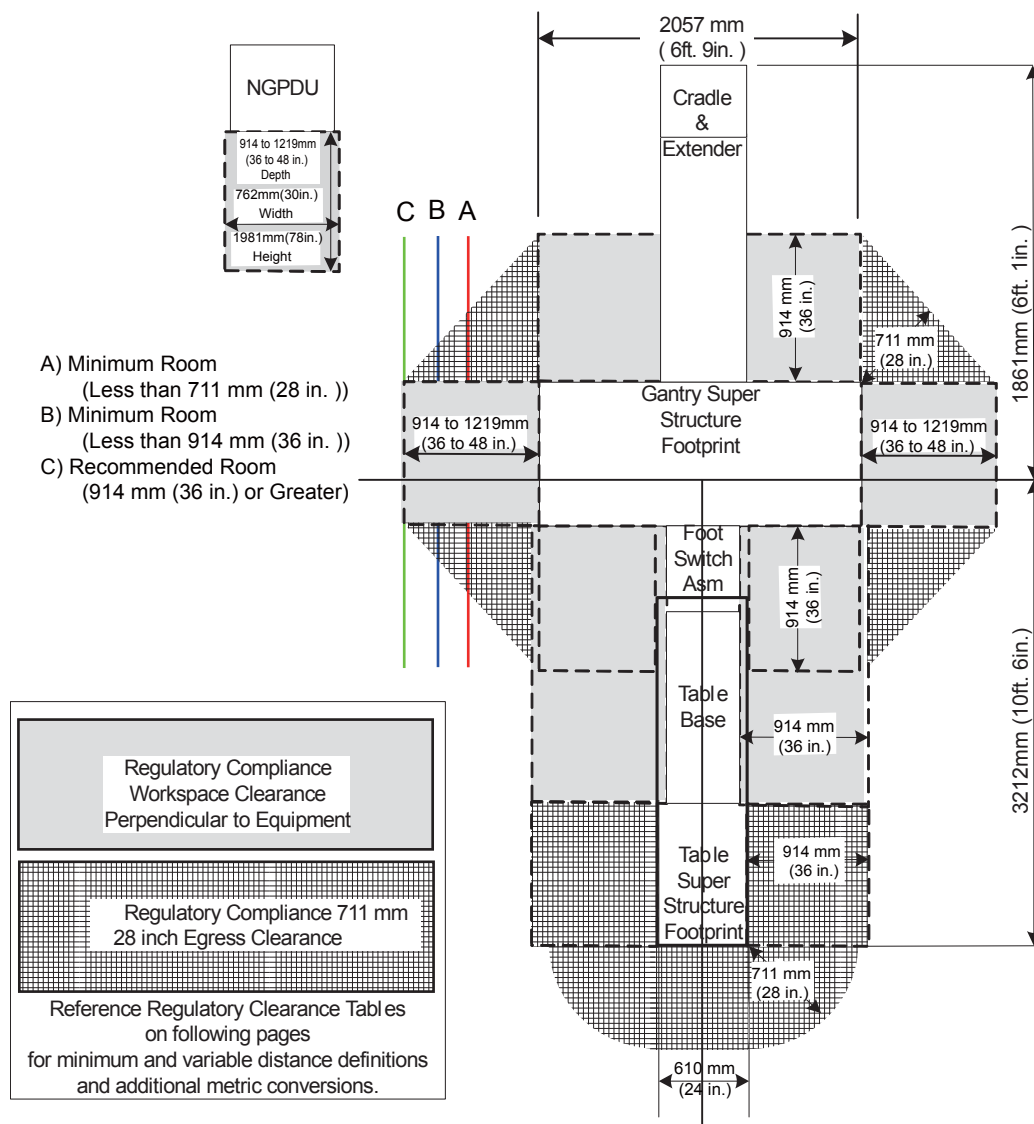


Figure 4-1 Regulatory Clearance Requirements for BrightSpeed System Configurations

Note: See [Section 3.0](#) for Service Clearances

2.1.1 Regulatory Code Description

Egress: 29 CFR 1910 Subpart E (OSHA) and NFPA 101 (Life Safety Code) define the minimum requirements for means of egress. The requirement most applicable to equipment installation and room layout is minimum width of exit access. Under OSHA 1910.37(f)(6), the minimum width of exit access shall in no case be less than 28 in. from any potentially occupied point in the room.

Under NFPA 101 (2006 edition) 7.3.4.1, the minimum width of any means of egress is 36 in. However, NFPA allows this to be reduced to 28 in. around furniture or equipment, provided that a 36 in. clearance would otherwise be available without moving permanent walls.

Electrical Clearance: 29 CFR 1910 Subpart S (OSHA) and NFPA 70E (Standard for Electrical Safety in the Workplace) define minimum clearance requirements for the workspace around electrical equipment. Under both OSHA 1910.303(g)(1) and NFPA 70E (2004 edition) 400.15, a minimum clear space of 36 in. depth (with minimum 30 in. width and 78 in. height) must be provided in front of electrical equipment with parts operating at 600 volts or below and likely to require examination, adjustment, servicing, or maintenance while energized.

This safety clearance requirement applies to all GEHC equipment. Although 36 in. is the minimum clearance for most installations, the standards require an increased minimum clearance distance where parts operate above 150 volts (but still below 600 volts) under the following circumstances:

- If the wall or surface directly facing the electrical equipment is grounded (e.g. brick, concrete, or tile) or includes grounded protrusions (such as medical gas ports, metal door or window frames, water sources and metallic sink structures, metallic cabinetry, electrical disconnects or emergency off panels, air conditioners or vents), then a 42 in. clearance depth is required.

If the possibility exists of exposed and unguarded live parts on both sides of the workspace (for example if a power distribution unit were positioned on the wall directly facing the GEHC equipment), then a 48 in. clearance depth is required.

2.1.2 Regulated Minimum Working Clearance by Major Subsystem

Requirements apply to equipment operating at 600V or less, where examination, adjustment, servicing, or maintenance is likely to be performed while live parts are exposed.

Direction of Service Access is defined as perpendicular to the surface of the equipment being serviced.

- Required regulatory clearance distances must be maintained and may not be used for storage. This includes normal system operation as well as service inspection or maintenance..

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access: front of console	914 mm (36 in.)	There are no exposed live part hazards with the cover in place. If the console is placed under a counter, the front edge of the console must be even with the vertical edge of the console workspace. Note: This component is typically serviced from the front with access to the rear.
Service access width: Front of console	762 mm (30 in.)	This is the width of the workspace in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 4-3 Console Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Head clearance	1981.2 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981.2 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 4-3 Console Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of NGPDU)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 4-4 NGPDU Subsystem

- For the gantry and table, distances are measured from the enclosure, not the finish covers.

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (All Sides)	914 mm (36 in.)	If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) on both sides of workspace with the operator between is required. If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.

Table 4-5 Gantry Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 4-5 Gantry Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Table Head or Foot)	914 mm (36 in.)	There are no exposed live parts hazards with the cover in place. This component is typically serviced from all four sides. this is the width of the workspace on each side of the equipment. A minimum of 914.4 mm (36 in.), or the width of the equipment, whichever is greater, is required.
Direction of Service Access (Table Sides)	914 mm (36 in.)*	*This distance can be reduced to 711 mm (28 in.) provided a written and signed approval is obtained by the local team from the local AHJ (Authority Having Jurisdiction). The signed document must be on file with GE.
Direction of Service Access (Table Foot)	711 mm (28 in.)	For the front gantry cover removal, a minimum of 457 mm (18 in.) is allowed only if an unobstructed egress space of 711 mm (28 in.) is maintained around the equipment for room exit. This also means no trip hazards exist along the path of egress.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.

Table 4-6 Table Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of UPS)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Right side and length of UPS)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 4-7 UPS Subsystem

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of A1 Disconnect)	914.4 mm (36 in.)*	There are no exposed live part hazards with the cover in place. This component is typically serviced from the front with access to the rear. *If exposed live parts of 151 - 600 volts are present, 1219 mm (48 in.) is required on both sides of the workspace with the operator between. *If the opposite wall is grounded and exposed live parts of 151 - 600 volts are present, 1067 mm (42 in.) is required.
Service Access Width (Right side and length of A1 Disconnect)	762 mm (30 in.)	This is the width of the working space in front of the equipment. A minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater, is required.
Head Clearance	1981 mm (78 in.)	This is the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). A minimum of 1981 mm (78 in.) or the height of the equipment, whichever is greater, is required.

Table 4-8 A1 Disconnect Subsystem

2.1.3 Terms and Definitions

Egress: The path of exit from within any room. U.S. regulations require a minimum of 28 in. (711.2 mm) of continuous and unobstructed space, including trip hazards along the path of exit.

Workspace: The dimensional box required for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. The U.S. regulation minimum is 36 in. (914.4 mm), but additional conditions can increase the minimum dimension requirement. GEHC defines this as the envelope of the component superstructure with the external covers in place.

Service Access Width: The width of the workspace in front of the equipment. A minimum of 30 in. (762 mm), or the width of the equipment, whichever is greater.

Head Clearance: The height dimension of the workspace. The height of the workspace measured from the floor at the front edge of the equipment to the ceiling or overhead obstruction(s). 78 in. (1981.2 mm), or the height of the equipment, whichever is greater.

Grounded Wall: Any wall that can be electrically conductive to earth ground. Masonry, concrete, and tile are considered conductive. Additional commonly found aspects of a wall should also be considered grounded. This is not an all-inclusive list:

- Medical gas ports and plates
- Metal doors and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinetry
- A1 main disconnect panel
- Equipment Emergency Off panels
- Industrial equipment (such as air conditioners and vents)
- Expansion joints
- Surface raceway
- Exposed wall conduits
- Floor outlets boxes

The following are not considered as grounded elements of a common wall:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks
- Ceiling tile grids

2.2 Additional Regulatory Clearance Information

2.2.1 Minimum Room Size (Limited Access)

The CT Gantry Left Side Limited Access Initiative provides the capability to reduce the minimum room size for CT Systems while still meeting all installation requirements and specifications. This adds left side flexibility, allowing the CT system to be sited in rooms with widths 559 mm (22 in.) smaller than the current minimum room width. Left-side access and egress may be restricted. Refer to your site's installation print for your room's detail.

If you are using the square meters (square feet) to determine compliance, remember that the front cover clearance is wider than the regulatory clearance along the table length, and that the cover park position is behind the table in the home position. The cover is removed on tilting dollies and can be moved side-to-side to reach the park position of 457 mm (18 in.), if this is not an egress route.

Wall duct and conduit on walls within the regulatory clearance areas shall be 1067mm (42 in.),

measured from the covers to the obstruction. Servicing of the CT System can be safely performed within the regulatory envelopes, however sufficient space must be maintained to remove system covers, and replace large system components. To achieve this clearance for the gantry, clear space must be available to maneuver the gantry covers mounted on the service dollies. Surface floor raceway cannot be used in the egress route areas. OSHA ramps are available. The FE lifting the rear or front cover to avoid floor obstructions is not an EHS-approved service procedure. One Service Engineer shall be able to accomplish all service component replace tasks listed without the need for special tools or equipment, such as a tube change, detector change, and HV tank.

2.2.1.1 Regulatory Caution

Site prints are required for all system installations including relocation and moves. CT room layout, as shown on your site print, shall meet all regulatory requirements as described in the installation manual. Additional room components, such as cabinets, reduce room size. Equipment not shown on the site print may void the caution statement, making the room non-compliant. Actual site measurements before installation will be taken to determine room size and compliance.

2.2.1.2 Egress Clearance

Egress requires a clear, unobstructed route out of the room, either around the back of the gantry or around the back of the table. If your egress route is not around the back of the table, maintain 457 mm (18 in.) of clearance between the back of the table, with a continuous width of 3200 mm (126 in.), 1600 mm (63 in.) on each side of the table center line, on each side to any obstruction so that the front cover can be removed. Refer to the Pre-Installation manual for more details on service clearances.

Exceptions

Rooms smaller than 3353 mm x 6096 mm (11 ft. x 20 ft.), require construction to meet the minimum requirements. The design center or your GE PMI may have additional recommendations for your room size.

2.2.1.3 Operational Caution

In a minimum room layout 356 mm - 686 mm (14 in. - 27 in.), the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, there may be limited equipment access on the gantry left side when loading patients or when positioning patient equipment in the room between the gantry and the wall. Detailed customer installation tasks are detailed in the product Pre-Installation manual, Chapters 1-4.

2.2.2 System Specifications (BrightSpeed)

2.2.2.1 Recommended Room Size

This room configuration offers the most flexibility for future upgrades. It has sufficient workspace and space to add millwork, while meeting all regulatory requirements. This room would be compatible with most two-step future installations.

Recommended Room Size

3962 x 6299 mm (13ft. x 20ft. 8in.)

Same Regulatory requirements apply

2.2.2.2 Typical Room Size

This room size and configuration is commonly found on most sites and allows for some future upgrades. It has sufficient workspace but limited space to add millwork and meet all regulatory requirements. This room may be compatible with some two-step future installations.

Typical Room Size

3912 x 5868 mm (12ft. 10in. x 19ft. 3in.)

Same Regulatory requirements apply

2.2.2.3 Minimum Room Size

This room size and configuration is the smallest functionally acceptable space for this product. Size and layout often eliminate these rooms for future upgrade considerations. Due to limited size, equipment and regulatory requirements, this room often has limited workspace, and very limited or no in-room millwork, but still meets all regulatory requirements. This room is not compatible with two-step future installations

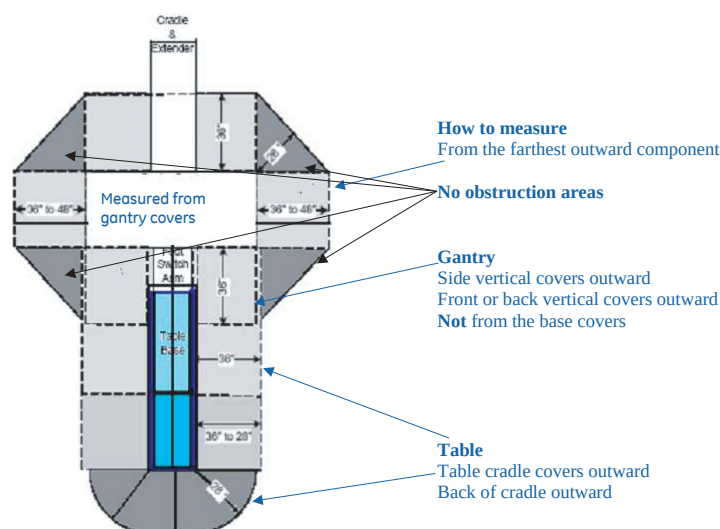
If you are using the square footage (Sq M) to determine compliance, remember that the front and rear cover clearances are wider than the regulatory clearance along the table length, and that the cover park position is behind the table in the home position. The rear cover is also on wheels and must be able to be removed from behind the gantry for service.

Minimum Room Size

3353 x 6096 mm (11ft. x 20ft.)

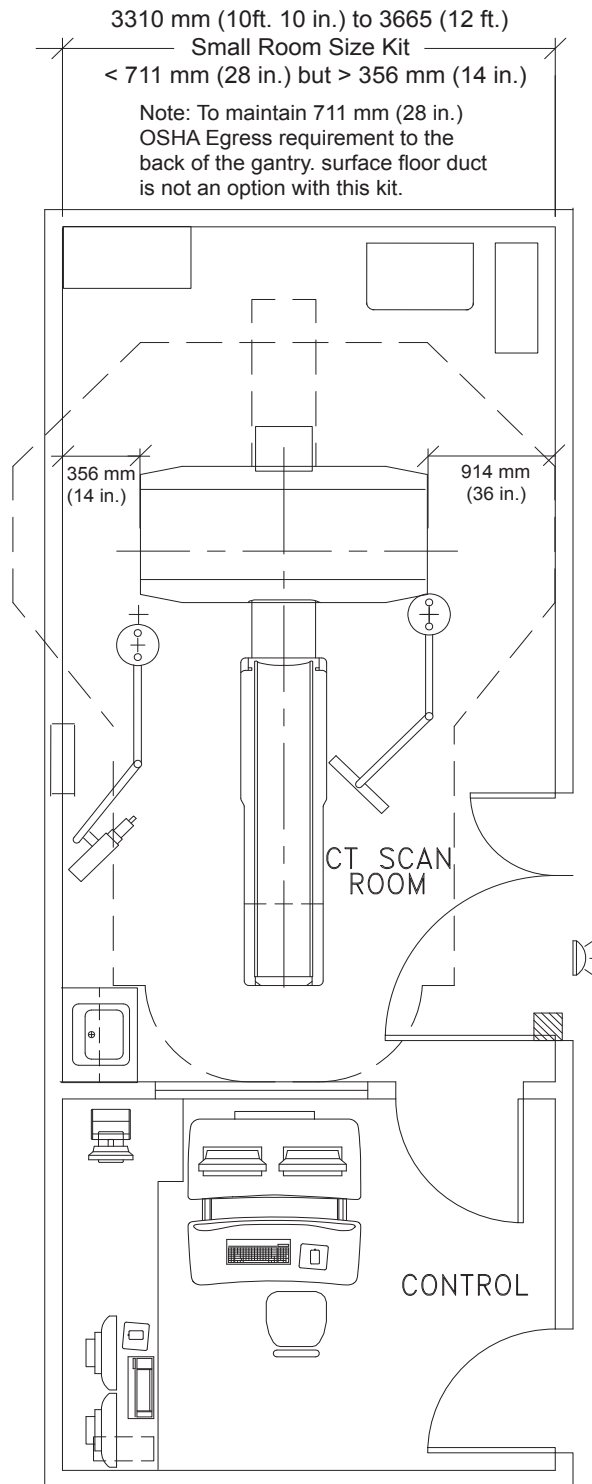
Same Regulatory requirements apply, with the addition of no energized left side service.

2.2.3 How to Measure

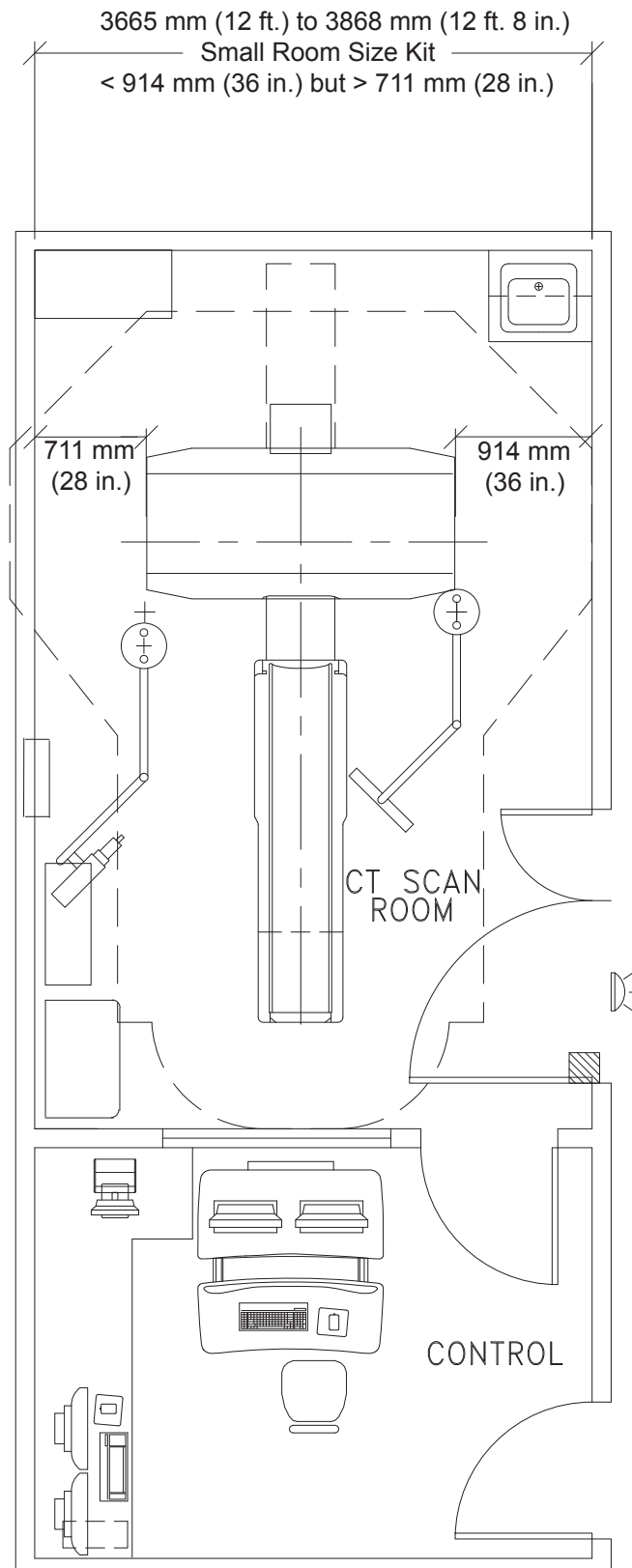


2.2.4 Minimum Room Size & Requirement Layouts

Room A - Less than 711 mm (28 in.) but greater than 256 mm (14 in.) measured from the covers to the left sidewall. In this configuration service, egress and workspace are compromised around the gantry.

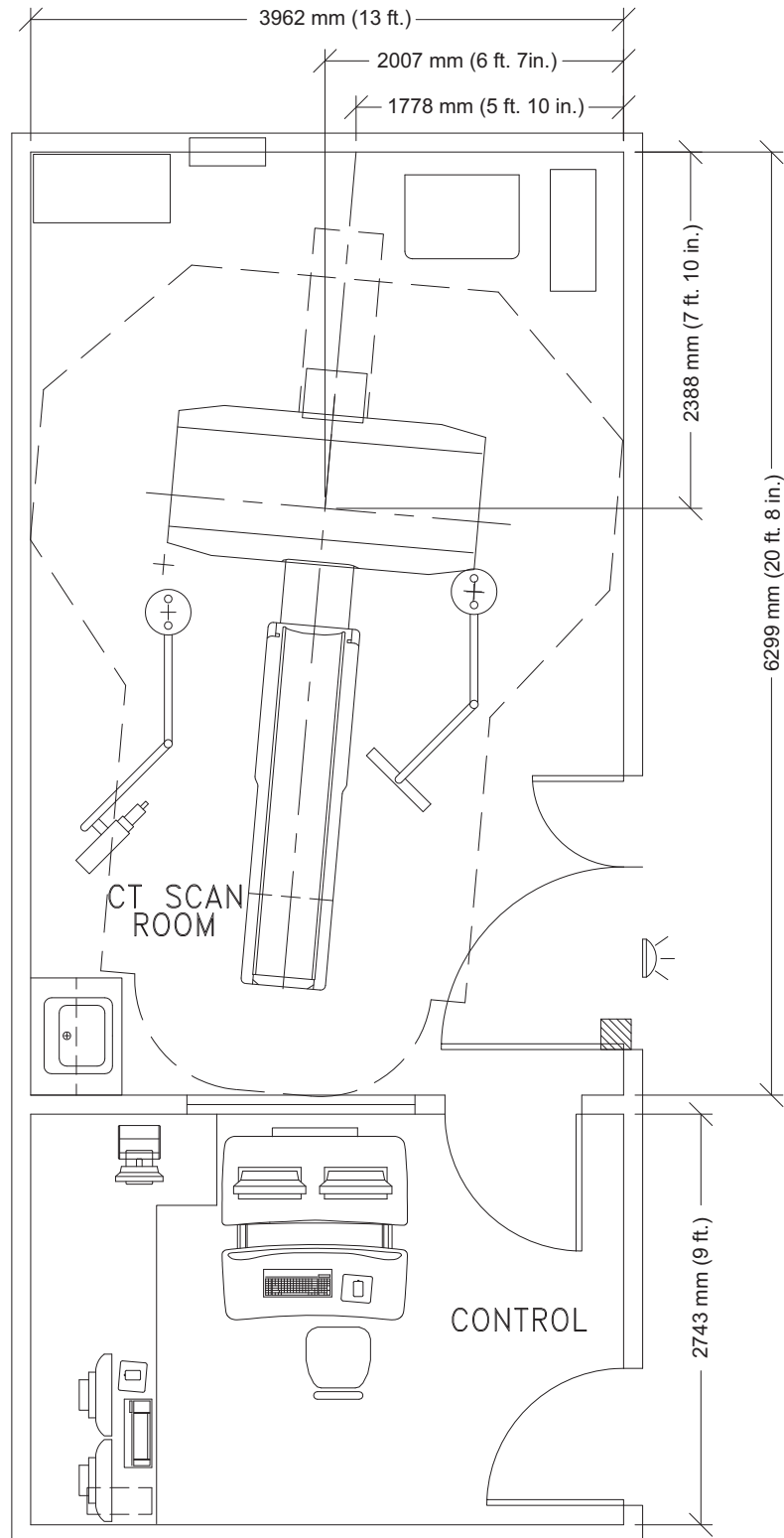


Room B - Less than 36 in. but greater than 28 in. measured from the covers to the left sidewall. In this configuration service, egress and workspace are acceptable around the gantry.



2.2.5 Recommended Room Size & Requirement Layouts

Note: Your room layout may meet the Recommended or Typical room requirements but look different than the room shown below. Contact your sales person to have a detail room layout completed for your site.



Section 3.0 Service Clearances

Servicing of the CT System can be safely performed within the regulatory envelopes defined in [Section 2.1](#), however sufficient space must be maintained to remove the covers from the system. To achieve this clearance for the gantry, clear space must be available to maneuver the gantry covers mounted on the service dollies. One Service Engineer can accomplish this.

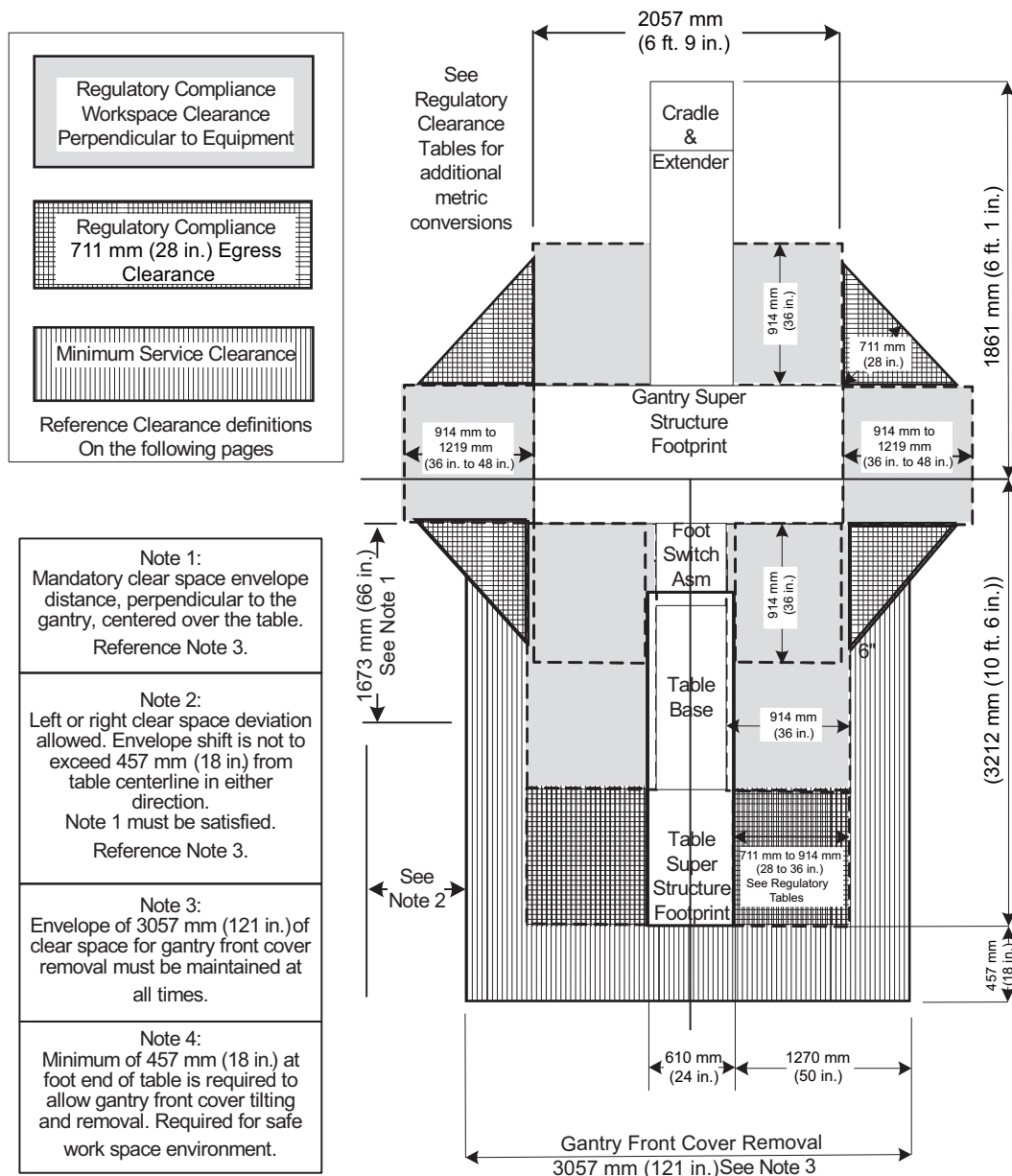


Figure 4-2 Minimum Service Clearance

3.1 Service Clearances for Single Service Engineer

- Gantry front cover removal requires the use of the “Tilting Cover Dollies”. These dollies allow the Service Engineer to separate the cover from the gantry, tilt the cover 90 degrees, roll the cover to the foot end of the table, and then tilt the cover an additional 90 degrees such that the front cover is now upside down relative to the normal system mounted condition. [Figure 4-2](#) illustrates the minimum clear space required to achieve this operation. The gantry front cover must be removed to a position that satisfies the minimum regulatory clearances.
- The gantry rear cover, with service dollies installed, requires a width 2244 mm (88 in.) and a depth of 610 mm (24 in.) of clearance for removal. Sufficient space must be calculated to move the cover either straight back or to a side of the table to satisfy the minimum regulatory clearance shown in [Figure 4-2](#). This means the rear cover cannot violate the workspace on the rear or either side of the gantry.
- If gantry service requires both the front and rear covers be removed, then these covers must be positioned within the room in such a manner as to not violate the regulatory clearances on any side of the gantry. This may necessitate removing the covers from within the suite. This should be discussed with the customer and provisions made to accommodate this potential event.
- A single Service Engineer can safely perform servicing of the table. Sufficient clear space must be available to maintain regulatory clearances when the table covers or cradle are removed.
- In your room layout design, service shall have clear and unobstructed access to the gantry tube change area for all major component replacements. These components must be able to reach the service area without lifting or rigging by one service engineer. Major components include:
 - CT X-ray tube in crate
 - High voltage tank(s) in crate
 - Slipring in crate
 - Detector assembly

Be aware of cabinet placement, and how surface floor ducts are used in room configurations.

3.2 Power Distribution Unit (NGPDU) Service Clearance

Positioning of this component must be considered for regulatory compliance as defined in [Section 2.1](#) , Regulated Minimum Working Clearance by Major Subsystem.

See Regulatory Tables.

3.3 Console Service Clearance

The console does not present an exposed live parts hazard. However, a minimum working space depth of 1219.2 mm (48 in.) and full width of the console be maintained at all times for service activity. Additionally sufficient space needs to be provided for repositioning of the console and side clearance for rear service access. Egress as well as other service requirement shall be considered when siting the console. See [Figure 4-5](#) for a Recommended control room layout.

Section 4.0

Short Footprint Consideration

4.1 Introduction of Short Footprint function

If the site room length cannot satisfy the requirements for standard mode. Short Footprint mode can be considered.

Short Footprint mode is to set limitation to table cradle scannable limitation(A) so that cradle cannot move in out of the limitation. The scannable range is limited accordingly.

Short Footprint features are as follows:

- Cradle Movement limitation can be set at any position.
- Table Height Limitation can not be set (no change for footprint at Gantry Front side).
- Scannable range is depends on the Gantry Rear space (distance to the wall), but need to consider the Service Clearance and country's local regulation for Gantry Rear space.

4.2 Instruction of using Short Footprint function



NOTICE

Cradle limitation must comply with country or local regulatory clearance requirements.

Cradle movement limitation set for short footprint must be approved by customer during pre-installation.

- 1.) Refer to [Figure 4-3](#) , use floor template with ruler to prearrange the layout and calculate the cradle scannable limitation (A).
- 2.) Make GE siting print to meet regulatory and service clearance requirements.
- 3.) Record the distance from cradle limitation to wall (X) and cradle scannable limitation (A) for installation.

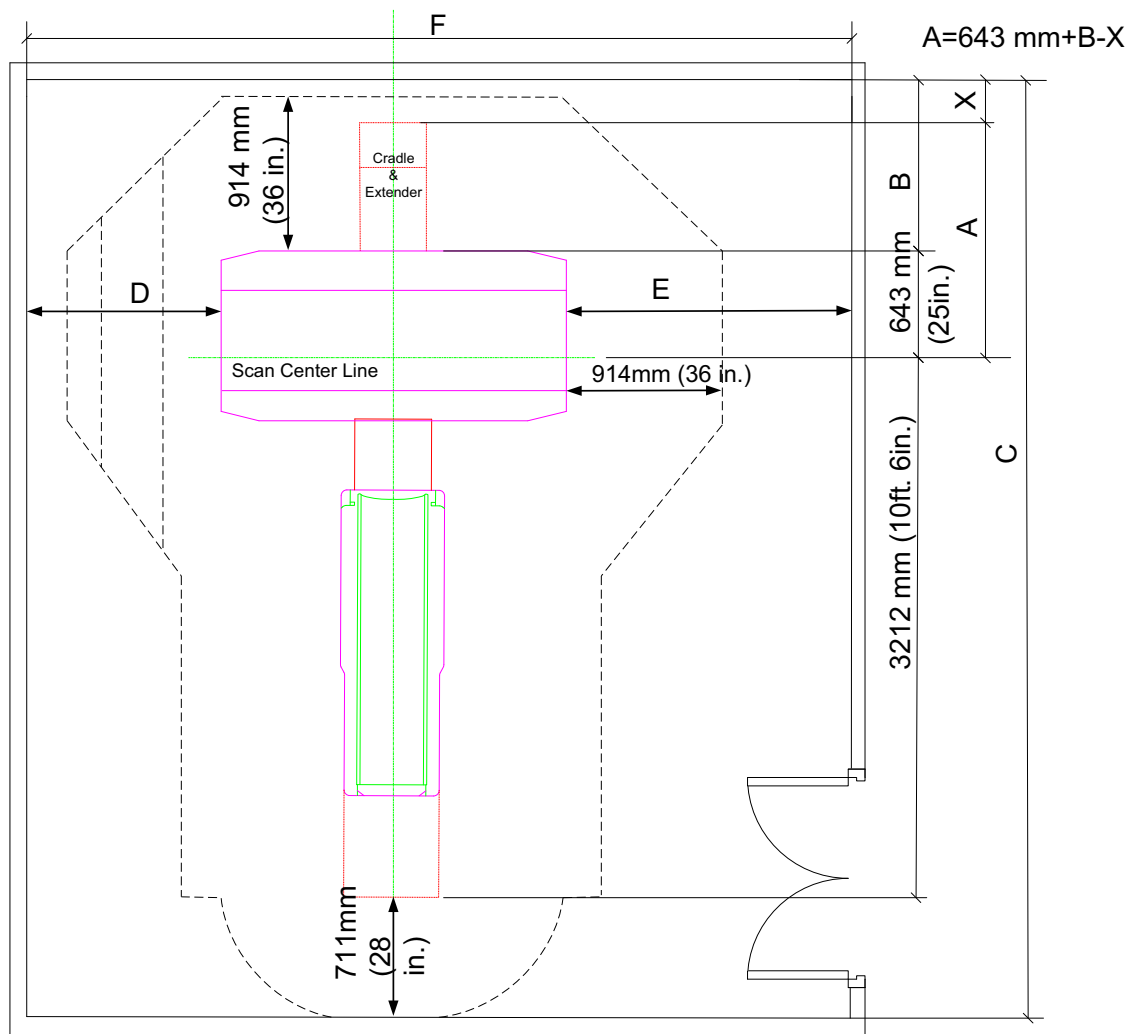


Figure 4-3 Short Footprint Calculation diagram

A: Cradle Scannable Limitation, the value to be set using Short Footprint function

$A(\text{Scannable Range(approx.)}) = 643\text{mm}(\text{Scan Center Line to Gantry Rear Cover})$
 $+ B(\text{Gantry Rear Cover to Wall}) - X(\text{Safety Clearance to prevent hitting cradle to wall}).$

B: Gantry rear cover to wall, no less than 914 mm (36 in.)

Note: When the length from Gantry rear cover to wall is 914mm, cradle scannable limitation would be 1457mm (Shown in [Figure 4-4](#)).

D: Gantry left side to wall, no less than 356mm (14 in.). Refer to Section 3.0 for details about Limited Access.

E: Gantry right side to wall, no less than 914 mm (36 in.). If Short Footprint is set for a short room length, bigger gantry side clearance is required to remove Gantry rear cover, or no removal of rear cover is accepted.

X: Distance from cradle limitation to wall.



NOTICE

It is Suggested that safety clearance from cradle IN-limit to wall should be no less than 100 mm.

Figure 4-4 shows minimum room layout with Short Footprint function (Gantry rear cover can be removed).

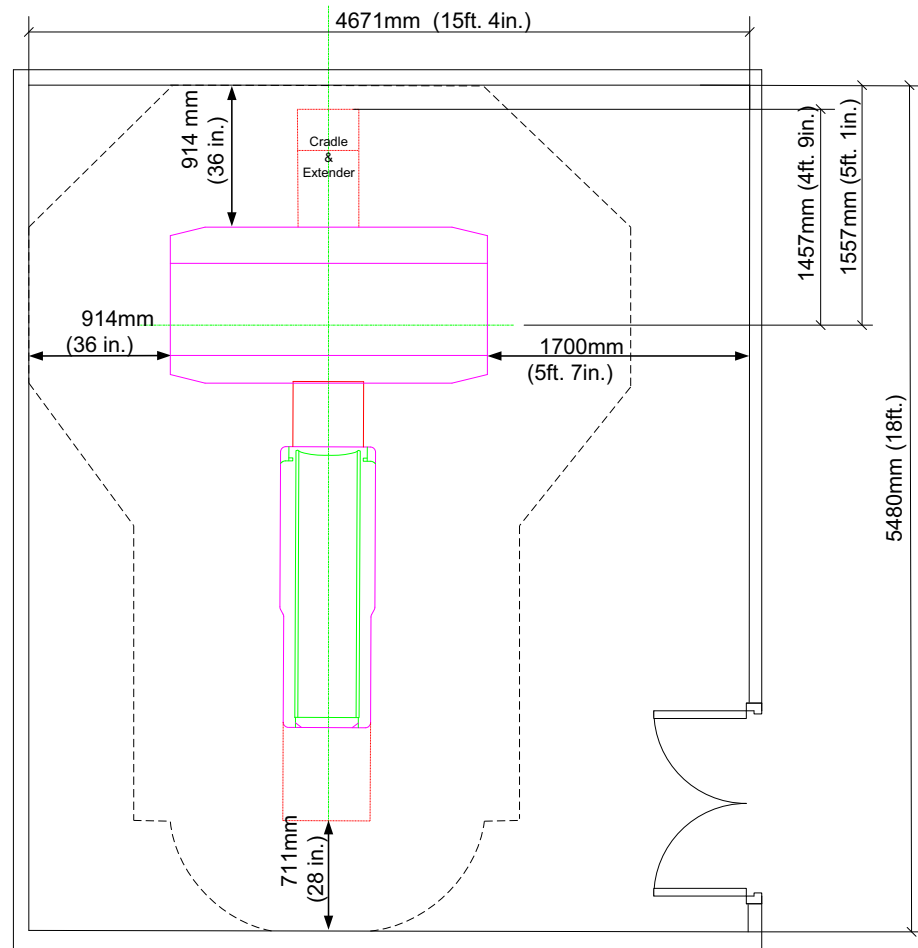


Figure 4-4 Minimum Room layout for Short Footprint function

Section 5.0

Recommended Layouts

5.1 Control Room Considerations

Two kinds of Console configurations are available:

- Standard Console: Console + Single Piece Desktop;
- Console with FWS: Console + Freedom Workspace Table (See [Figure 4-5](#)).

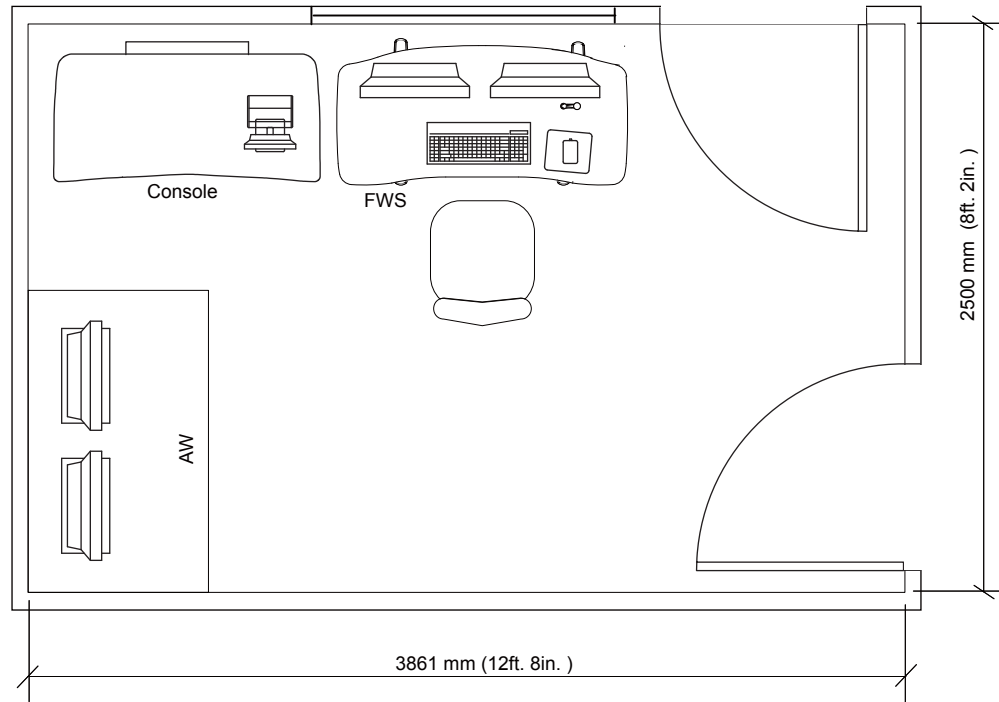


Figure 4-5 Recommended Control Room Layout with Freedom Workspace (FWS)



NOTICE

If a Freedom Workspace (FWS) console table is ordered, it **MUST** be placed **side by side** with the console. It can be placed on either side of the console. The Extended Cables Kit is shipped in FWS Console Table Assembly. The extended cable length is **3 m** (see Chapter 9 [Section 3.2](#) for details).

- The control room must provide a suitable operating environment for the console electronic, and operator working comfort.
- The console cannot be dismantled of, have components removed or rearranged in configurations other than as shipped.
- If operationally possible, the monitor desktop and user desktop components may be removed and placed on a counter-top, providing the cable lengths shipped are not altered or changed. The operator console cabinet then can be remote mounted, provided the cooling requirements are met. Maintain 152mm (6 in.) on all sides; and venting is required.
- A suitable work area, which is within reach of the operator's console, should be provided for placement of the injector control. Injector controls differ in dimensions depending on the brand selected.
- A PACS, workstation, image printer, or filming device are often placed in the operator console

control room area, and sometimes may be directly linked to the operator console.

- Additional components although linked via network or ethernet cable, are not powered from the CT operator console.
- Additional room power and network connection should be considered when reviewing the console work space.

5.2 Storage Cabinet

A storage cabinet is provided by GE Healthcare to store all supplied service equipment (see [Table 4-9](#) for equipment list). This storage cabinet (457 mm D x 914 mm W x 1067 mm H) (18 in. D x 36in. W x 42in. H; ~90 lbs) should be located in the scan room suite area, for easy service access.

ITEM	SIZE	WEIGHT (TOTAL)	
QA Phantom (water filled)	20 x 15 cm (7.9 in. x 5.9 in.)	5.5 kg	12 lb
35CM Phantom	35 x 7 cm (13.8 in. x 2.8 in.)	8.2 kg	18 lb
48CM Phantom	48 x 7 cm (18.9 in. x 2.8 in.)	11.4 kg	25 lb
Phantom Holder	25 x 25 cm (9.8 in. x 9.8 in.)	3.6 kg	8 lb
FE Box	30 x 38 x 30 cm (11.8 in. x 15 in. x 11.8 in.)	6.8 kg	15 lb
Rear Cover Dollies	158 x 82 cm (62.2 in. x 32.3 in.)	11.4 kg	25 lb
Front Cover Dollies	85 x 20 cm and 85 x 15 cm (33.5 in. x 7.9 in. and 33.5 in. x 5.9 in.)	15.9 kg	35 lb
Install Support Kit (box)	30 x 30 x 38 cm (11.8 in. x 11.8 in. x 15 in.)	9.1 kg	20 lb
Tube Hoist Kit	77 x 8 cm and 38 x 15 cm (30.3 in. x 3.1 in. and 15 in. x 5.9 in.)	9.1 kg	20 lb
Balance Weight Kit		13.6 kg	30 lb

Table 4-9 Equipment to be stored in storage cabinet

5.3 Advantage Workstation (AW)

- Refer to Pre-Installation Manual 2111833 and Installation/Service Manual 2111831. For document access, please refer to GE Healthcare's Common Documentation Library website.

Section 6.0

Component Dimensions

DESCRIPTION	WIDTH		DEPTH		HEIGHT	
	INCH	MM	INCH	MM	INCH	MM
Color printer	23	584	18	457	7	178
Remote Color Monitor (LCD)	16.25	413	8	203	16	406

Table 4-10 Dimensions of Accessories

6.1 Table and Gantry

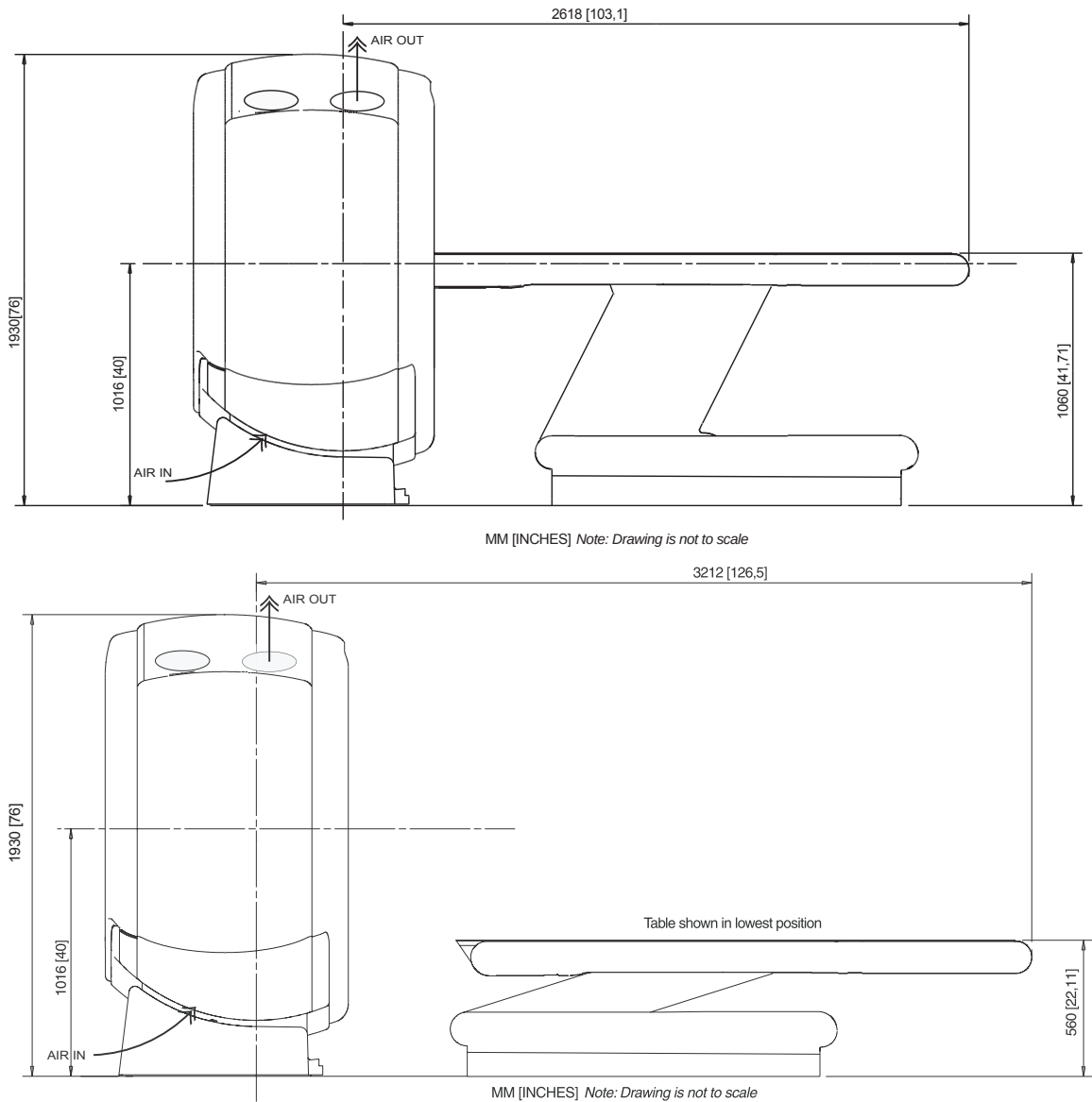


Figure 4-6 Table and Gantry (Side View)

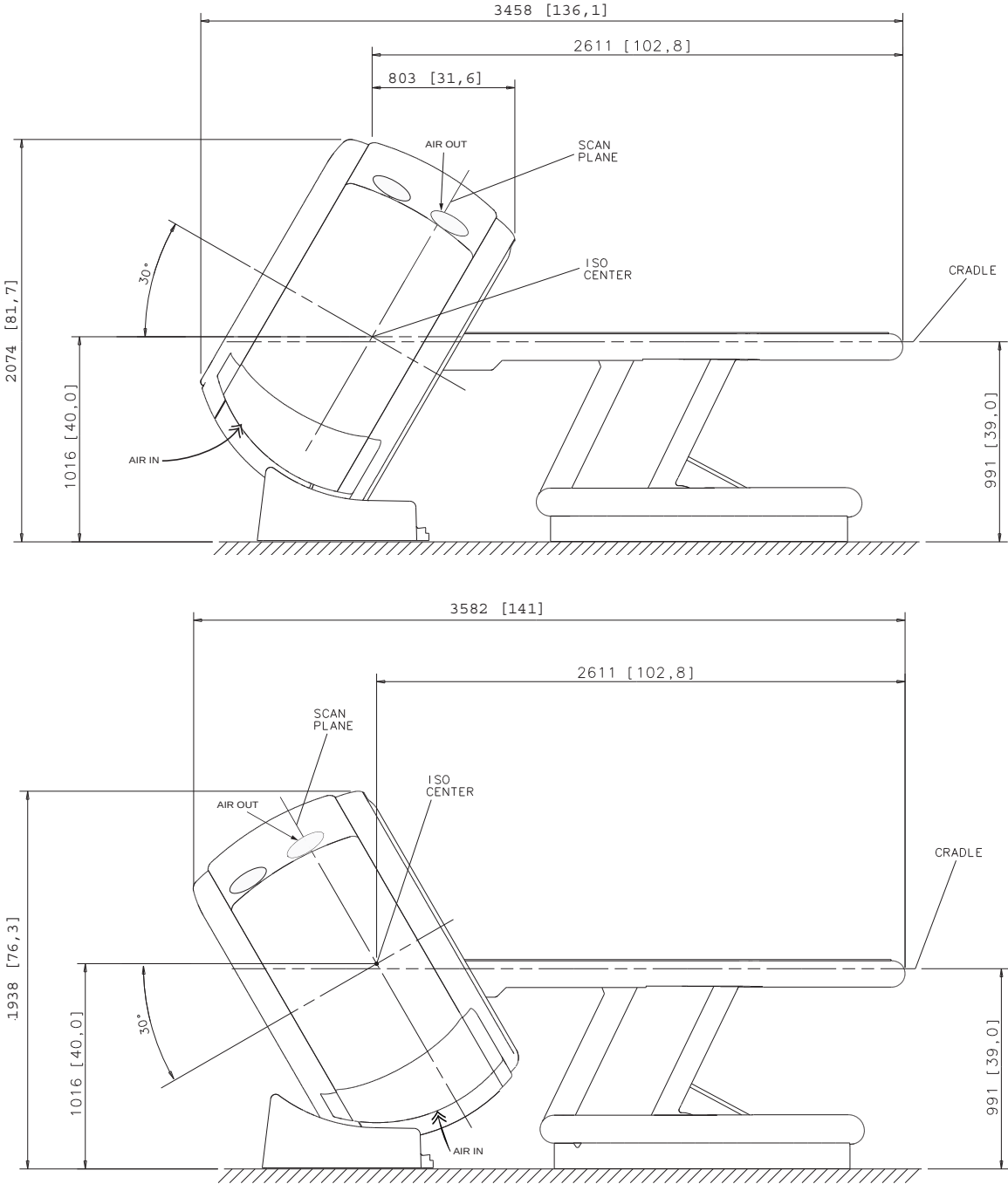


Figure 4-7 Gantry shown tilted +30° (top) and -30° (bottom)

6.2 Power Distribution Unit

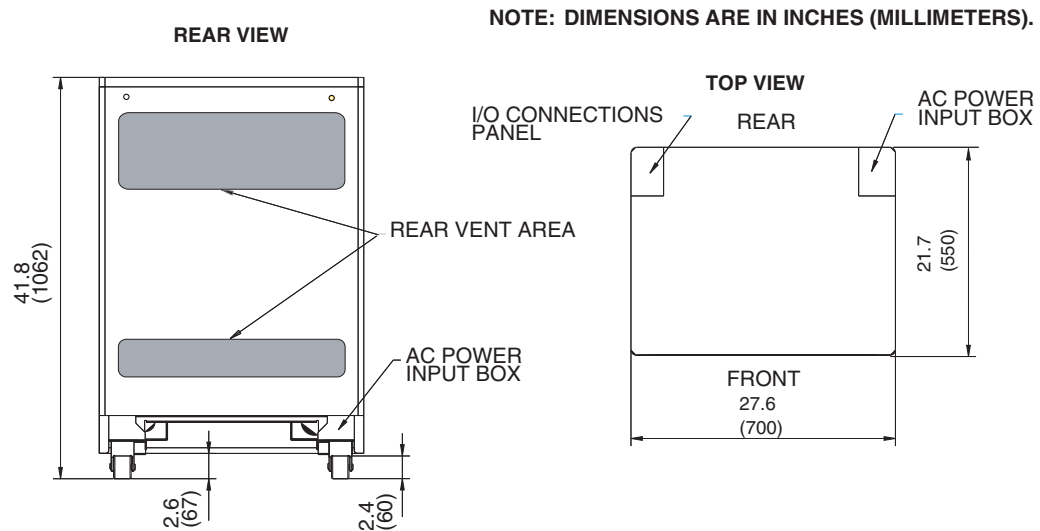
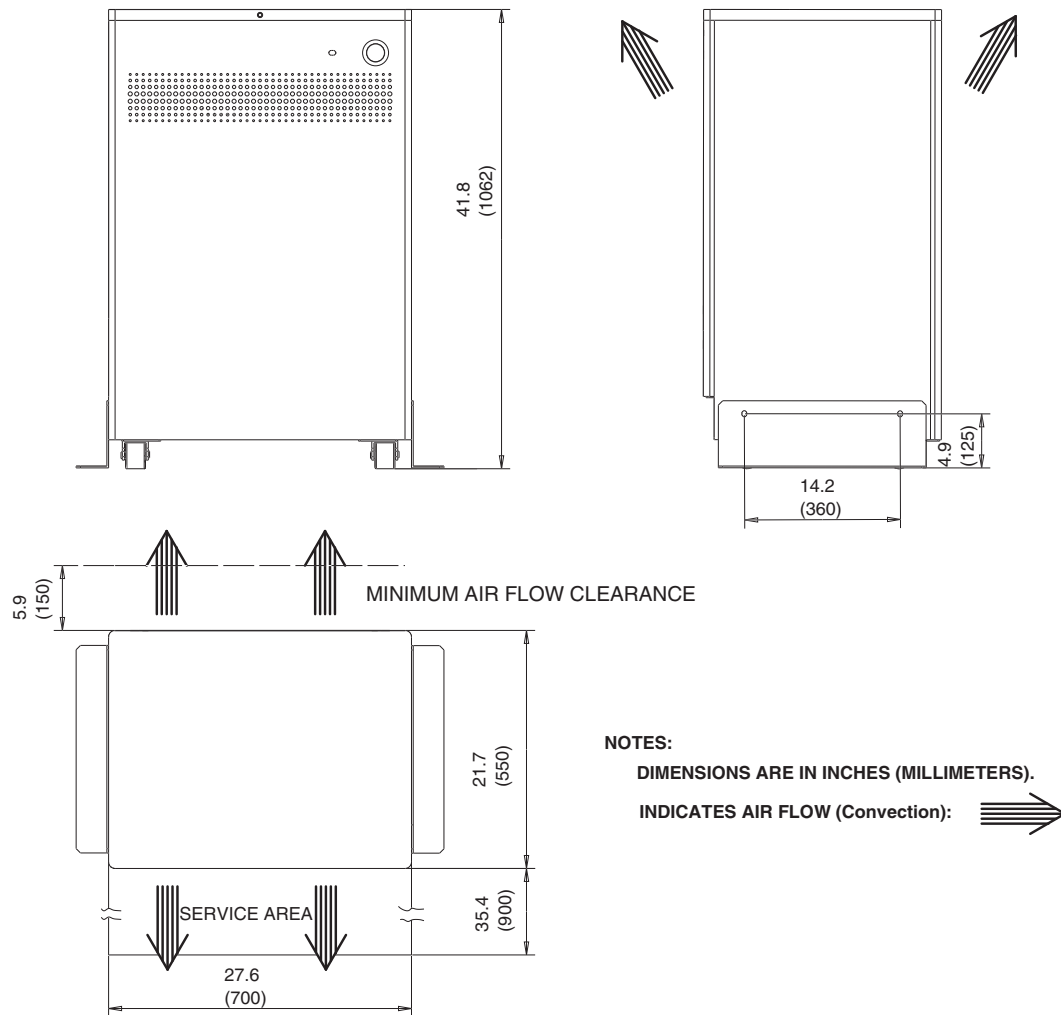


Figure 4-8 Power Distribution Unit (NGPDU)



NOTES:

DIMENSIONS ARE IN INCHES (MILLIMETERS).

INDICATES AIR FLOW (Convection):

Power Distribution Unit (NGPDU)

6.3 Operator's Console with Standard Table

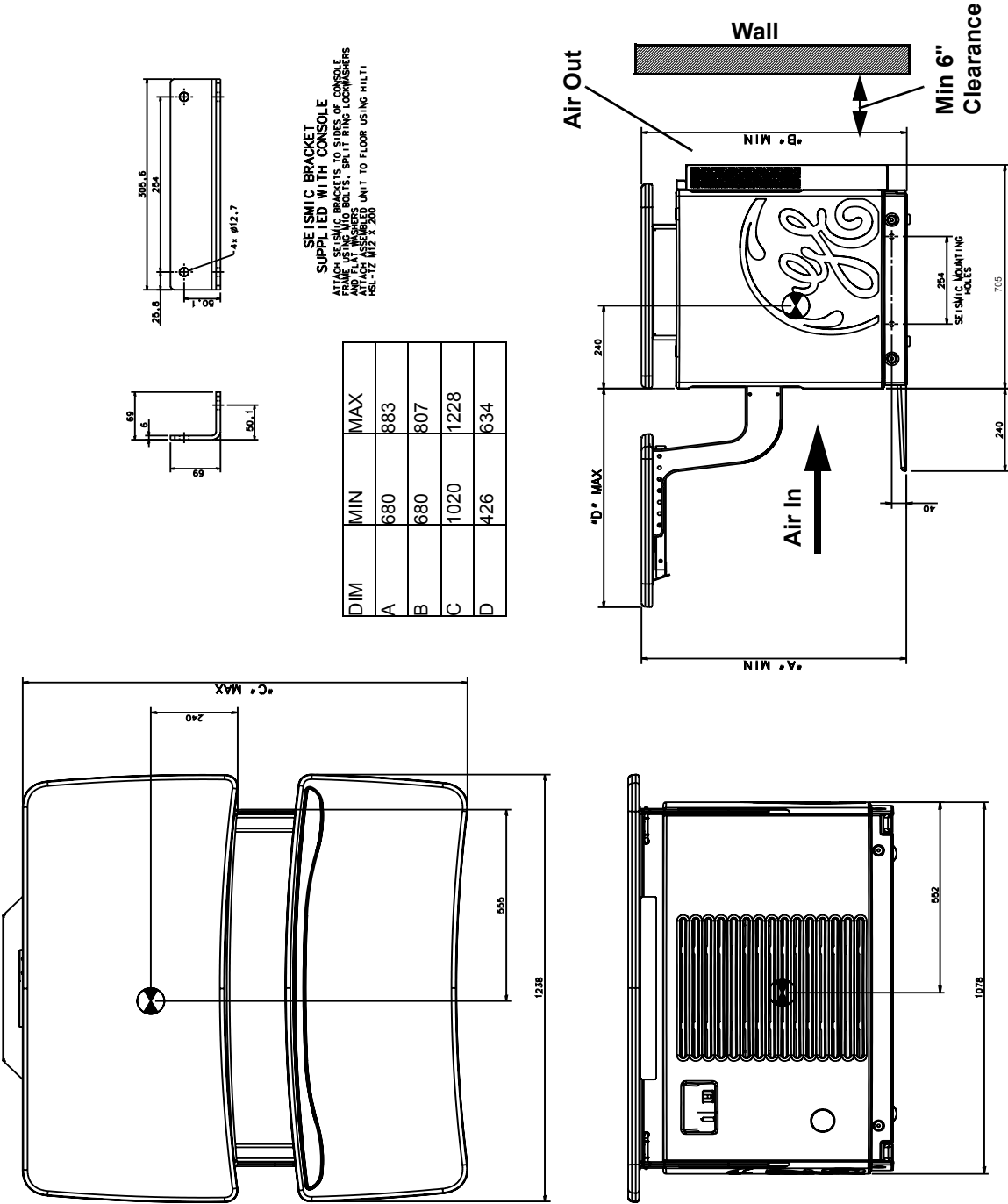


Figure 4-9 Operator's Console

6.4 Freedom Workspace Table

Height of Freedom Workspace is adjustable.

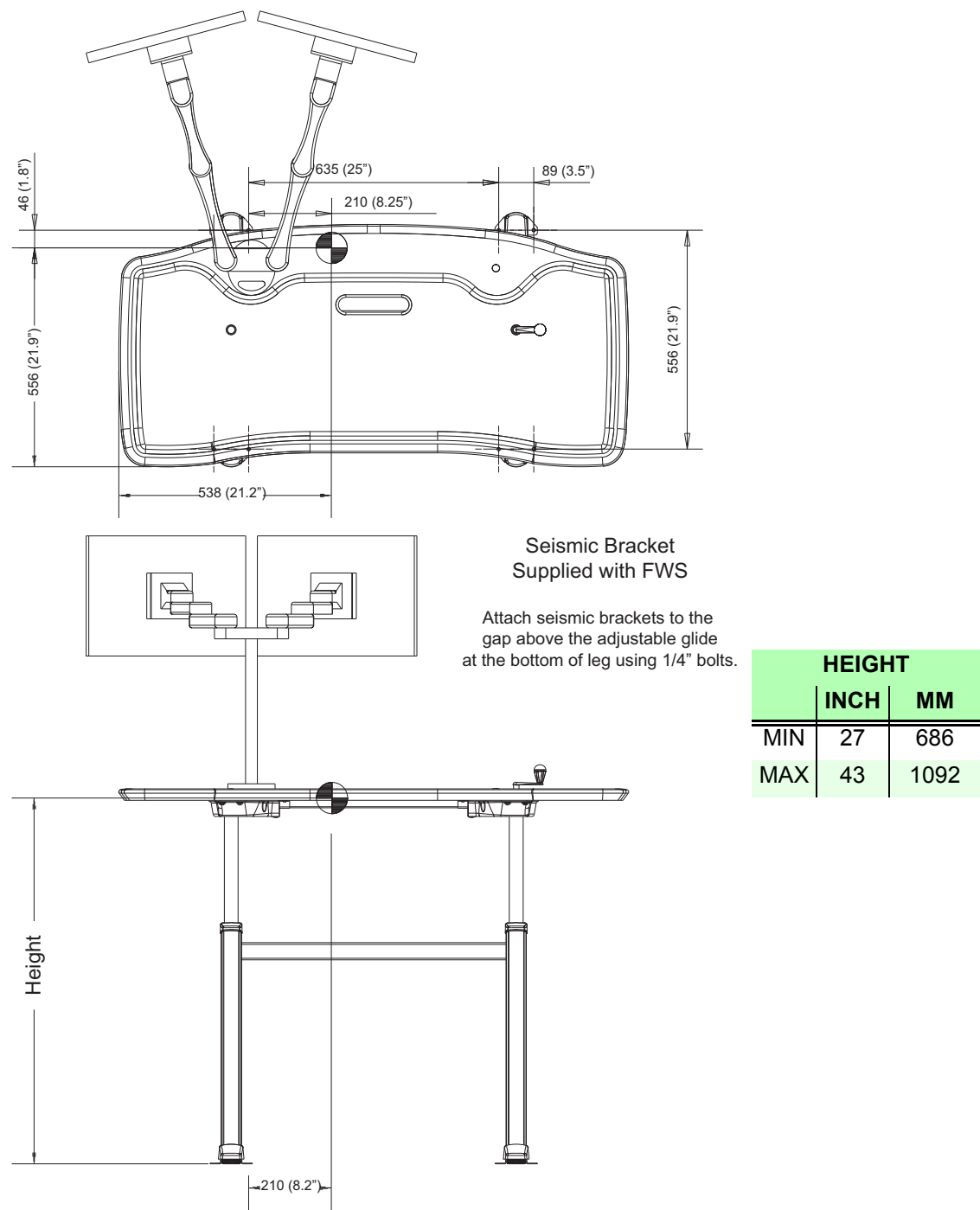


Figure 4-10 Freedom Workspace Table (part 5160735)

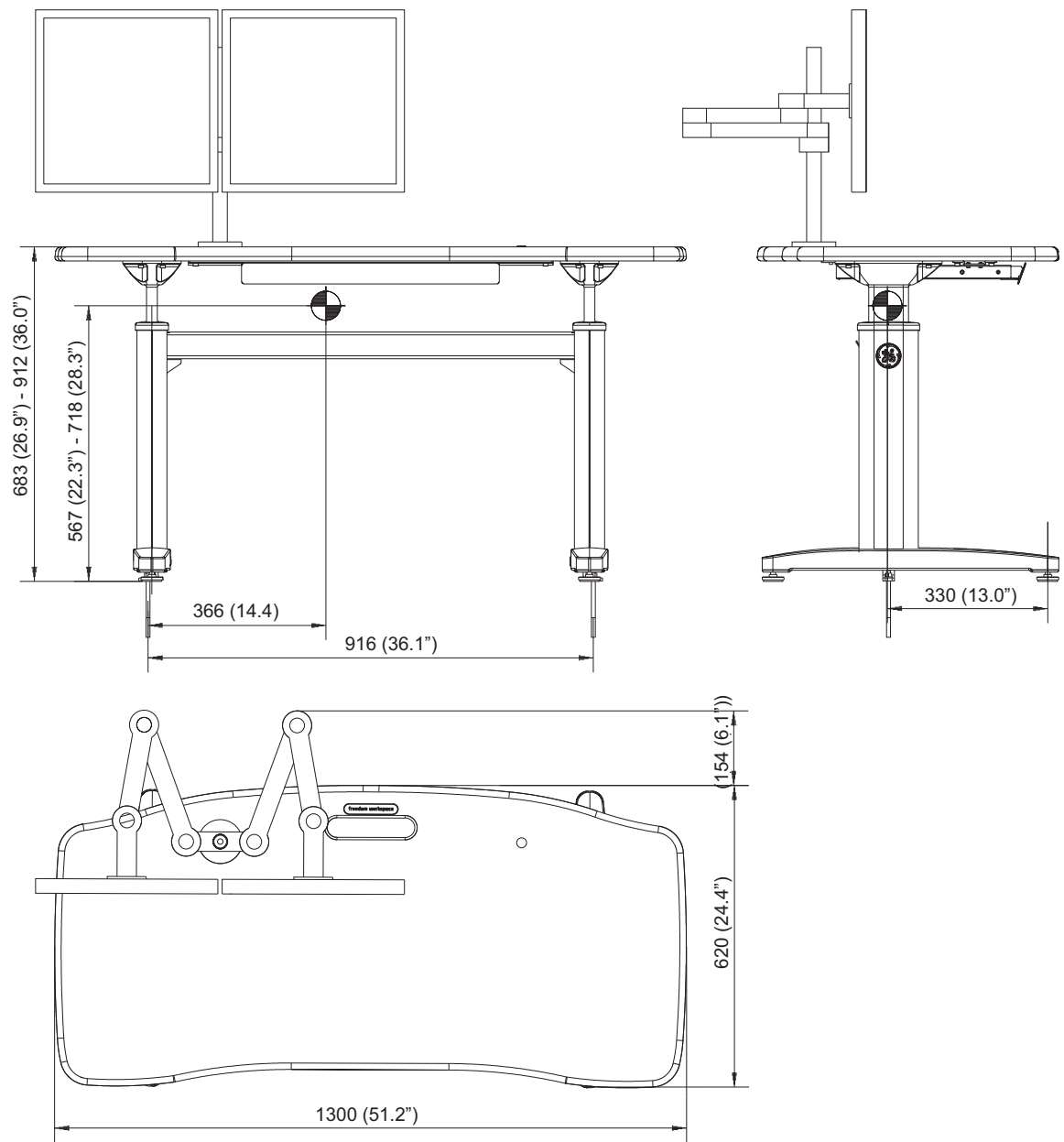


Figure 4-11 Freedom Workspace Table (part 5168666)

Section 7.0

Minimum Dimensions and Clearances

7.1 Options

Ceiling Pedestal mount lowest point to floor (Injector or Monitor) 2438.5mm (96.0 inches)

7.2 System Operation

• Finished ceiling to floor (recommended)	2743.0mm	(108.0 inches)
• Finished ceiling to floor (minimum)	2438.5mm	(96.0 inches)
• Table max extension head end w/extender from Center Line	2030.0mm	(80.0 inches)
• Table extension head end w/extender to obstruction	152.0mm	(6.0 inches)
• Table in lowest pos. w/cradle @home pos. to Center Line	3209.0mm	(126.5 inches)
• Back of Console to wall	152.0mm	(6.0 inches)
• Back of PDU to wall	152.0mm	(6.0 inches)

7.3 Injector Control

A suitable work area, which is within reach of the operator's console, should be provided for placement of the injector control.

Wall mounted, ceiling mounted and pedestal units need cables to be routed from the gantry area to the console area. Injectors AC power is supplied with Accessory I/F hardware kit.

Note: Injector cables should not be routed with the system cables.

Mounts are available in different configurations and lengths. Refer to Injector documentation for detailed installation instructions.

Section 8.0

Structural Requirements

8.1 Table and Gantry Mounting Requirements



WARNING **POTENTIAL FOR PATIENT INJURY.**
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.



NOTICE It is the purchaser's responsibility to provide an approved support structure and mounting method for all floor types other than those listed. General Electric is not responsible for any failure of the support structure or method of anchoring, including seismic requirements.
GE is not responsible for methods other than those listed.

Table and gantry mounting dimensions are shown in [Figure 6-2](#), [Figure 6-3](#), [Figure 6-4](#). Refer to [Chapter 6](#) for additional details of floor loadings, component weights, and Gantry and Table installation and anchoring.

Anchor gantry and table to floor by a means that will maintain their relative alignment and meet applicable building and other local codes, including seismic structural mounting requirements.

Floor structure must be capable of withstanding the occupied weight of table and gantry, and the individual contact area loading of these components. Refer to [Section 6.0](#) for each of the three (3) major components of the BrightSpeed system.

Support areas of the patient table and gantry must rest on solid concrete, not resilient tile or carpeting which will slowly yield over a period of time and disturb alignment of table to gantry.

Factors that could cause misalignment between gantry and table due to floor sag should be considered. The cradle can potentially carry a 205 kg (450 lb) patient. Center of gravity changes as cradle cantilevers.

Take into consideration all other moving weights such as gurneys or personal equipment. Refer to [Chapter 6](#) for gantry and table mounting details.

No part of floor surface within table and gantry, nor the two interface areas between table and gantry, should be higher than the support area for table and gantry.

8.2 Floor Anchors

Provided floor anchors are designed for use ONLY on concrete floors that meet the 102 mm (4 in.) concrete floor requirements. All other anchoring methods (on floor types other than the 4-inch concrete minimum) must be determined to meet the stated GE minimum load requirements, at the customer's expense, by their structural contractor. The customer's contractor is responsible for the installation of all anchors other than those shipped with the system.

8.3 Floor Strength

Concrete floors must have a minimum strength of $f'_c = 2000$ psi (1.4×10^7 Pa) at 28 days (curing time) for mounting floor anchors. It is the responsibility of each customer to have appropriate tests performed to determine and measure concrete strength.

Note: If installing the GE BS scanner on a floor type thinner than a 102 mm (4 in.) concrete floor, the purchaser, at their expense, shall provide acceptable anchoring and mounting methods that meet all structural specifications provided in sections 8.1 through 8.5 of this Pre-Installation Manual.

8.4 Floor Levelness

The CT Room floor levelness requirement is important for accurate patient positioning. Floor levelness in the Scan Room must not be greater than 6 mm (0.25 in.) between depression and high spots over any 3048 mm (120 in.) distance within the area of the gantry/table template (see [Figure 2-1, on page 36](#)).

Note: Floor must meet levelness specification to properly align the table gantry. Minimum gantry height at this specification will be 1/2" (15 mm) to prevent cable crushing.

Table level may not be achievable if overall floor levelness is greater than the specification.

Overall floor level must be 0" to use under gantry cable entrance. Minimum gantry height will be 3/4" 20 mm with this option to prevent cable crushing.

The use of floor shims is not suitable to achieve floor levelness. It is recommended that the concrete be leveled to meet this requirement.

8.5 Floor Vibration

The CT equipment may be sensitive to vibration in the frequency range of 0.5 to 20 Hz depending on the amplitude of the vibration. It is the customers responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specific limits, However, it is ultimately the customer/architect/engineer responsibility to design the site solution.

8.5.1 Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed 10^{-3} m/s² rms maximum single frequency above ambient baseline from 0.5 to 80 Hz (measured in any 1 hour during a normal operating period).

8.5.2 Transient Vibration

The behavioral characteristics must be such that any measurable transient disturbance must also be minimized to less than 0.01 m/s² peak-to-peak.

8.5.3 Equipment Location

To minimize the interference, the CT equipment should be placed on a solid floor, located as far as possible from the following vibration sources:

- Parking lots
- Roadways
- Subways
- Trains
- Hallways
- Elevators
- Heliports
- Hospital power plants containing pumps, motors, air handling equipment and air conditioning units

8.6 Walls: Scan Window

The recommended patient viewing window dimensions are 1219 mm wide by 1067 mm high (48 in. x 42 in.). The location of the window is dependent on the position of Operator Workspace position. Consult [Section 10.0](#) of this chapter and a **qualified radiological health physicist** for radiation protection requirements of glass (lead content and thickness).

Note: The operator at the Operator Workspace must be able to view the patient during a scan.

8.7 Ceiling Requirement

The weight of "Boom in a room LCD monitor" assy is 50 kg., and ceiling shall have enough strength against the specification.

Section 9.0 Network Connections

Broad-band is the standard network connection and is required for BrightSpeed Series systems. (A dial-up modem is optional.) A 1000 baseT high-speed network is desired, with 100 baseT network service acceptable. Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

The CT system is connected to the network through the Console.

- A patch cable (not to exceed 3.05 m (10 ft.) should be provided by the customer, and it is used to connect the console to a wall box. (See Notes on [Figure 9-3](#))
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera and console.
- The run from the hospital switch to the CT wall outlet must not exceed 88m (290 ft.). Bandwidth performance is degraded when the length reaches 91m (300 ft.) or greater.
- For the optional modem: **Two phone lines should be provided by the facility.** One line is for use with a modem and must be an analog line. The second line is a voice only line.

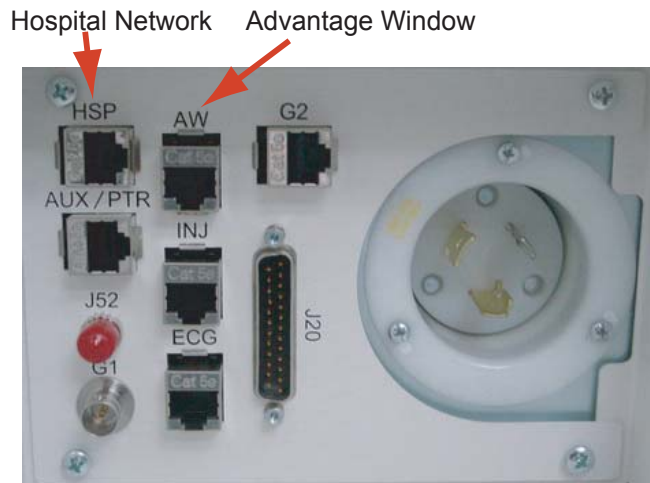


Figure 4-12 Console Rear Bulkhead

9.1 US Broad Band Process Overview

The United States network connectivity requirement for this product is broad-band. The US process relies on the Project Manager Installation to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

9.2 Customer Broad-Band Responsibilities

Provide GE Healthcare Installation Project Manager with an accurate site address, telephone number, contact name, and e-mail address for the:

- Customer Champion
 - Coordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments
 - Act as a focal point in assuring site broad-band infrastructure meets GE Healthcare requirements for connection as determined by a mutual assessment with the GE Healthcare Connectivity team.
- IT Contact
 - Complete an equipment assessment with GE Healthcare Connectivity team to determine site readiness for broad-band
- Contact your Installation Project Manager, for the name of the zone Broad-Band Specialist
 - Work with the Customer Champion to complete any identified infrastructure changes
 - Provide IP addresses for new CT equipment
 - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption
 - Utilize an Internet Service Provider that supports static routing



GE Medical Systems

Remote Service Broadband - Customer Site Assessment

Site Name: _____ FE Name: _____
City, State: _____ FE Phone: _____
Date: _____ FE Email: _____

- | | Yes | No |
|---|---|--------------------------|
| 1. Does your site currently have a persistent (24x7) Internet connection? | ☐ | ☐ |
| 2. Is the GEMS Diagnostic Imaging equipment on the Local Area Network and will it be accessible to the Internet? | ☐ | ☐ |
| 3. Does your site have a VPN device today? | ☐ | ☐ |
| 4. Is the VPN device one of the models below? <i>If Yes, please select the model from the options below.</i> | ☐ | ☐ |
| ☐ a) Cisco Pix Firewalls
☐ b) Cisco Routers
☐ c) Cisco 3000 Series (Altiga acquisition)
☐ d) Checkpoint Firewalls Software Version 4.1 and higher
☐ e) Nortel Contivity Software Version 3.2 or higher
☐ f) Redcreek | ☐ g) Symantec (Raptor) firewalls
☐ h) Firebox
☐ i) Linux S/WAN
☐ j) Sidewinder
☐ k) Netscreen
☐ l) None
☐ m) Other _____ | |

**If No, the GEMS Connectivity Support Team can help determine device compatibility.*

5. Does your VPN device support "triple DES" Encryption? ☐ Yes ☐ No
6. Has approval been given to install this VPN connection? ☐ Yes ☐ No
- Site Approver's Name _____
7. Provide your VPN Installer information, this is the person who will be contacted to schedule the VPN install.

Customer Installer Name: _____
Installer Telephone Number: _____
Installer e-mail address: _____

Notes: _____

Field Engineer needs to provide compatible system information:

System ID	IP Address	Gateway Address
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

[Additional System and IP Address Spaces Available on Page 2](#)

If you have questions or need assessment support contact your Zone Champ or:

Joe Gracz - HQ Support 1-262-524-5261
Joseph.Gracz@med.ge.com

Once you have completed both pages of this form, please send it to:

- a) Judy Heyer judy.heyer@med.ge.com
b) Judy Heyer Fax# 414-918-4707

[Use the send button on page 2](#)



GE Medical Systems

System ID List & Compatibility Matrix

MR	Software Version	CT	Software Version
Include all Signa(LX, Ovation, OpenSpeed and CV/I) MR's System Ids and IP Addresses in assessment.		HiSpeed CT/e	Minimum 6.0
		HiSpeed CT/e dual	Minimum 2.06
		HiSpeedNX/I	Minimum 5.5
		LightSpeed QX/I	Minimum 1.3
Prior to completing an InSite checkout systems must be upgraded to Version 9.0 or 9.1 unless shown in the compatible list below.		LightSpeed Plus	Minimum 2.1
Signa TwinSpeed	Release 9.0	LightSpeed Ultra	Minimum 3
Signa	Release 9.1	HiSpeed X/I	
Signa	Release 10.x	CT/I	Minimum 6.2
Signa CV/I	CNV4	Lxi, Dxi, Fxi, Zxi	Minimum 6.01
Signa 3T		BrightSpeed	TBD
Signa Profile	Minimum 7.66		
Signa Contour	Minimum 7.66		
Magnet Monitor	Minimum 2.3 software		
Network Products	Software Version	Nuclear/Pet	Software Version
AW 4.0	Version 4.0 or Above	Advance	Minimum 5.1
AW 4.0P		eNTEGRA	Minimum 2.03

X-Ray

Cardiac

INNOVA	All
--------	-----

Mammography

Seno2000D	All
-----------	-----

Digital Radiology

Revolution XQ/I Version 10.12.5 or above

Revolution XR/d Version 18.0

Multi Vendor

MR	PSI	CT	PSI
Philips ACS 1.5T	MZP 400	Picker MX-TWIN	CZM 400
Philips NT 2000	MZP 401	Picker MX 8000	CZM 401
Philips NT 3000	MZP 402	Picker PQ	CZM 500
Philips NT ACS	MZP 403	Picker PQ 2000	CZM 501
Picker Outlook 0.23T	MZM 301	Picker PQ 5000	CZM 502
Picker Edge 1.5T	MZM 800	Picker PQ 6000	CZM 503
Picker Vista 1.0T	MZM 801	Picker PQS	CZM 504
Picker Eclipse 1.5T	MZM 900	Siemens AR STAR	CZS 100
Picker Polaris 1.0T	MZM 901	Siemens AR.C	CZS 101
Siemens Harmony 1.0T	MZS 100	Siemens AR.HP	CZS 102
Siemens Impact 1.0T	MZS 200	Siemens AR.SP	CZS 103
Siemens Impact Expert 1.0T	MZS 201	Siemens AR.T	CZS 104
Siemens SP 1.0T	MZS 400	Siemens PLUS-4 EXP	CZS 400
Siemens SP 1.5T	MZS 401	Siemens PLUS-4 POW	CZS 401
Siemens SP4000 1.5T	MZS 403	Siemens PLUS-4 VOL	CZS 402
Siemens Symphony 1.5T	MZS 500		
Siemens Vision 1.5T	MZS 600		

Field Engineer needs to provide compatible system information:

[illegible]

Once you have completed this form, please send it to:

Judy Heyer
Judy Heyer

email
Fax #

judy.heyer@med.ge.com
414-918-4707

Section 10.0 Radiation Protection



NOTICE Scanner-room shielding requirements should be reviewed by a qualified radiological health physicist taking into consideration:

- Scatter radiation levels within the scanning room (see [Figure 4-14](#))
- Equipment placement
- Weekly projected work-loads (# patient/day technique (kvp*ma))
- Materials used for construction of walls, floors, ceiling, doors, and windows
- Access to surrounding scan room areas
- Equipment in surrounding scan room areas (e.g., film developer, film storage)

[Figure 4-13](#) depicts measurable radiation levels within the scanning room while scanning a 32 cm CTDI phantom (body) and a 20 cm water phantom (head) with the technique shown. The mAs, kV and aperture scaling factors shown in [Table 4-11](#) can be used to adjust exposure levels to the scan technique used at the site.

Note: Actual measurements can vary. All measurements have an accuracy of $\pm 20\%$ because of measurement equipment, technique, and system-to-system variation.

Use the correction factors shown in [Table 4-11](#) to adjust exposure levels to the usual scan technique at your site.

CHANGED PARAMETER	MULTIPLICATION FACTOR
mAs	new mAs/100
80 kV	0.21
120 kV	0.71
140 kV	1.0
4 x 3.75mm images	0.82
16 x 0.625 LD	0.59
8 x 1.25 LD	
4 X 2.5 LD	
Fluro 5mm	
4 x 1.25 LD	0.40
5mm (1i)	
Fluro 2.5 mm	
1 x 1.25mm images	0.20
2 x 0.625 LD	0.10
1 x 1.25	

Table 4-11 Shielding Requirements Scaling



NOTICE The units of measure used for radiation levels have been changed in this publication, from mR (millirads) to μGy (micrograys). The conversion factor is:

$$1 \text{ mR} = 8.69 \mu\text{Gy}$$

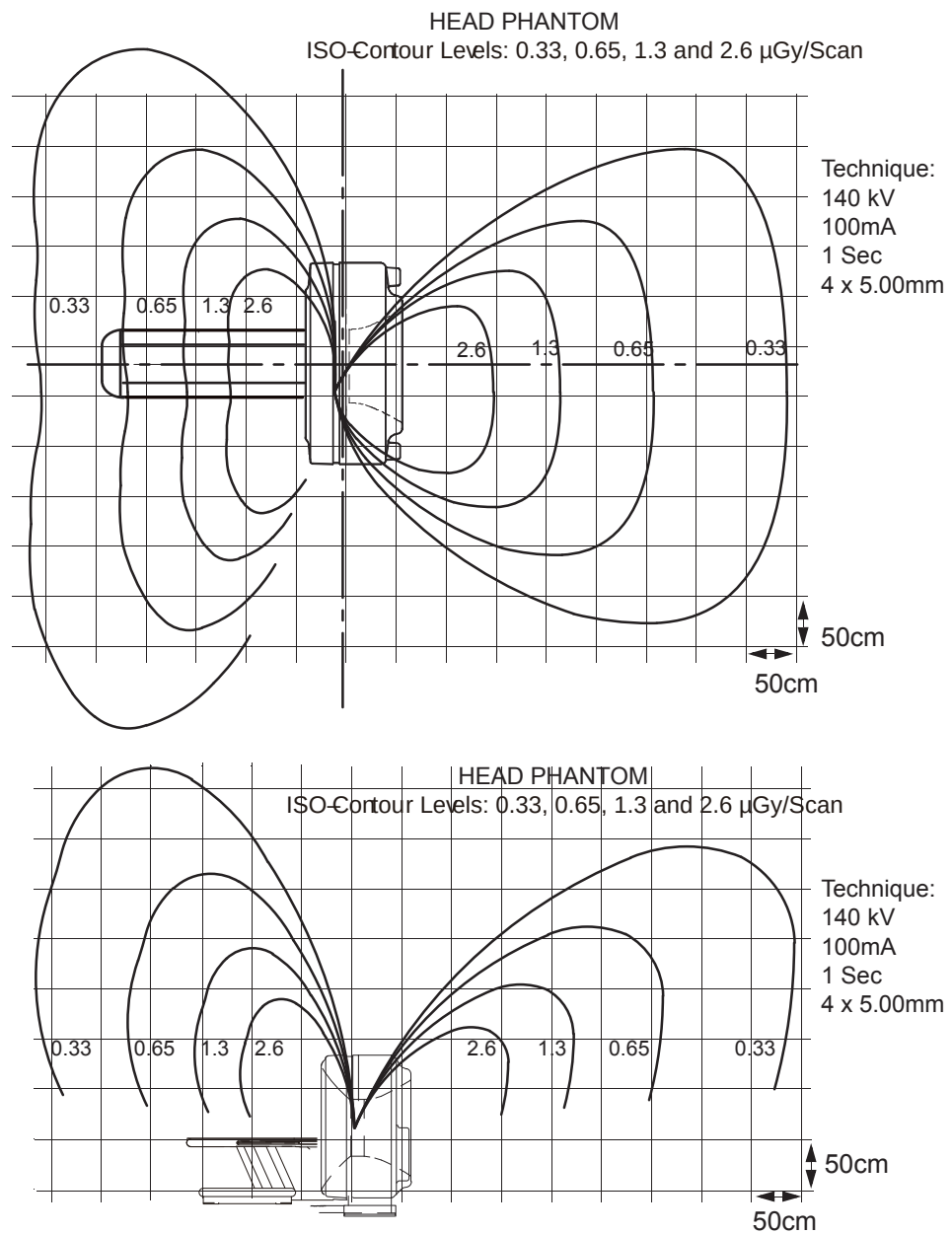


Figure 4-13 Typical Scatter Survey (Head Filter)

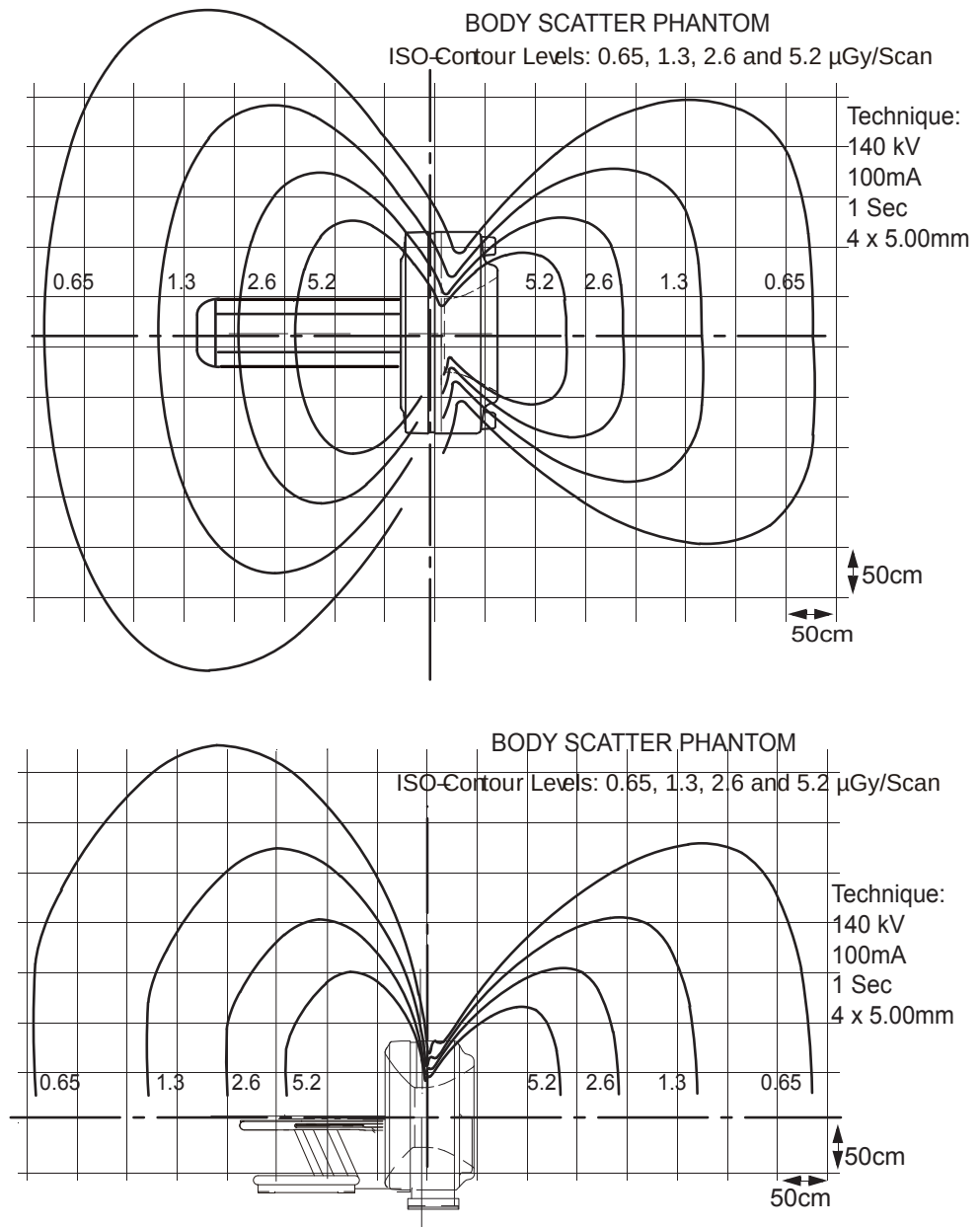


Figure 4-14 Typical Scatter Survey (Body Filter)

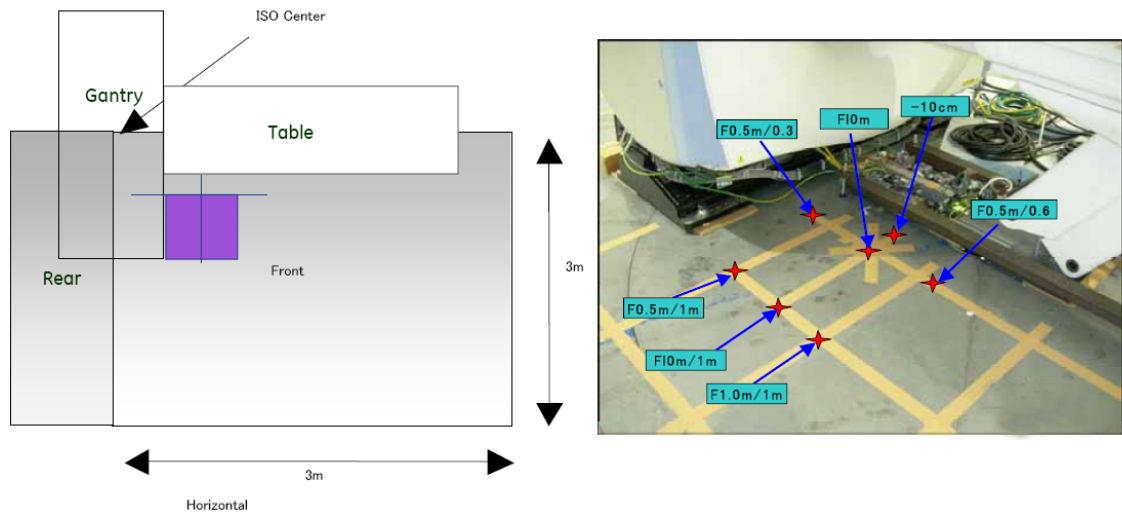


Figure 4-15 Fluoro Stand Position

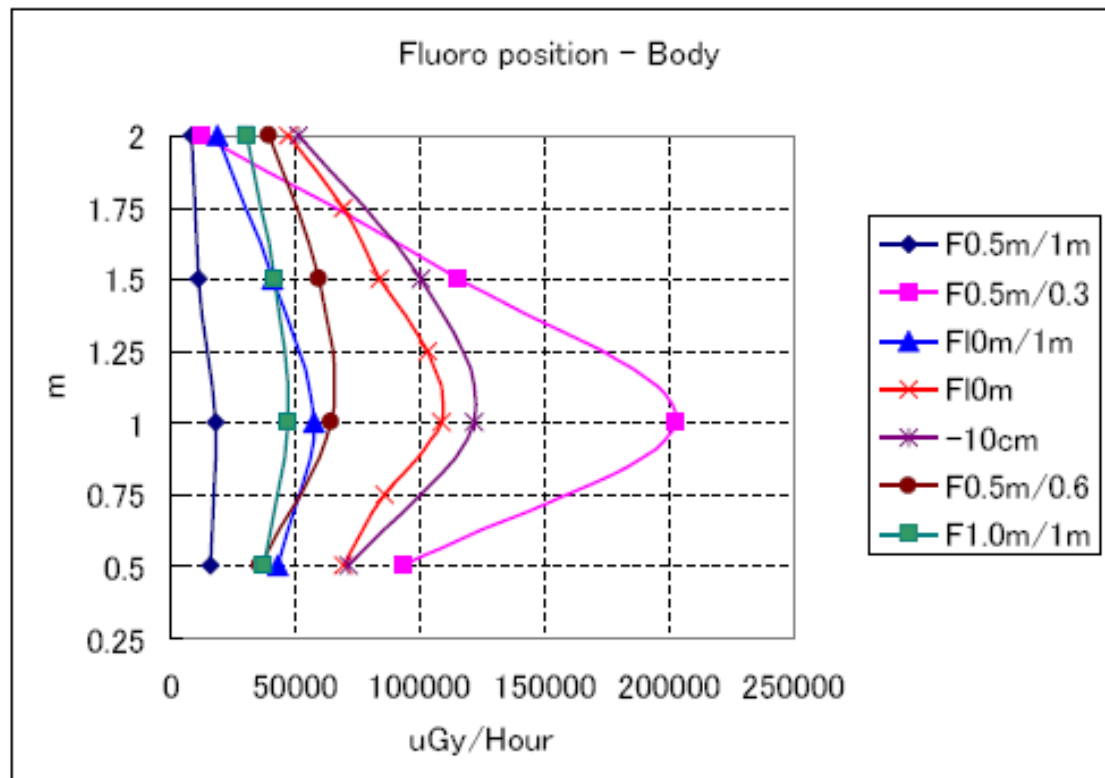


Figure 4-16 Scatter Plot of Fluoro - Body

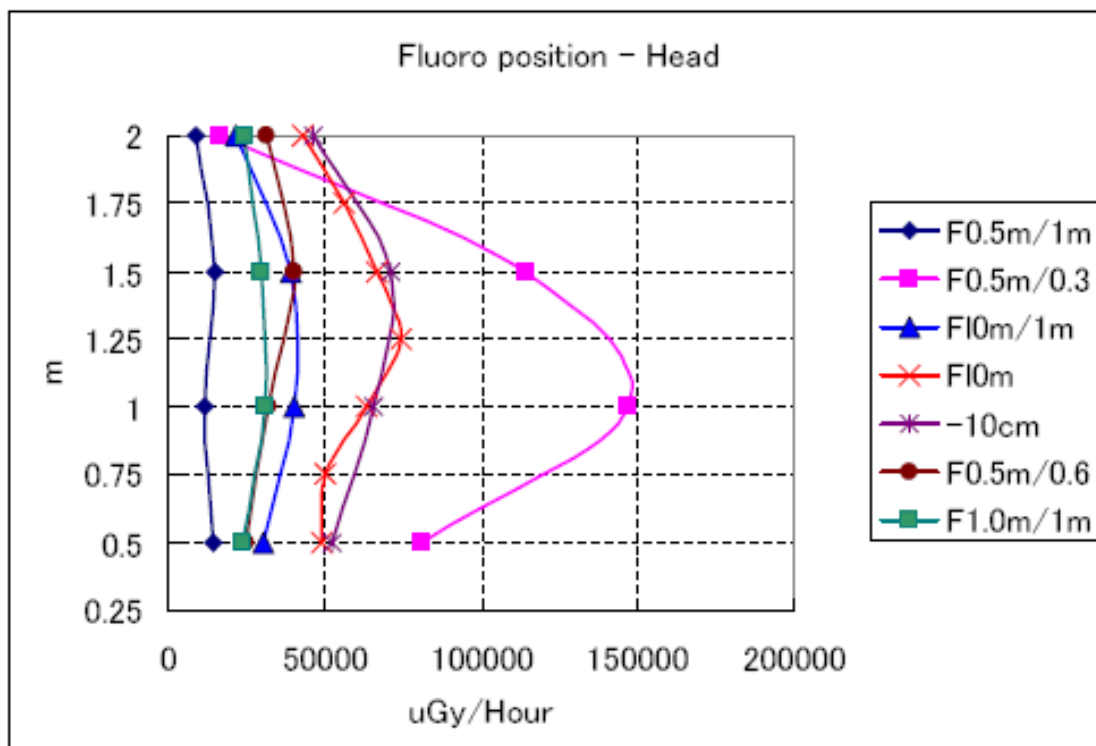


Figure 4-17 Scatter Plot of Fluoro - Head

Chapter 5

Environmental Conditions

Ratings and duty cycles of CT subsystems apply if site environment meets the standards of this section. Maintain environmental conditions listed below at all times – including, for example, overnight, weekends and holidays. Shut down the CT system if air conditioning is not working.

When system is shut down for major repair, air conditioning may be shut down also.



NOTICE

Silver, copper, gold films used in the CT system are especially sensitive to the presence of sulfide, chloride and nitrate contaminates. Sulfur is the most damaging. If high levels of contaminates exist site may want to consider appropriate air filtration systems.

Section 1.0

Temperature and Humidity Specifications

Environmental specifications apply to the table, gantry, power distribution unit, and operating console.



NOTICE

System Operation and Image Quality may be affected, if environmental specifications are exceeded.

1.1 Temperature (Scan Room & Control Room)

Maximum allowable ambient room temperature:	26° C (79° F)
Recommended ambient room temperature:	22° C (72° F)
Minimum allowable ambient room temperature:	18° C (64° F)
Maximum allowable ambient room temperature rate of change:	3° C/hr. (5° F/hr.)

Equipment Room: If a separate equipment room is used to house the PDU, the allowable temperature range is 64° - 79° F (18° - 26° C).

1.2 Humidity (Scan Room & Control Room)

Maximum allowable non-condensing relative humidity:	60%
Minimum allowable non-condensing relative humidity:	30%
Maximum allowable relative humidity rate change:	5% per hour

1.3 Other Guidelines

- To help determine the hospital room environmental conditions, a temperature and humidity recorder may be temporarily installed close to where the system will be installed. Note readings before and after installation, to verify the true temperature and humidity in your environment.
- Consider heating, ventilating, air conditioning (HVAC) needs and redundancy. An air conditioner with two compressor units, rather than one, may prevent system downtime. A back-up (redundant) air conditioner permits CT system operation during an extended repair of the primary air conditioner.

Section 2.0 Cooling Requirement

Use [Table 5-1](#) to assist in cooling requirements planning. Over half the cooling used by your scanner is required for gantry operation. Locate a wall air-conditioning vent at floor level beside and behind gantry to meet both gantry cooling needs and provide patient comfort. Do not locate any cooling vents directly above the gantry. Air returns above the gantry are recommended.

SYSTEM COMPONENT	BTU/HR	WATT
1. Gantry recommended (See NOTE 1)	18,766	5,500
2. Table	682	200
3. Power distribution unit	5,118	1,500
Recommended Scan Room Subtotal (see notes):	24,566	7,200
4. Operator's console w/ 1 IG, 2 monitors & SCSI Tower	8,189	2,400
MOD Drive (Option)	682	200
IG Node, each (Option)	1,024	300
Remote Monitor (Option)	222	65
DASM (Option)	290	85
Recommended Control Room Subtotal (w/o option):	8,189	2,400
System Total (Recommended) (See NOTE 1)	32,755	9,600
Option: Advantage Windows	256	75
Option: UPS	5,200	1,200
ROOM TOTAL (SEE NOTE 2)		
NOTE 1: With 75 scan rotations per patient: Recommended BTU/hr. provides for up to six patients per hour. It is also needed during calibration of the system. NOTE 2: Cooling requirements do not include cooling for room lighting, personnel or non-CT equipment.		

Table 5-1 Cooling Requirements (Worksheet)

Please refer to [Figure 4-6](#), and [Figure 4-9](#) in [Chapter 4, Section 6.0](#), for component air flow requirements.

Section 3.0 Altitude

System operating altitude is from -492 ft. to 7,875 ft. (-150 meters to 2400 meters).

Section 4.0

Electro-Magnetic Interference (EMI)

Note: If power sub-stations exist under or above the scan room, or near the control room, consider EMI testing to determine if your proposed room meets the published acceptable EMI room limits. This also includes high voltage lines under the scan or control room floor.

4.1 Gantry and Table

Locate gantry in ambient static magnetic fields of less than 10^{-4} tesla (1,000 milligauss) to guarantee specified imaging performance. Ambient AC magnetic fields must be below 10^{-6} tesla (10 milligauss) peak.



WARNING REQUISITES FOR ROOM

GANTRY & TABLE ARE ALLOWED TO BE INSTALLED ONLY IN X-RAY PROTECTED ROOMS, WHICH PROVIDE AN ATTENUATION OF AT LEAST 12dB FOR RADIO DISTURBANCES FROM 30 MHz TO 1 GHz.

(The Warning statement above is specified by IEC 60601-1-2, First Edition, 1993)

4.2 Color Monitor

Locate color monitors in ambient static magnetic fields of less than 5×10^{-5} tesla (100 milligauss) to guarantee color purity and display geometry. See [Figure 5-1](#).

4.3 Console / Computer Equipment

Locate computer equipment in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss) to guarantee data integrity. See [Figure 5-1](#).

4.4 Magnetic Media

Locate magnetic media in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss).

4.5 PDU

The PDU produces an electromagnetic field that radiates outward from its cabinet in all directions. Do not place sensitive electronics (e.g., console or computer equipment - the UPS is not classified as sensitive electronics) within one-and-a-half meters (1.5m) of the Power Distribution Unit., in any direction (including above or below). See [Figure 5-1](#).

4.6 EMI Reduction

If fields of excessive EMI are known or suspected to be present, consult GE Medical Systems Sales & Service for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from source of magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- High-powered radio signals are a source of EMI.

Maintain good screening of cables and cabinets.

4.7 UPS

The Uninterruptable Power Supply (UPS) provides a consistent power supply to various electrical components of the system. Also, it continues to provide electrical power to components during a site-wide power outage so components can be safely shut down. The UPS should be kept at least one meter (1m) away from sensitive electronics (the PDU does not include sensitive electronics).

4.8 Equipment EMI “Envelopes”

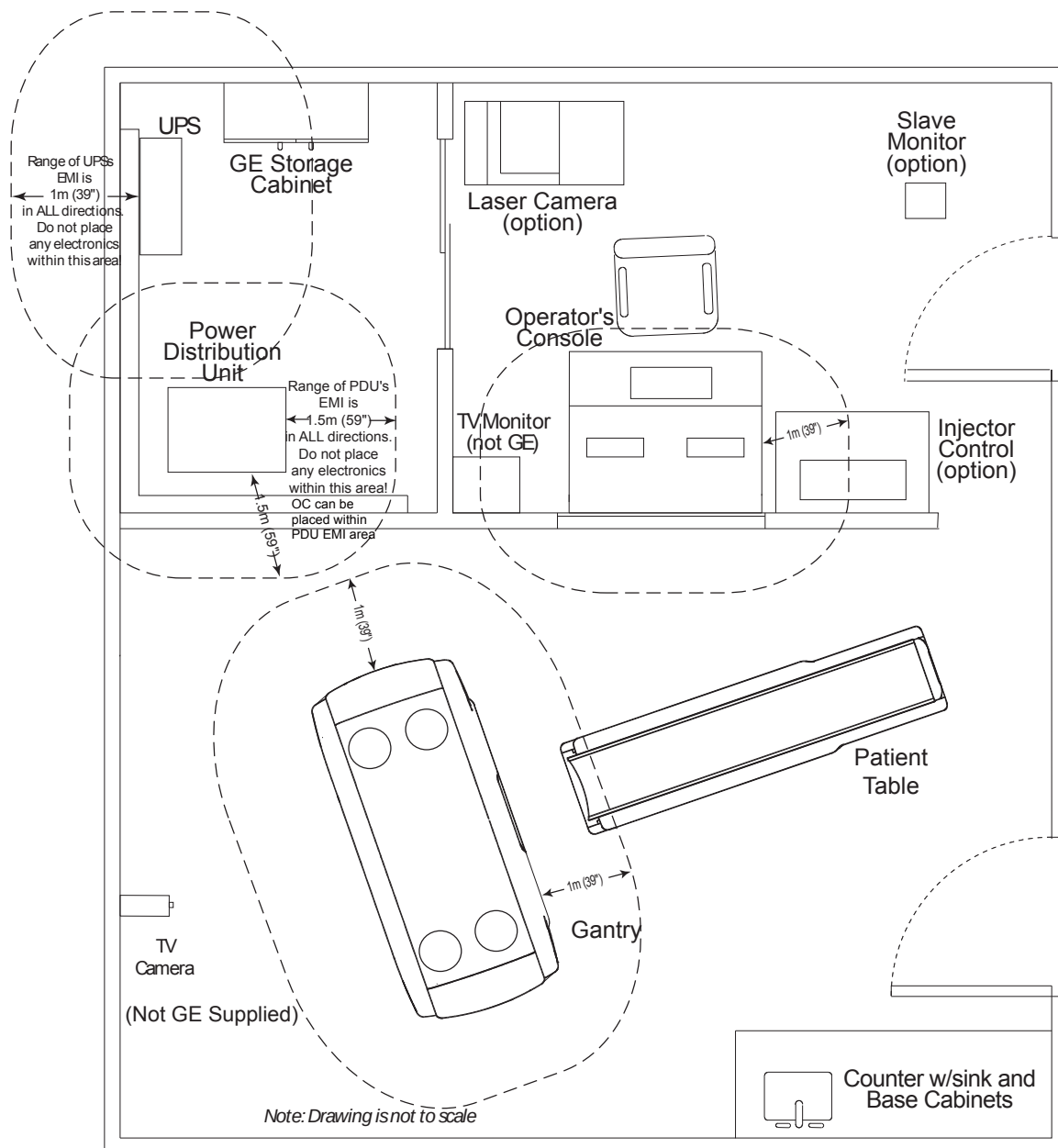


Figure 5-1 Sample Room Layout, showing approximate EMI requirements

Chapter 6

Floor Loading and Weights

Section 1.0 Floor Loading

The BrightSpeed system has a total floor load of approximately 6340 lbs (2880 kg). About 5088 lbs (2310 kg), including patient (450 lbs (205 kg)), is concentrated in the table-gantry assembly.

Table 6-1 lists CT components with weight, size, floor loading and normal mounting requirements.

ITEM	NET WEIGHT LB(KG)	OVERALL W X D INCH (MM)	WEIGHT/AREA LB/SQ. FT. (KG/M ²)	LOAD PATTERN IN. (MM)	NORMAL METHOD OF MOUNTING IN. (MM) (GE SUPPLIED) ¹
Gantry (w/ covers)	~3899 (~1770)	81 x 40.1 (2057 x 1018)	263 (1286)	Rectangular base plate 27.6 X 77.4 (700 X 1966) with four round pads, each 2.5 (63.5) in contact with floor. Individual pad loadings are: 910 lb, 960 lb, 1040 lb, and 1090 lb (see Figure 6-3).	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at four leveling pads into concrete floor.
Dollies (each)	250 (114)				
Top Cover (each)	15.4 (7)				
Side Cover (each)	11 (5)				
Front Cover	92.5 (42)				
Rear Cover	92.5 (42)				
Patient Table	1201 (545) Includes 450 (205) Patient	24.3 x 94 (616 x 2387)	175 (852)	Rectangular base 17 X 57.5 (432 X 1460) with five round pads, each 2.5 (63.5) in contact with floor. Worst-case cantilever pull on any bolt is 1235 lb (561 kg).	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at three leveling pads into concrete floor.
Dolly	215 (98)				
Power Distribution Unit	~770 (~350)	28 x 22 (700 x 550)	185 (909)	Four Casters support area of 28 X 22 (711 X 559).	Casters are for positioning and service. Set on floor. May be anchored to floor using angle brackets ² in seismic zones.
Console w/ single-piece desktop & w/o monitors	Elite: 430 (195) Edge/Excel: 397 (180)	48.8 x 48.6 (1240 x 1235)	Elite: 140 (681) Edge/Excel: 134 (652)	Four Casters or Leveling Feet support area of 46 X 19 (1168 X 483).	Casters are for positioning. Set on floor. Console may be anchored to floor using angle brackets ²
Console w/o single-piece desktop & w/o monitors	Elite: 353 (160) Edge/Excel: 320 (145)		Elite: 116(563) Edge/Excel: 110 (533)		
All-In-One Console w/ single-piece desktop & w/o monitors & w/o IG	363 (165)		113 (567)		

Notes:

- 1.) Use the GE Supplied mounting hardware ONLY IF APPROVED by qualified personnel.
[See statements in 8.1 - Table and Gantry Mounting Requirements, on page 65.]
- 2.) Angle brackets are included on the shipping skid, and are also available in the OPTIONAL Seismic Kit (R4390JC).

Table 6-1 BrightSpeed System Floor Loading

ITEM	NET WEIGHT LB(KG)	OVERALL W X D INCH (MM)	WEIGHT/AREA LB/SQ. FT. (KG/M ²)	LOAD PATTERN IN. (MM)	NORMAL METHOD OF MOUNTING IN. (MM) (GE SUPPLIED) ¹
All-In-One Console w/o single-piece desktop & w/o monitors & w/o IG	286 (130)	48.8 x 27.8 (1240 x 705)	89 (447)		
Console w/ single-piece desktop & w/o monitors & w/ 3 IG (Elite only)	496 (225)	48.8 x 48.6 (1240 x 1235)	163 (789)		
Console w/o single-piece desktop & w/o monitors & w/ 3 IG (Elite only)	419 (190)	48.8 x 48.6 (1240 x 1235)	138(666)		
Monitor - LCD (each)	20 (9)	16.5 x 9.7 (420 x 247)			
Freedom Workspace (part 5160735)	86 (39)	51 x 26 (1295 x 662)			
Freedom Workspace (part 5168666)	99 (45)	51 x 24 (1300 x 620)			

Notes:

- 1.) Use the GE Supplied mounting hardware ONLY IF APPROVED by qualified personnel.
[See statements in [8.1 - Table and Gantry Mounting Requirements](#), on [page 65](#).]
- 2.) Angle brackets are included on the shipping skid, and are also available in the OPTIONAL Seismic Kit (R4390JC).

Table 6-1 BrightSpeed System Floor Loading

Section 2.0

Mounting Data, Including Seismic

The following pages show center-of-gravity information for system components:

- Gantry: [Figure 6-2](#)
- Table: [Figure 6-4](#)
- Power Distribution Unit: [Figure 6-5](#)
- Operator's Console/Computer: [Figure 6-6](#)
- Freedom Workspace Table: [Figure 4-10](#) and [Figure 4-11](#)

Floor mounting hole locations for components that don't have templates are also in this section. Customer is responsible for seismic mounting. Refer to all applicable codes in your area. GE provided floor anchors ([Figure 6-1](#)) are designed to be used ONLY on concrete floors that meet the concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and shall be set to a minimum depth of 3 inches at each anchor point. Any anchors having more than 1 inch of thread showing above the nut, when torqued to 55 ft.-lbs, shall have a second anchor installed in the closest adjacent hole. the second anchor shall be installed to the standard depth and torque specifications.

MOUNTING REQUIREMENTS	GANTRY	TABLE
Minimum Floor Thickness:	102 mm (4 in.)	102 mm (4 in.)
Recommended Drilling Depth:	95 mm (3.75 in.)	95 mm (3.75 in.)
Average Anchor Embedment:	89 mm (3.5 in.)	89 mm (3.5 in.)
Minimum Anchor Embedment:	76 mm (3 in.)	76 mm (3 in.)
Available Alternate Anchor Locations:	Yes	Yes
Shipped Anchor Size:	76 mm (3 in.)	76 mm (3 in.)
Alternate Anchoring Methods:	Yes (see notes, below)	Yes (see notes, below)
FLoor Levelness Requirement:	3 mm (0.125 in.) over 3 m (10 ft.)	3 mm (0.125 in.) over 3 m (10 ft.)

Table 6-2 Gantry and Table Mounting Requirements

- Note: All other anchoring methods on floor types other than the concrete minimum must be determined at the customer's expense by a structural engineering contractor, and anchoring method must be certified to meet the stated GE minimum load requirement and torque specification.
- Note: If installing the GE BS scanner on a floor type other than a 4 inch concrete floor, all structural specifications in this document must be reviewed and met.

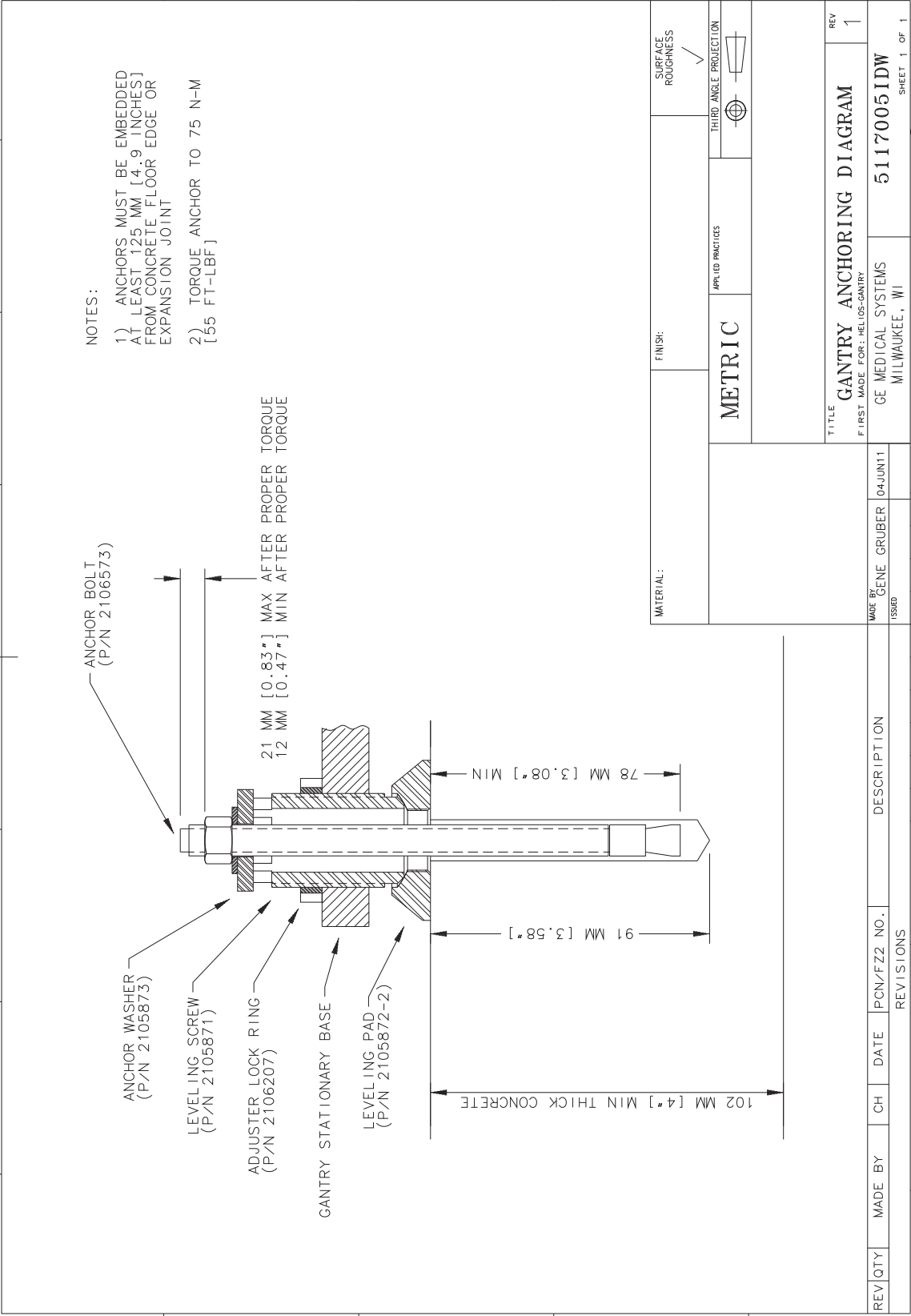


Figure 6-1 Typical Floor Anchor, Gantry and Table

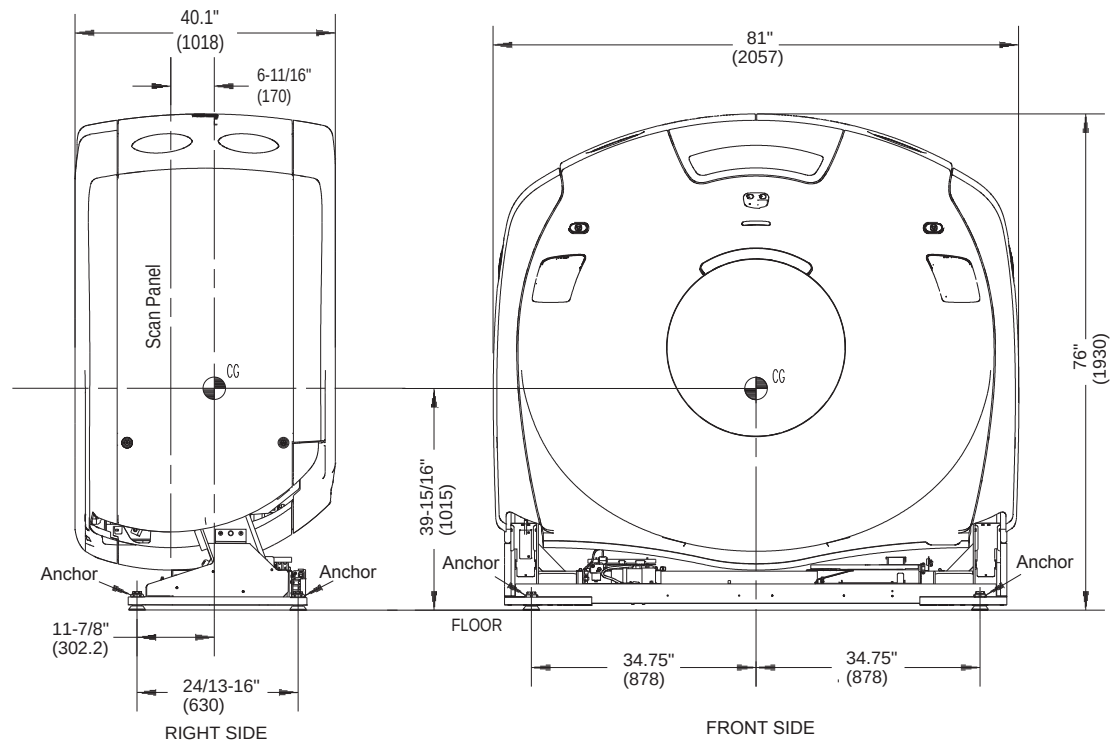
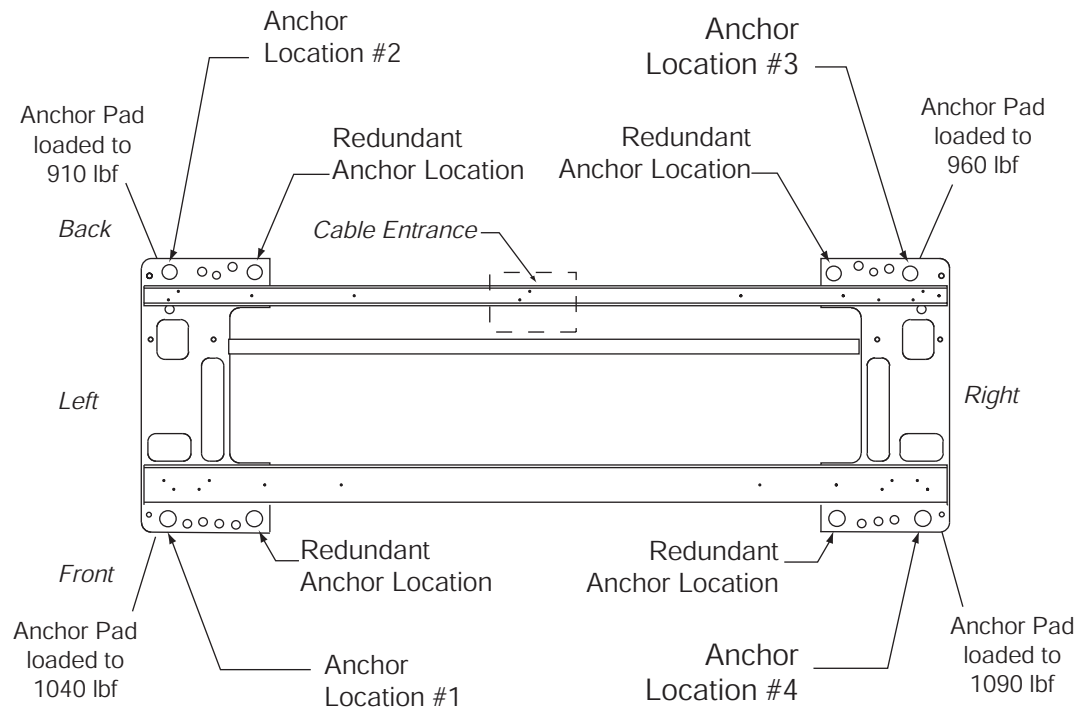
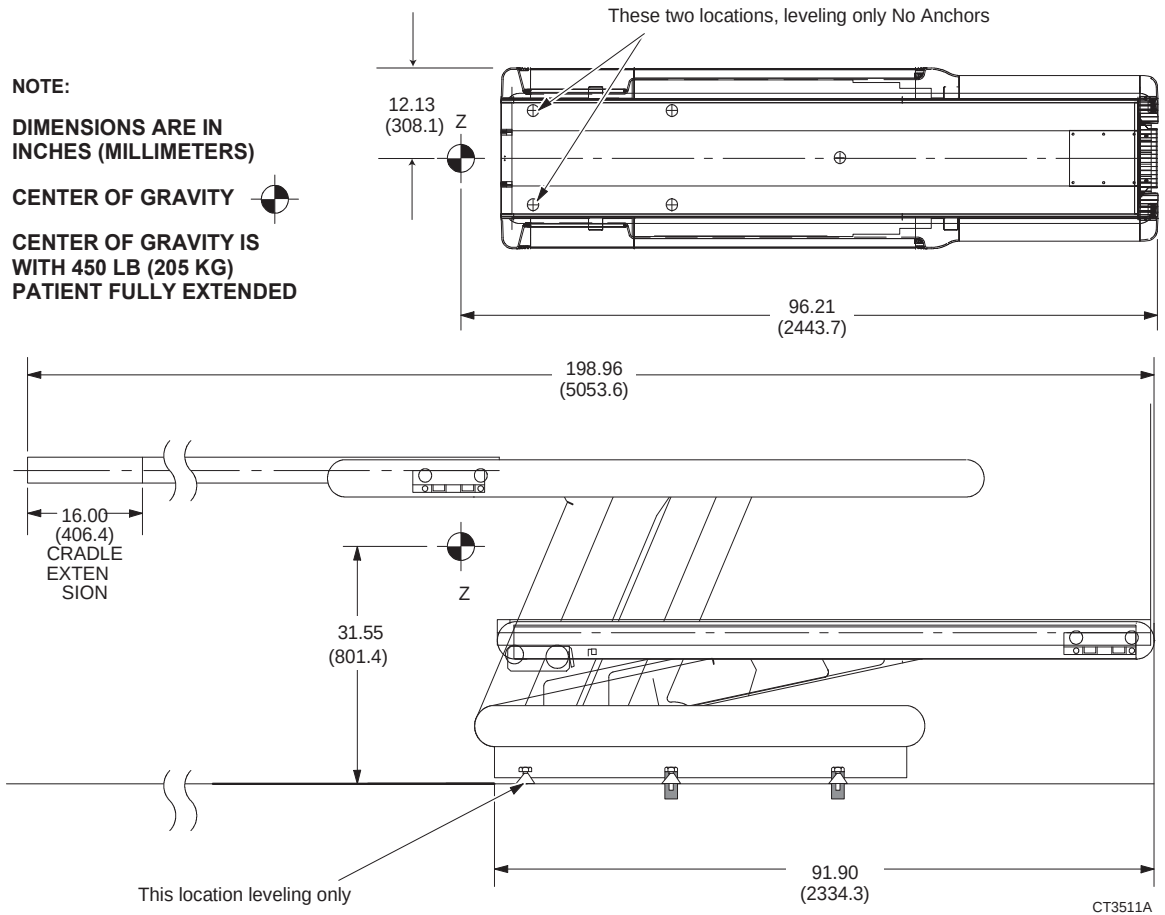


Figure 6-2 Gantry (CT2)



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 6-3 Gantry Anchor Locations

NOTE:**DIMENSIONS ARE IN
INCHES (MILLIMETERS)****CENTER OF GRAVITY** **CENTER OF GRAVITY IS
WITH 450 LB (205 KG)
PATIENT FULLY EXTENDED****Figure 6-4 Patient Table (CT1)**

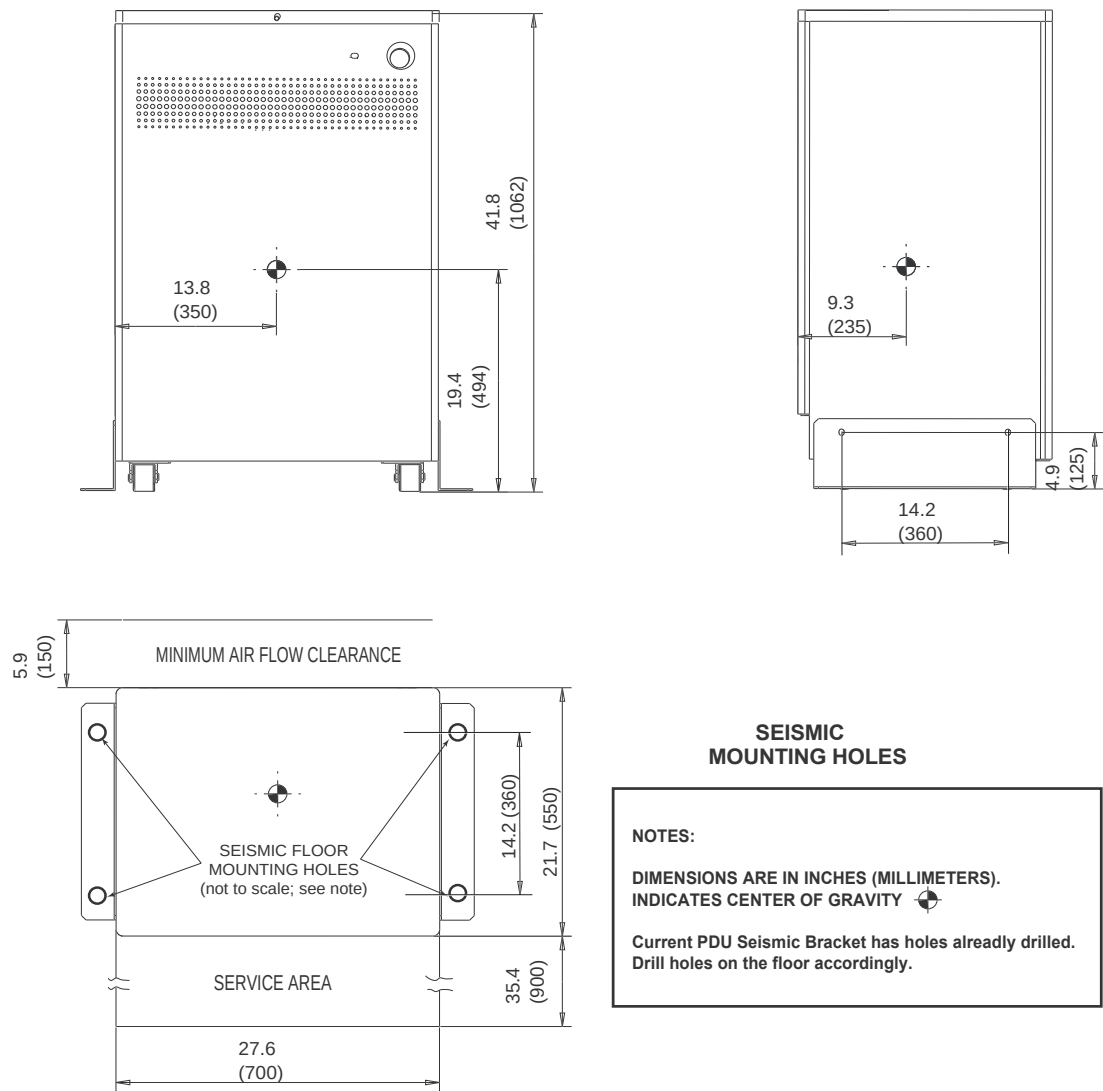


Figure 6-5 Power Distribution Unit (NGPDU)

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Chapter 7

Delivery Data



WARNING SOME ASSEMBLIES ARE TOP-HEAVY. BE CAREFUL NOT TO TIP.

Section 1.0 Van Delivery

The CT system is packed for van shipment with minimum tear-down of components. It consists of approximately 10 shipping containers.

Section 2.0 Delivery/Shipping Requirements & Considerations

The BrightSpeed System is not designed to tolerate excessive mishandling, including dropping, shock, vibration, tipping or hoisting.

The Gantry, Console, Table and PDU must NEVER be dropped. A drop from a height greater than one half inch ($\frac{1}{2}$ ") may induce structural damage to the frame or other major components. Damage resulting from a drop (e.g., bent frame, misalignment) may not be obvious until after system installation is complete.

To avoid dropping the Gantry, Dock to Dock shipment is recommended. Other methods are acceptable, provided that the system is not dropped or otherwise mishandled. For example, the system may be moved via flat-bed wrecker or by rolling it across SMOOTH sidewalks or other paved surfaces.

When moving Gantry off of a flat-bed wrecker, attach the straps to the lowest point possible on the dolly. Lower the Gantry at the slowest reasonable rate. See [Figure 7-1](#).

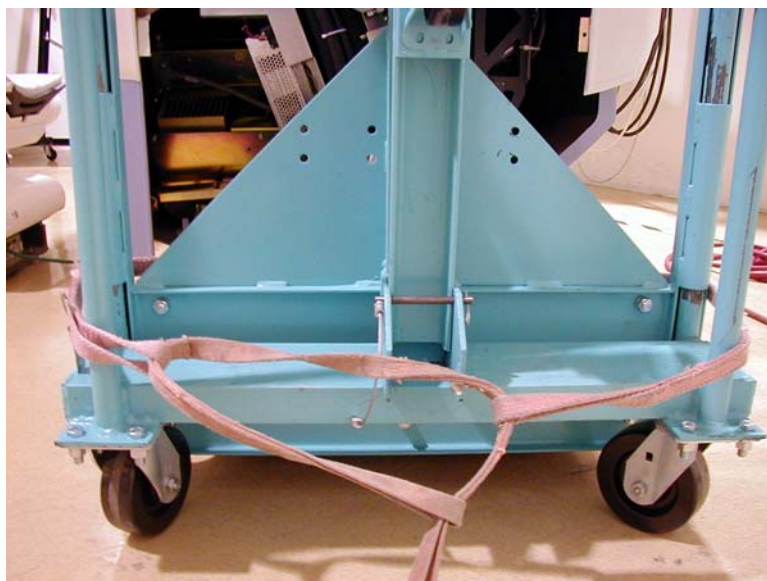


Figure 7-1 Gantry Strap Location

The BrightSpeed System—including Gantry, Console, Table and PDU—is not designed to tolerate the excessive shock or vibration that may occur during unloading. For example, rolling the Console across a “washboard” style ramp may vibrate components to the extent of loosened or broken connections, etc. Damage resulting from shock or vibration (e.g., monitor, CD-ROM, hard-drive or octane failure) may not be evident until after system installation is complete.

All system components must remain upright at all times, and must not be tipped. Nor should the Gantry be hoisted. The Gantry should be moved by rolling it on its dollies only. Movement through hallways, doorways, elevators, etc., must be done without tipping or lifting the Gantry.

Protection for flooring along the move path (from dock to scan room) is advised.



NOTICE Lifting a gantry requires engineering approval for each occurrence. Your Project Manager Installation should contact engineering for all special lifting requirements.

- **Do Not lift the gantry with a Forklift**
- **Unauthorized gantry lifting will cause Gantry Bearing Damage**

Engineering Contact Information:

Email: Xiang.Zhang@med.ge.com

Phone: 86-10-58068625

Section 3.0

Site Environmental Considerations

3.1 Dust/Dirt Contamination

BrightSpeed systems (consisting of: Console, PDU, Table and Gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site. Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.



NOTICE The act of installing a GE CT scanner in a construction (i.e., unfinished) site will likely result in the following adverse effects:

- **Increased installation time**
- **Decreased installation quality**
- **Increased scanner downtime, due to increased service calls**

3.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of BrightSpeed CT scanner components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration's should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

Section 4.0

Storage Requirements

SHORT TERM STORAGE (LESS THAN 6 MONTHS)

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 40° to +80° F (4° to +27° C).
- Maintain relative humidity (non-condensing) between 20% and 60%.
- Maximum relative humidity rate of change is 5%/hr.
- The maximum temperature rate of change is 5° F/hr. (3° C/hr.)
- Air pressure should be between 700hPa and 1060hPa.



NOTICE Between delivery is considered short-term storage.
Van storage must meet the same specifications as above.

LONG TERM STORAGE (6 MONTHS OR MORE)

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 50° to +90° F (10° to +32° C).
- Maintain relative humidity (non-condensing) between 20% and 70%.

If you are storing a system longer than six (6) months, but less than 12 months, contact CT Engineering for installation start-up instructions. Storage longer than 12 months is not recommended.



Figure 7-2 Package Symbols (Storage)

Section 5.0

Extreme Temperature Transportation and Deliveries

Extreme temperatures should be avoided during system transportation and delivery.

Extreme temperatures are defined as below -29 degrees Fahrenheit (-34° C) or above 140 degrees Fahrenheit (60° C), without humidity control.

Section 6.0

System Transportation

When transporting the CT system, ensure that the system is not exposed, for an extended period of time, to temperatures or humidity outside of the following specifications.

Temperature: -29° to +140° F (-34° to +60° C)

Humidity: 5% to 95%



NOTICE Component Freezing occurs if CT system is exposed to temperatures below 0° F (-18° C) for a period longer than two days.
Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature, prior to installation.

Section 7.0

Gantry Considerations

The gantry is mounted on the pallet. Use dolly elevating casters to lift gantry off its base and roll it into position. See [Figure 7-3](#).



Figure 7-3 Gantry with dollies

Door Openings. Clear door openings for moving equipment into building must be 1067 X 2083 mm (42 X 82 in.) minimum. 2439 mm (8 ft.) corridor width is helpful.

Elevator requirements. To move gantry from receiving location to scanning room, consider elevator capacity and size. By removing side rails and one dolly after gantry is placed in elevator, gantry width/length and elevator depth requirements are reduced. Contact a representative of elevator manufacturer if gantry weight exceeds elevator capacity.

CONFIGURATION	LENGTH CM (IN)	WIDTH CM (IN)	HEIGHT CM (IN)
Dollies On, Side Rails On	281 (111)	129 (51)	200 (79)
Dollies On, Side Rails Removed	281 (111)	100 (39.4)	200 (79)
Dollies Off, Covers Off	197 (77)	86 (34)	185 (73)

Table 7-1 Size of Gantry & Dollies, with and without Side Rails

Minimum hallway and door size for gantry with covers and dollies attached, but side rails removed, is 101.6 cm (40 in.).

For alternative lifting arrangements and instructions, contact GE Installation Support Services.

Dollies: Dollies can be purchased for international shipments (B7850LD) or US shipments (B7850GD) to be used at the customer site. After the system has been removed from the crates, dollies shipped with international shipments **only** are not to be shipped back to GE in Milwaukee, WI, USA, but to be retained to the local GE office or warehouse.

Gantry shipping dimensions are:

2440mm (96 in.) long, 1370mm (54 in.) wide, 2180mm (86 in.) high

Section 8.0 Table Considerations

The table is shipped with 2 crates. The mainbody of table is put in crate without dollies. Bottom Cover and accessories are shipped in another crate.

Table shipping dimensions are

2620 mm (103 in.) long, 900 mm (35 in.) wide, and 1300 mm (51 in.) high.

1700 mm (67 in.) long, 770 mm (30 in.) wide, and 750 mm (30 in.) high.

Section 9.0 Console Considerations

The console is shipped without the keyboard table installed. The keyboard table is shipped with the console. See [Figure 7-4](#).

The dimensions of the console alone (as shipped) are 1170mm (46 in.) deep, 1340mm (53 in.) wide, and 1150mm (45 in.) high.



Figure 7-4 Console packing

Chapter 8

Power Requirements

Section 1.0

Introduction

The power distribution unit (PDU) supplied with the BrightSpeed system transforms and distributes power to all system components. The PDU is the only power entry point required to operate system. To minimize voltage regulation effects, power wiring between the facility main distribution panel and the PDU should be kept as short as possible.

When routing the power wiring all three phase wires and ground must be run in the same conduit or raceway duct. Power wires should be routed separately from system control and signal cables, using a separate conduit or trough in raceway duct.

Metallic conduit, floor duct or surface raceway may be used for running cables, depending upon local codes and practices. However, cable passageways should be large enough to install any cable with all other cables already installed. Use of non-metallic conduit is not recommended.

Section 2.0 System Input Power

2.1 Power Source Configuration

The BrightSpeed CT Scanner is designed to operate on a three-phase, four-wire wye power source. A solidly grounded wye source is preferred. The neutral wire does not need to be run to the system, i.e., four-wire connection. If a neutral wire is run, then it should be terminated in the A1 box.

A dedicated feeder from the nearest Main Distribution Panel (MDP) should supply power to the scanner. In accordance with the National Electric Code (U.S.), and similar applicable national and local codes, a protective disconnect device must be provided in the power line supplying the PDU. It must be located within 10 m (32 ft.) of the PDU, visible to PDU service personnel, and must have lockout /tagout provisions. This disconnect device is identified as "A1" in the interconnection schematic diagrams.

2.2 Rating

BrightSpeed Elite (16 slice), Edge (8 slice): 90kVA

BrightSpeed Excel (4 slice): 75 kVA is standard rating, 90 kVA is an optional rating.

The BrightSpeed system operates on three-phase power meeting the following specifications.

(Note: The value in the brackets is for 75kVA.)

Voltage	200 to 480 VAC
Capacity	90 kVA (75kVA)
Frequency	50 or 60 Hz +/- 3 Hz

- Maximum power demand = 90 kVA (75kVA) @ 0.85 PF at a selected technique of 140 kV, 380 mA (300mA).
- Average (continuous) power demand at maximum duty cycle = 20 kVA.

The "A1" disconnect device referenced above must provide overcurrent protection for the system and facilitate multi-point remote "Emergency Off" control of system power. A disconnect utilizing undervoltage release control is preferred over shunt trip devices. The rating of the "A1" disconnect device depends on the nominal line voltage at the site. Refer to [Section 3.0: Recommended Power Distribution System](#), for minimum rating requirements and suggested disconnect devices.



NOTICE The electrical rating is described on the system rating label attached on the gantry; not on the PDU.

2.3 Regulation

Total load regulation as measured at the PDU input terminals must not exceed 6%. The capacity of the facility transformer and size & length of feeder wires directly affect the load regulation presented to the system. Refer to [Section 3.0: Recommended Power Distribution System](#), for recommended single-unit installation specifics.

2.4 Phase Imbalance

The difference between the highest line-to-line voltage and lowest line-to-line voltage must not exceed 2% of the lowest line-to-line voltage.

2.5 Sags, Surges & Transients

Sags and surges of the power line must not exceed the absolute range limits shown in Table 8-1. Maximum transient voltages should be limited to 1500V peak.

2.6 Grounding

Metal conduit, raceway or the armor of armored cable used to power the system should be bonded to the PDU cabinet. However, in addition to such mechanical grounding, a dedicated 1/0 (55 mm²) or larger insulated copper ground wire must be run with the phase wires from the main distribution panel to the PDU.

Note: The shield or armor of armored cable is not sufficient for this purpose.

The ground wire should be bonded to intermediate distribution panels through which it passes in accordance with local codes. The resistance between the PDU ground and the facility earth ground must not exceed 0.5 ohm. In addition, the total resistance between the PDU ground and earth must not exceed 2 ohms.

Section 3.0

Recommended Power Distribution System

A dedicated feeder run from the facility main isolation transformer is recommended to power the BrightSpeed CT scanner. If the scanner must be powered from an existing distribution transformer and secondary feeder, such as the equipment distribution panel of an X-ray department, installation with other X-Ray equipment that use rapid film changers should be avoided. These changers use a large number of high powered, closely spaced exposures, which may coincide with the CT scan and produce image artifacts.

If a dedicated distribution transformer is provided for the scanner, the minimum recommended transformer size is 112.5 kVA, rated 2.4% regulation at unity power factor. For this configuration, the minimum recommended feeder size and overcurrent protection device based on line voltage is shown in [Table 8-3](#) Minimum Feeder Wire Size.

In all cases, qualified personnel must verify that the transformer and feeder, at point of take-off, plus the run to the BrightSpeed CT scanner meet all the requirements stated in this document.

SYSTEM CHARACTERISTICS:

Note: [\(xxx\):The value in the brackets is for 75kVA.](#)

- Maximum power demand = 90kVA [\(75kVA\)](#) @ 0.85 PF: at a Selected Technique of 140 kV, 380 mA [\(300mA\)](#).
- Continuous (average) power demand at maximum duty cycle = 20kVA [\(16.7kVA\)](#).
- Maximum allowable total source regulation is 6%.
- Minimum recommended transformer size: 112.5 kVA [\(93.75kVA\)](#), with 2.4% rated regulation at unity power factor. Resultant maximum allowable feeder regulation is 3.4%.

The nominal line voltage must fall within one of the ranges listed below

Nominal Line Voltage	200	220	240	380	400	420	440	460	480
Hi-Line Limit, +10%	220	242	264	418	440	462	484	506	528
Lo-Line Limit, -10%	180	198	216	342	360	378	396	414	432
Continuous Line Current	58	52	48	30	29	27	26	25	24
Momentary Line Current	260	236	217	137	130	124	118	113	108
Maximum Line Current	289	262	241	152	144	137	131	126	120
Minimum Recommended Circuit Protection Rating	150	150	150	110	110	100	100	90	90

Table 8-1 Nominal Line Voltage

OPTION	CAT NUM
225 kVA Transformer	E4500AW
Isolation Transformer	E4500BC

Table 8-2 Purchasable Options

FEEDER LENGTH (MDATO A1) FEET (METERS)	MINIMUM FEEDER WIRE SIZE, AWG OR MCM (SQ. MM)/ VAC								
	200 VAC	220 VAC	240 VAC	380 VAC	400 VAC	420 VAC	440 VAC	460 VAC	480 VAC
50 (15)	1/0 (55)	1/0 (55)	1/0 (55)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
100 (30)	2/0 (70)	1/0 (55)	1/0 (55)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
150 (46)	4/0 (100)	3/0 (85)	2/0 (70)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
200 (61)	5/0 (125)	4/0 (100)	4/0 (100)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
250 (76)	6/0 (170)	5/0 (125)	5/0 (125)	1 (45)	1 (45)	2 (35)	2 (35)	2 (35)	3 (30)
300 (91)	7/0 (215)	6/0 (170)	5/0 (125)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	2 (35)	2 (35)
350 (107)	8/0 (275)	7/0 (215)	6/0 (170)	2/0 (70)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
400 (122)	8/0 (275)	7/0 (215)	7/0 (215)	2/0 (70)	2/0 (70)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)

Table 8-3 Minimum Feeder Wire Size

SUB-FEEDER LENGTH (A1 TO PM) FEET (METERS)	MINIMUM SUB-FEEDER WIRE, AWG OR MCM (SQ. MM)								
	200 VAC	220 VAC	240 VAC	380 VAC	400 VAC	420 VAC	440 VAC	460 VAC	480 VAC
32(9.7536)	1/0 (55)	1/0 (55)	1/0 (55)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)

Table 8-4 Minimum Feeder Wire Size

- 1.) [Table 8-1](#), [Table 8-3](#), and [Table 8-4](#) above are based on the use of copper wire, rated 75 C and run in steel conduit. Ampacity is determined in accordance with the National Electrical Code (NFPA 70), Table 310-16 (2002)
- 2.) The minimum feeder size is determined by the ampacity of the circuit protection device listed above, except where a larger size is necessary to meet total source regulation limits.
- 3.) A 1/0 (55 sq. mm) ground wire is recommended in all cases.

Section 4.0 Ground System

The BrightSpeed CT Scanner has been designed to use an equal potential grounding system. The required ground system is shown in [Figure 8-1](#). There are three primary grounding points:

- A system power ground point located in the PDU.
- A reference ground point located between gantry and table base.
- A patient ground point located at the front of the table base.

All exposed metal surfaces in the patient vicinity are grounded to the reference ground point.

For additional information, refer to Electrical Safety Equipment, Direction 46-014505.

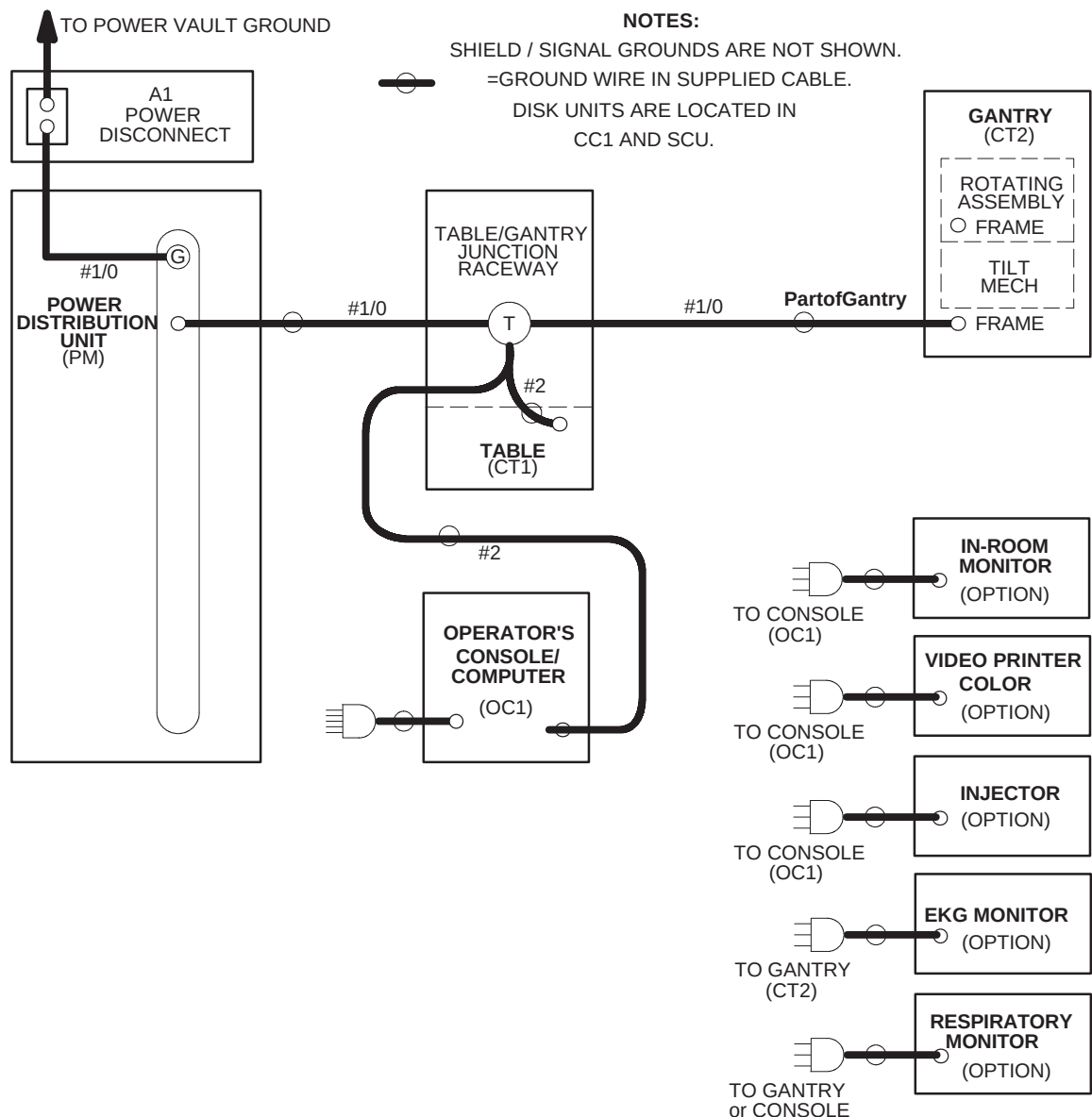


Figure 8-1 System Ground Map

Chapter 9

Interconnection Data

Section 1.0 Introduction

Figure 9-3 shows interconnection runs for a 50/60 Hz system.

Table 9-1 shows component designators for supplied equipment and options and wall power outlets.

Table 9-6 lists customer-installed wiring and supplied cables. Actual length of each run is less than the length of supplied cables to allow for routing inside equipment. Cable diameters and sizes of connectors are provided to aid in sizing conduit and access plates.

Table 9-2 and Table 9-4 list details for connection to BrightSpeed Series equipment, using short length and long length cables, respectively. Details are listed for the following types of runs as appropriate:

- Flush-floor duct
- Computer floor
- Through-wall bushing
- Junction box
- Surface floor duct
- Through-floor duct
- Wall duct
- Conduit

Need for additional junction boxes is minimized by use of either a cable raceway system or a raised computer floor. BrightSpeed Series systems use prefabricated cables with large plugs. Therefore, conduit or pipe is not recommended for cable runs.

Long cable set (B71182CA) and short cable set (B71172CA) are both optional. Order the cable set that is required for your site.

Section 2.0 Component Designators

DESIGNATOR	APPLIES TO	SOURCE
A1	Primary power disconnect	Contractor supplied
CT1	Patient table	System
CT2	Gantry	System
ITL	InSite telephone lines	Contractor supplied
LP	Line printer	Option
OC1	Operator's console/computer	System
PM	Power distribution unit	System
SEO	System emergency off	Contractor supplied
SM	Slave monitor	Option
WL	"X-ray on" warning light	Contractor supplied
DS	Door Interlock Switch	Contractor supplied
XCVR	Ethernet transceiver	System

Table 9-1 Component Designators

Section 3.0 Interconnect Runs, Wiring and Cables

3.1 GEMS Supplied (Optional, Short Run)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	28 (20)	8.5 (6.1)	2343529-2	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (.751)	3	(2) 4 (1) 8	22 (0.87) Dia
051	28 (20)	8.5 (6.1)	2343530-2	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15.3 (.604)	4	14	11.2 (0.44) Dia
052	28 (20)	8.5 (6.1)	2343528-2	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
053	65 (60)	19.8 (18.3)	2343531-2	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12.3 (.483)	3	10	56.4 (2.22) Dia
054			n/a	LVAC, Gantry to Table	1015		600	120VAC			3	14	
055	28 (20)	8.5 (5.97)	2371450-2	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (0.62) Dia
056	68 (57)	20.8 (17.4)	2371450-4	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (0.48) Dia
100	32.5 (20)	9.9 (6.1)	5120646-2	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
101	71 (60)	21.7 (18.3)	5120645-2	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
102	71 (63)	21.7 (19.3)	2373436-3	Signal (LAN), Gantry to OC			1900	<30VDC		5.9 (.234)	8	24	15 (0.59) Dia
103	68 (60)	20.7 (18.3)	2117848-7	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (0.39) Dia
104			n/a	Signal, Gantry to Table		FT-4	300		80		25	22	

Table 9-2 GEMS Supplied Cables (Optional, Short Run) - UL Information

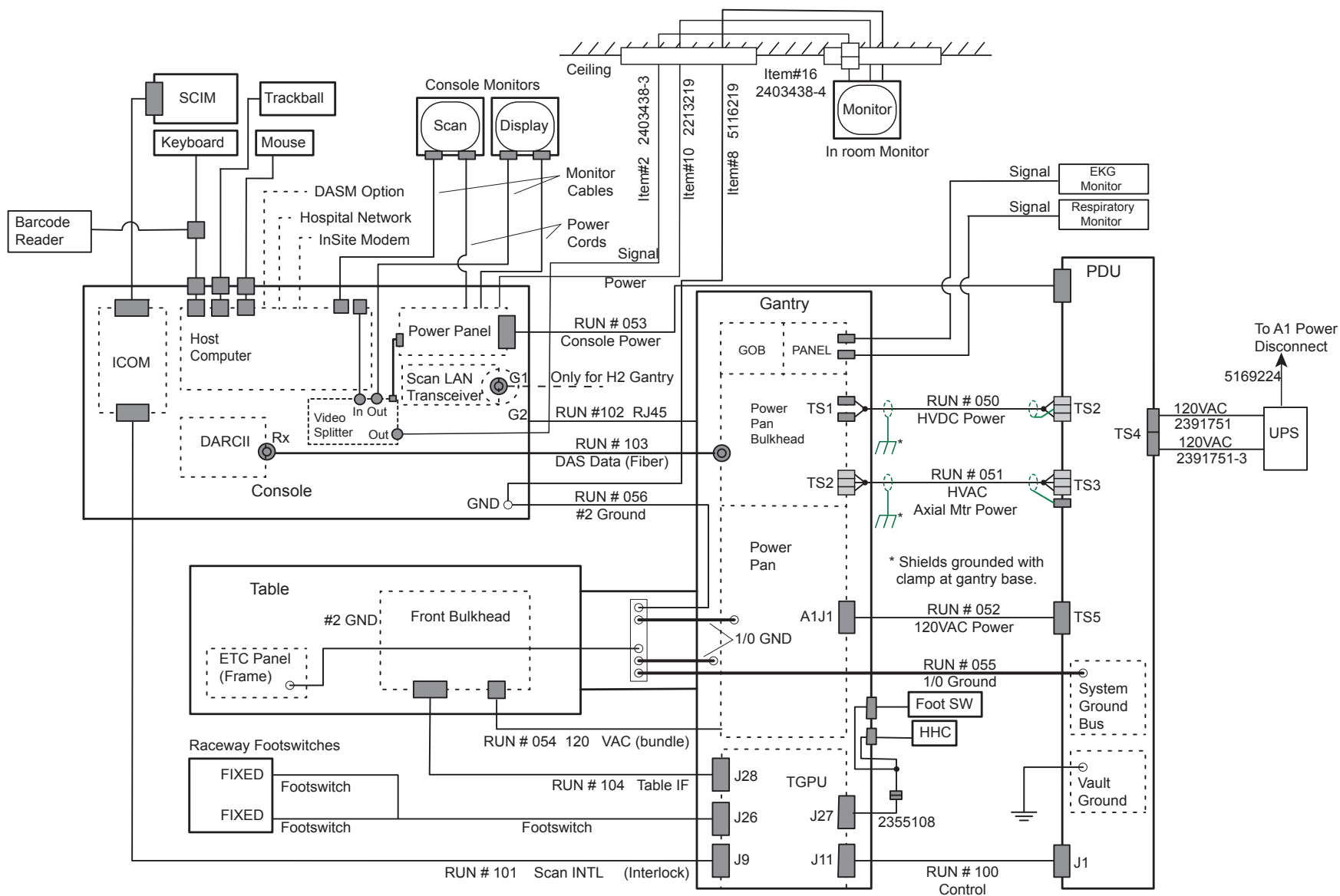
3.2 GEMS Supplied (Freedom Workspace Table Extended Cables Kit)

Extended Cables Kit (5160577) is used for FWS:

DESCRIPTION	QUANTITY	CONNECT TO	PART NUMBER	LENGTH	
				IN	MM
DVI-A HD15 Video Cable	1	Monitor	5160573	181.10+/-3.94	4600+/-100
DVI-A HD15 Video Cable	1	Monitor	5160573-2	181.10+/-3.94	4600+/-100
USB Extension Cable	1	Trackball	5160574	118.11+/-3.94	3000+/-100
PS_2 Extension Cable	1	Keyboard	5160575	118.11+/-3.94	3000+/-100
PS_2 Extension Cable	1	Mouse	5160575-2	118.11+/-3.94	3000+/-100
SCIM CABLE	1	SCIM	5160576	251.97+/-3.94	6400+/-100
Power Code	2	Monitor	5162253	216.54+/-3.94	5500+/-100

Table 9-3 Extended Cables

Note: Freedom Workspace Table is only used for BrightSpeed and BrightSpeed Select series Systems.



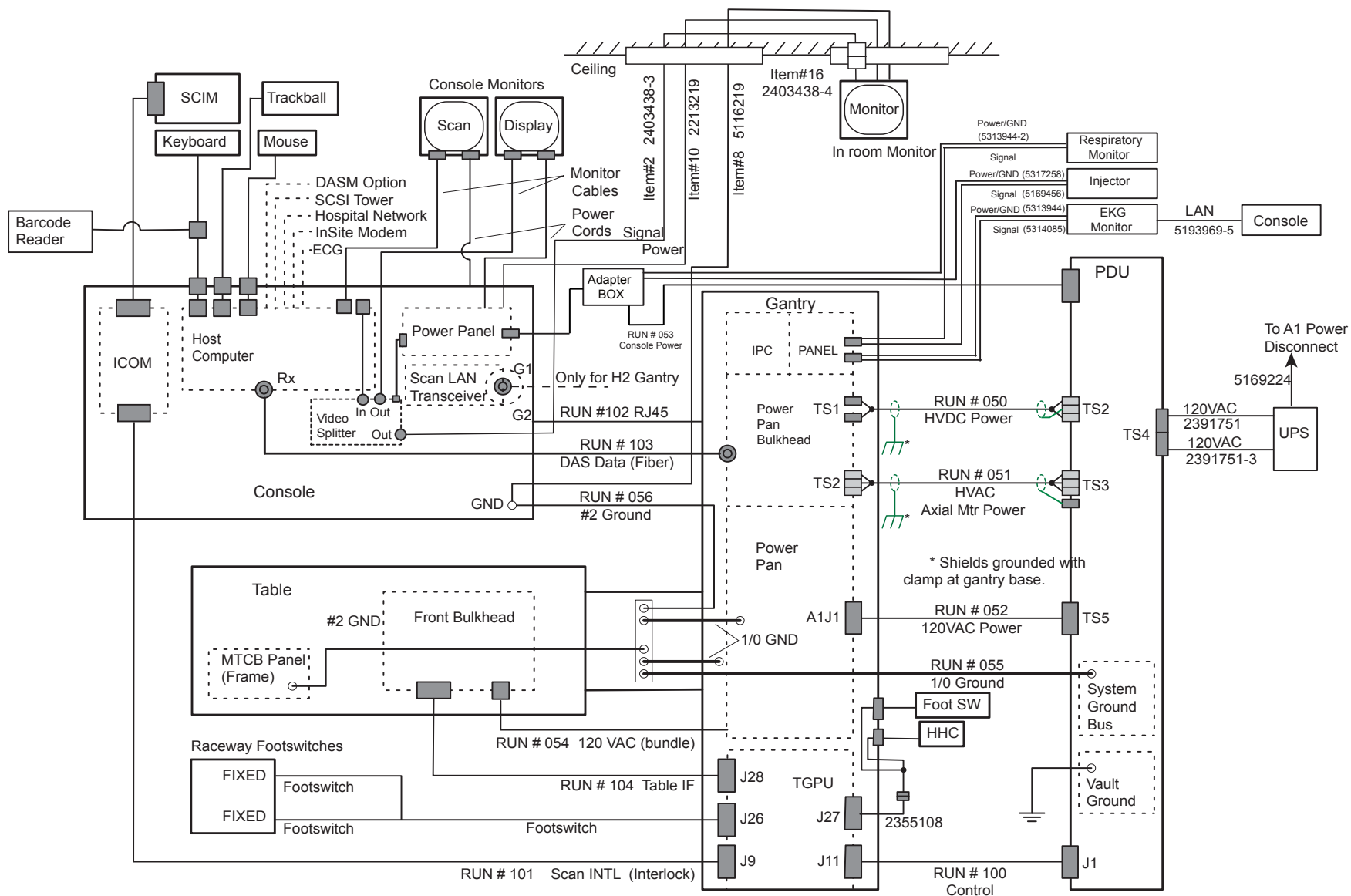


Figure 9-2 System Interconnect Diagram (With IPC Board)

3.3 GEMS Supplied (Optional, Long Run)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	63 (55)	19.3 (16.76)	2343529	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (.751)	3	(2) 4 (1) 8	22 (0.87) Dia
051	62.5 (55)	19 (16.76)	2343530	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15.3 (.604)	4	14	11.2 (0.44) Dia
052	63 (57)	19.3 (17.56)	2343528	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
053	80 (75)	24.5 (22.86)	2343531	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12.3 (.483)	3	10	56.4 (2.22) Dia
055	63 (55)	19.3 (16.76)	2371450	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 0(.62) Dia
056	83 (75)	25.5 (22.86)	2371450-3	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (0.48) Dia
100	70 (62)	21.4 (18.86)	5120646	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
101	86 (78)	26.35 (23.71)	5120645	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
102	86 (81)	26.3 (24.86)	2373436-2	Signal (LAN), Gantry to OC			1900	<30VDC		5.9 (.234)	8	24	15 (0.59) Dia
103	80 (75)	24.3 (22.86)	2117848-2	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (0.39) Dia

Table 9-4 GEMS Supplied Cables (Optional, Long Run) - UL Information

3.4 GEMS Supplied (Cables of Options)

OPTION	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
Fluoro	75	22.9	2403438-3	5 BNC MALE TO HD 15 MALE 75 FEET		FT4		1Vp-p	75	9.1 (0.358)	5	26	
	70	21.2	2213219	POWER CABLE FOR LCD-CONSOLE TO LCD		FT1	120	120VAC	105	9.3 (0.366)	3	14	
	71	21.5	5116219	Grounding Cable For LCD Console To LCD	1015	VW-1	600	0V	105		1	8	
	15	4.6	2403438-4	HD 15 FEMALE TO HD 15 MALE 15 FEET				1Vp-p	60	8.0 (0.315)	5	26	
	1.3	0.4	2355108	JUMPER CABLE FOR ADAPTING 2286150 TO WORK WITH H-POWER MSUB							8	22	
UPS	15	4.6	2391751	POWER CABLE, NGPDU TO UPS	2587	FT4	600	208VAC	90	5.8 (0.228)	5	8	
	15	4.6	2391751-3	POWER CABLE, UPS DISCONNECT PANEL TO NGPDU	2587	FT4	600	208VAC	90	5.8 (0.228)	4	8	
	45	13.6	5169224	UPS CONTROL CABLE	2587	FT4	600	120VAC	90	10.3 (0.406)	5	18	
Injector	100	30.5	5169456	GANTRY TO INJECTOR	1007	VW-1	300	<30VDC	80	1.57 (0.062)	3	22	45(1.78) Dia
	8.2	2.5	5317258	POWER CABLE INJECTOR TO CONSOLE	62	FT2	300	120VAC	60	9.4 (0.37)	3	14	36(1.41) Dia
Cardiac	30	9.1	5198566	GANTRY TO EKG MONITOR	2919	UL1685 UL loading	30	<30VDC	80	6.45 (0.254)	6	24	37(1.45) Dia

Table 9-5 GEMS Supplied Cables for Options - UL Information

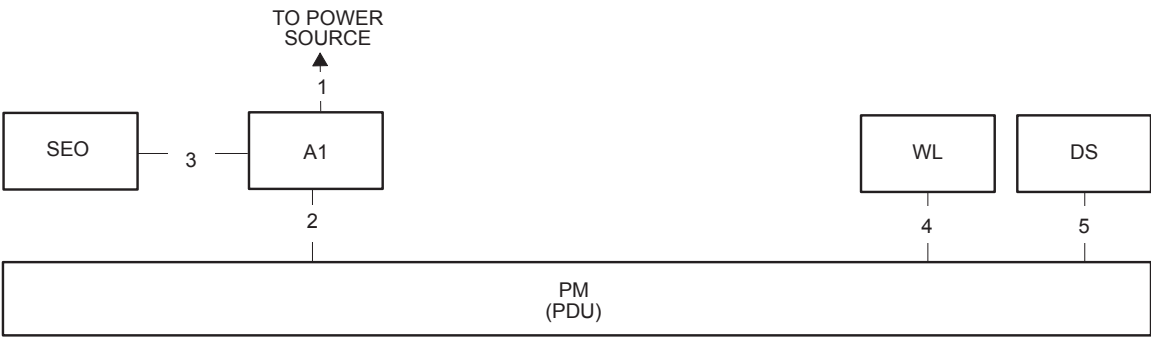
OPTION	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
Adv 4D Resp	100	30.5	5199717	GANTRY TO RPM UNIT	2464	FT4	300	<30VDC	80	6.6 (0.26)	4	22	37(1.45) Dia

Table 9-5 GEMS Supplied Cables for Options - UL Information

3.5 Contractor (Customer) Supplied

CUSTOMER INSTALLED WIRING		DESCRIPTION	CABLES SUPPLIED			PLUG PULLING DIMENSIONS		WIRE & CABLE PIGTAILS FT. (M.)	
QTY	SIZE AWG (MM ²)		PART NO	LENGTH FT. (M.)	DIA. IN (MM)	FROM	TO	FROM	TO
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1)									
Maximum Run Length *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
1	*	NEUTRAL (Not Required)						3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 4 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL A3J2-1,2,3,4							
RUN NO. 5 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK A3J6-1,2							
*	REFER TO Table 8-4 on page 101 FOR AWG (MM2) WIRE SIZES.								

Table 9-6 Runs 1, 2, 3, 4 and 5 Connections



- NOTES:
- 1) Used for remote diagnostics - Option
 - 2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
 - 3) Category 5 cable. Use one of the following patch cords:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m
 - 4) In order to avoid any violation of each National Regulation (NEC in USA, CCC in China, etc.), use of the complied cable/wire is recommended. For China market, China end-user shall purchase the power supply cable that has the CCC mark.

Only one phone connection is required for the system.

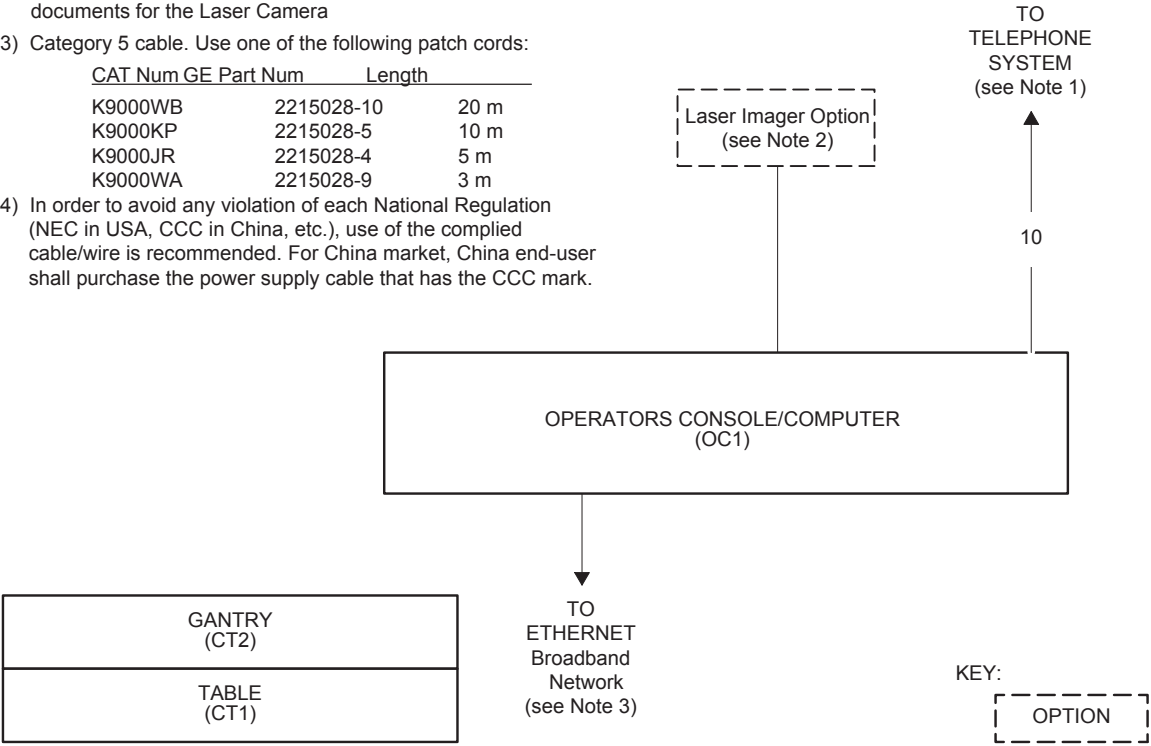


Figure 9-3 Interconnection Runs

Section 4.0

Contractor Supplied Components

REFERENCE	ASSOCIATED EQUIPMENT	MATERIAL/LABOR SUPPLIED BY CUSTOMER CONTRACTOR	USA VENDOR / CAT NO. GE CATALOG
A1 380 - 480V 50/60 Hz	Fusible Disconnect and Magnetic Contactor	3 Pole, 380V - 480V, Combination breaker with magnetic contactor. Includes control transformer, optional UPS interface, On/Off controls and auto-restart feature	Recommended:* • E4502AC (110A) • E4502AB (90A)
ITL	In-suite Telephone Lines	Supply 2 voice-grade telephone lines. One line must be a direct number from outside the facility – do not route this line through a telephone switchboard. Telephone line operating charges are paid by customer.	
BBNC (required)	Broad-Band Network Connection	Broad-Band network connection wall jack, located within 1m (39inches) of console location, for internal hospital networking and InSite Broad-Band connectivity.	
	System Components	Reference the system installation drawings supplied by Installation Support Services within your geographic area.	

*Refer to [Table 9-9 on page 114](#).

Table 9-7 Contractor-Supplied Components

Section 5.0

Fuse

ITEM	NUMBER	QTY	FRU CODE	DESCRIPTION/NAME
1	2351493	1.0	Yes	100A FUSE
2	2364059	1.0	Yes	GLASS FUSE
3	46-170021P50	1.0	Yes	FUSE 12 AMPS 250 VOLTS BUSSMAN MDA12 DUAL ELEMENT.
4	46-170021P15	1.0	Yes	Time Lag FUSE 8 AMPS 250 VOLTS (REVIEWED TK, 7/91).
5	2336517-2	1.0	Yes	FUSE 25 AMPS 700 VOLTS 2.5M SECONDS
6	46-170021P52	5.0	Yes	FUSE 3 AMPS 250 VOLTS (REVIEWED TK, 7/91).
7	46-170021P10	2.0	Yes	006.000A 0250V 3AG FAST UL/C (REVIEWED TK, 7/91).
8	46-170021P101	1.0	Yes	FUSE 20 AMPS 700 VOLTS 2~ LONG X 9/16~ DIA FAST BLOW
9	46-170021P106	1.0	Yes	FUSE 8 AMPS 250 VOLTS BUSSMANN ONLY
10	46-170021P31	2.0	Yes	1/2A, 250V SLO-BLO FUSE. TYPE 3AG, 1.25~ X 0.25~ GLASS BODY. (REVIEWED TK, 7/91).
11	2106993-5	2.0	Yes	FUSE 20 AMPS 500 VOLTS .3 SECONDS
12	46-327160P1	1.0	Yes	12.0A, 125VAC, DUAL ELEMENT TIME DLY FNM-12, 0.406 X 1.5 LG, FIBER BODY
13	2379651	1.0	Yes	FUSE - 700 V, 200 A, FAST SEMI
15	2238207-3	1.0	Yes	FUS FF 14X51 2A 700VDC C 50KA ULR
16	99183979	1.0	Yes	SLOW BLOW FUSE HCP 6X32 10A 250V UL

Table 9-8 FUSE KIT BS 90KVA (2385412-2 BOM, rev 2)

ITEM	NUMBER	QTY	FRU CODE	DESCRIPTION/NAME
17	U8009FF	2.0	Yes	FUSE 3.2A
18	U8008FF	1.0	Yes	FUSE 2.0A
19	U8006FF	2.0	Yes	FUSE 1.0A

Table 9-8 FUSE KIT BS 90KVA (2385412-2 BOM, rev 2) (Continued)

Section 6.0

UPS Interconnect

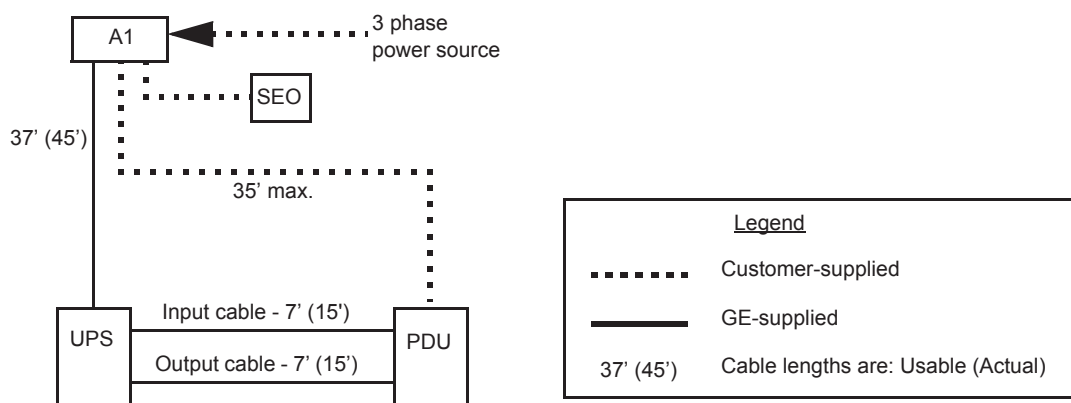


Figure 9-4 Typical UPS Interconnect



NOTICE UPS Kit B7999PP **REQUIRES** installation of one of the A1 Panels listed below.

PDU Type & Model #	Max. Mom. kVA Rating	Recommended Main Disconnect (A1) Cat #		Optional Partial UPS Kit Cat #
		Europe & Asia (380-400V or 420V)	North America (440V or 460-480V)	
NGPDU-3 2326492-3	90kVA	E4502AC (110A) (incl. Auto Restart & Integrated UPS Control)	E4502AB (90A) (incl. Auto Restart & Integrated UPS Control)	B7999PP (incl. 12.5KVA UPS&40A hardware kit) REQUIRES one of the A1 Panels shown at left

Table 9-9 BrightSpeed Series Partial UPS Back-up Options

Note: Conduit is required between:

- A1 and UPS
- UPS and PDU
- PDU and A1

Section 7.0

Typical Customer Supplied Wiring

7.1 Primary Power Disconnect

Requires LOTO compatible disconnect to install this system.

If a UPS is required, a GE Disconnect is strongly recommended for safe operation. The GE disconnect and UPS are designed to work together.



Figure 9-5 Primary Power Disconnect (A1)

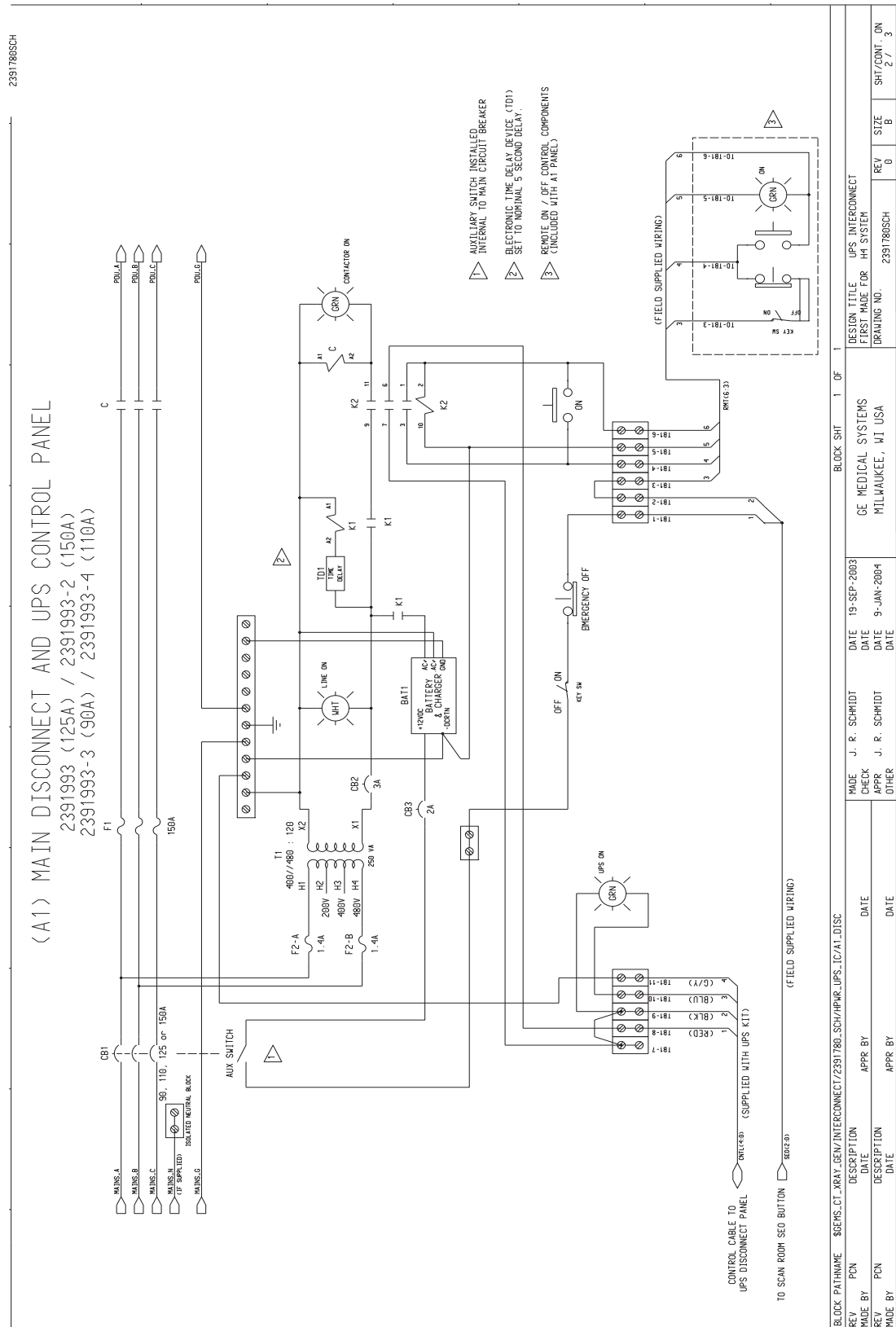


Figure 9-6 Primary Power Disconnect (A1) - Fusible Disconnect and Magnetic Contactor

7.2 Scan Room Warning Light & Door Interlock

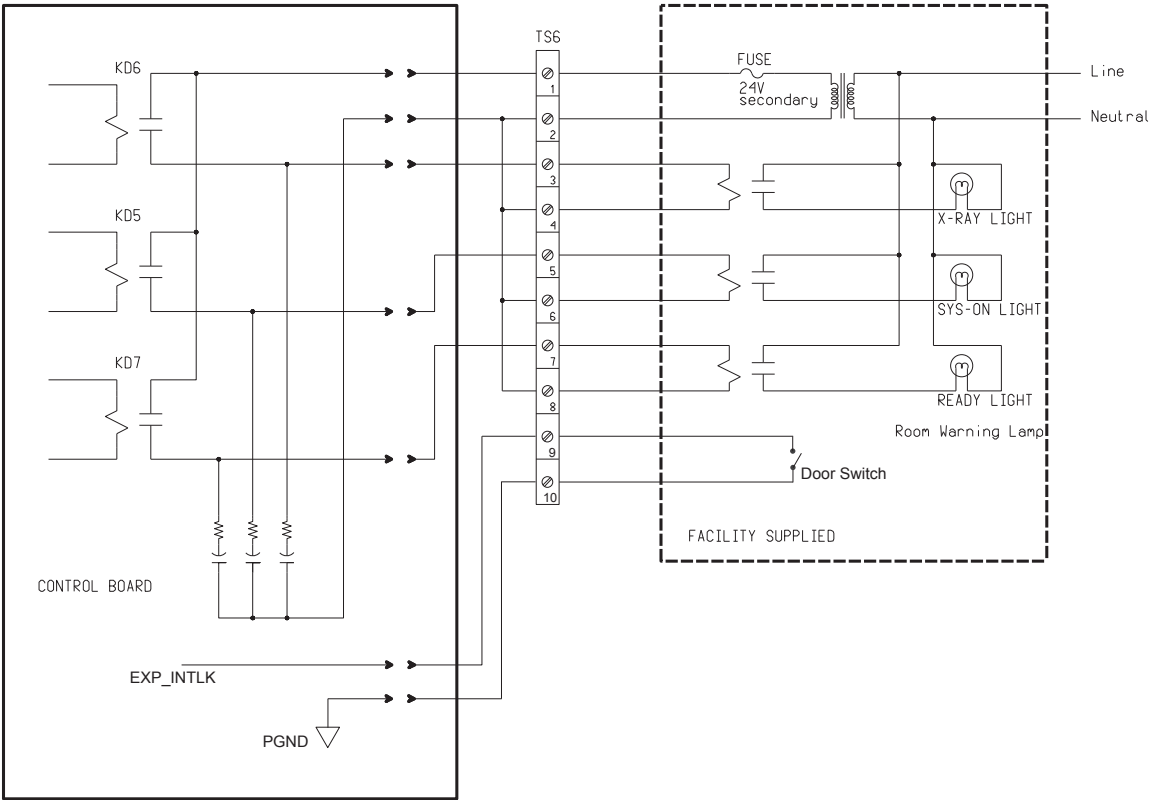


Figure 9-7 Typical TS6 Warning Light & Door Interlock Connections

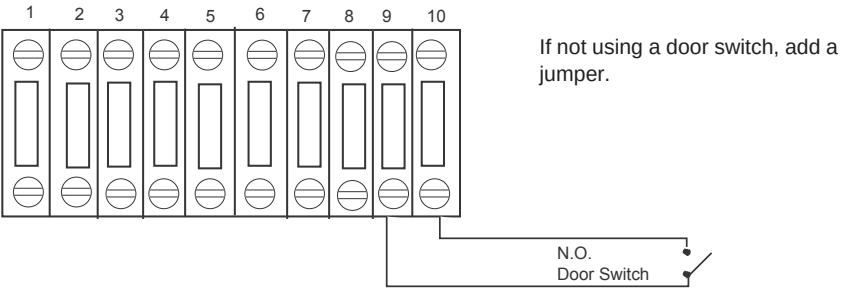


Figure 9-8 TS6 Room Door Interlock Connections - With a Door Interlock

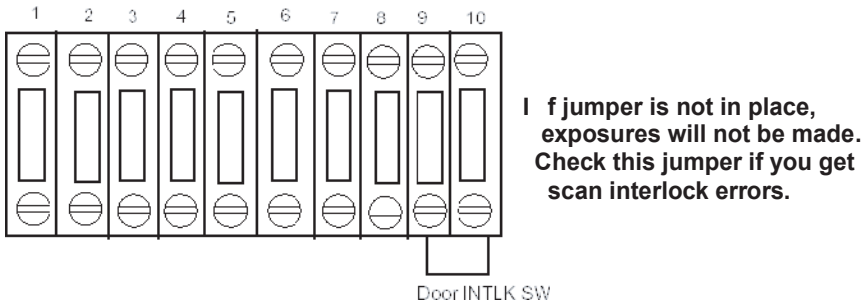


Figure 9-9 TS6 Room Door Interlock Connections - Without a Door Interlock

Appendix

CT Installation Site Ready Form

This appendix contains the CT Installation Site Ready Form, which is required for the Pre-Installation process.

CT Products Installation Site Ready Inspection Form**Required Site Information**

Customer Name Install Date Site Visit Date
 Order Number System ID: GEE Technician

IS Present: Select Customer Present: Select GE final site prints on site Select

This site ready visit must be completed at least 3 business days prior to the installation date

These items must be on site.

SRV Items

- 1) Actual room size must match the site print, including cabinets, sinks and accessories
- 2) Do Not continue this inspection without an approved final site print.

a) GE site prints Project/Rev # Product type Select

Check if present

- b) Quick start kit or PIM on site ☐ **required for this task**
 c) Pre-install video reviewed ☐ **required for this task**
 d) Floor template ☐ Optional order if required

Customer Checks**Customer Awareness Items (review only)**

Radiation Shielding Was shielding reviewed by a health physics for the product to be installed ☐
Structural Floor Vibration Checked ☐ Site EMI Reviewed ☐
Operational Clearances Will a bed fit in room ☐ Work space around table gantry reviewed ☐

Site Delivery Information

Delivery route inspection Doors and hallways check will Select tilt, or zero clearance dollies needed

Select One Select One Riggers needed Select Elevator size & load limits checked ☐

Site Information

System ID created Select One Network ID's address sheet completed Select One
 Camera setup information available Select One Equipment Order reviewed Select One
 Customer Contact Phone # Identify customer to accept system
 De-Installation Required /planned Date:

Requirements

Site Requirements Escalations required if specification are not met.

Network Communication Broad Band installed & Active Select One see page 2

Scan Room Requirements Met [90" min. ceiling ☐

Room size [w/GT1700 table min. 13'-6" x 19'-21"] ☐ [w/GT2000 table min. 13'-6" x 22'-0"] ☐

Structural Requirements Met

4" Floor thickness ☐ Meets floor level specification (+/- 1/8") ☐ Anchoring Select one

HVAC Requirements Met

Can maintain (70 degree F) minimum ☐ [Temp = 64° to 79°F, max. 5°F change per hr]

Can maintain minimum Humidity spec ☐ 30% minimum

Electrical Requirements Met 480 VAC 125 Amps ☐ 380 VAC 150 Amps ☐ LOTO Compatible Disconnect ☐

Power & Quality Audit Done Preliminary inspection must be completed Select One

All Regulatory Clearances Confirmed Select One **Note:** Must meet ALL **Regulatory requirements to pass**

Site condition:

Finished Ceiling ☐, Floor & Cabinets Installed ☐ Walls Painted ☐ ALL Construction in the area Completed? ☐

Note: The Area must be in move in condition. IF **NO** Reschedule

CT Products Installation Site Ready Inspection Form

Site ready Report

GE Energy Site Ready Report:

Site Issues: (Pro's & Con's) List all observed site issues that would present installation problems

GE PMI - Delivery Details:

Check if a short term storage location is required. Will floor protection be required for the delivery

Broadband:

Has the BB Specialist completed the BB installation GE setup and setup? Notify the zone BB specialist if there are questions. Broadband is expected to be complete at install.

Inspection Results

The site ready visit will result in a report to the project manager indicating one of the following:

Pass - All required items are present, completed and the site is ready for installation. Must meet all regulatory requirements, have a final site print that accurately matches the room to **pass**.

Conditional Pass - is issued when 80% of all of the required tasks are completed and all parties agree that the 20% will be completed by the installation delivery date.

If a "Conditional Pass" is granted on the inspection date, the project manager must present conclusive evidence that unfinished tasks are completed and that the site will be ready one business day prior to delivery. Must meet all regulatory requirements, have a final site print that accurately matches the room.

Fail - is issued when less than 80% of the tasks are completed, area is under construction and all parties cannot agree that the remaining work will be completed by the requested installation delivery date. Failed sites shall be rescheduled when all of the above items are met and the room is completed.

Re-inspection Date Are pictured attached?

Inspection Attendance

Zone LCT

Comments:

All US installations - E-mail to: brian.rendina@ge.com

All US site escalation should be sent to:

brian.rendina@ge.com; general site ready issues

bradley.bassler@med.ge.com or SRV, site prints, construction and delivery items

george.farrington@med.ge.com or charles.bruckl@med.ge.com regulatory and clearances issues

Note: Regulatory requirements can not be changes or deviated.

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